

# NxM Templates for SuperCDMS SNOLAB Run 4 using Ge Activation Data

K-line event selection · 1x1 and NxM (PCA) templates

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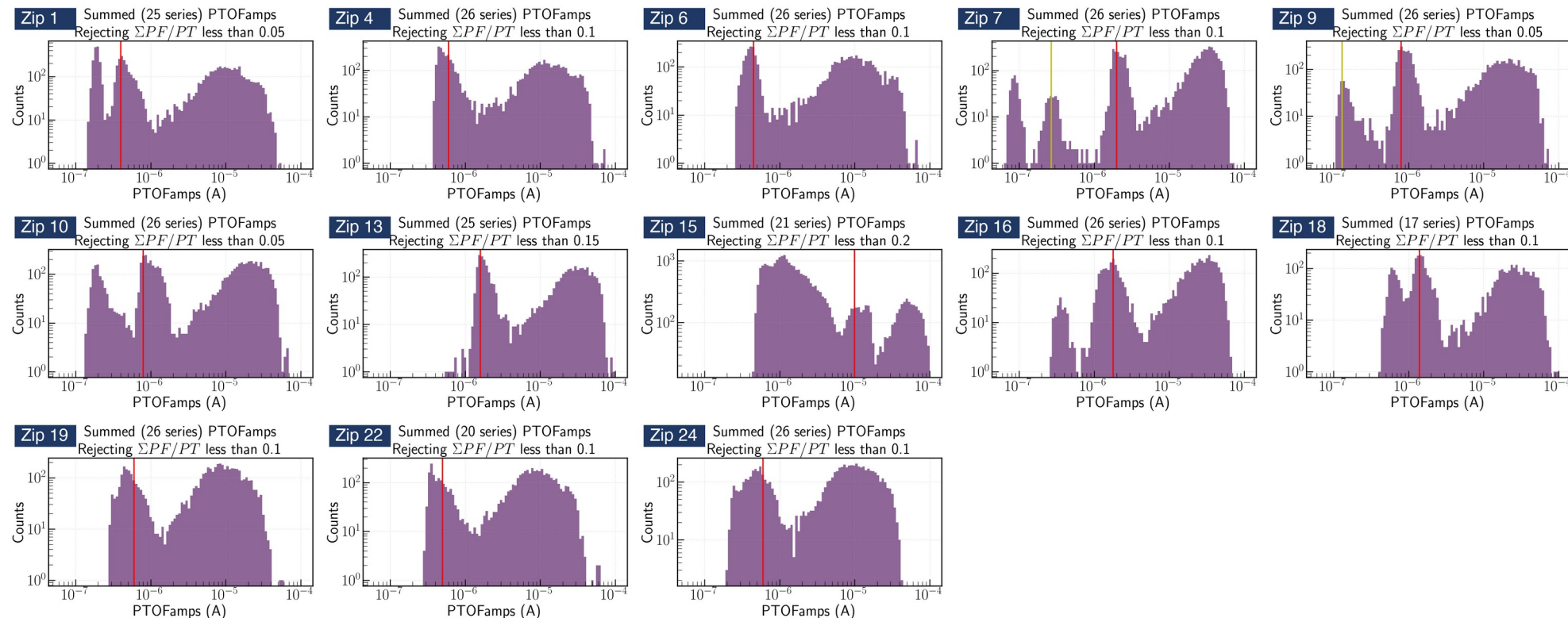
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- 1. Dataset:** Ge activation K-line event selection, raw trace cache (13 detectors, 23–30 series each,  $\approx$ 1–2 h per series)
- 2. Pulse fit:** low-pass  $\rightarrow$  fit  $\rightarrow$  align
- 3. Data cleaning:** fit\_ok, NRMSE valley cut (0.4), rise-time ceiling (0.3 ms)
- 4. Cross-checks with real pulses:** cuts remove noise, real pulses stay; slow-fall follow-up
- 5. Template generation:** 1x1 weighted template, NxM PCA templates
- 6. Template file and future steps:** cdmsbats ROOT files, ready for merge; three-, four-exponential fits
- 7. Backup:** weak detectors, rise-time cut details, full results for every zip

# Ge activation dataset

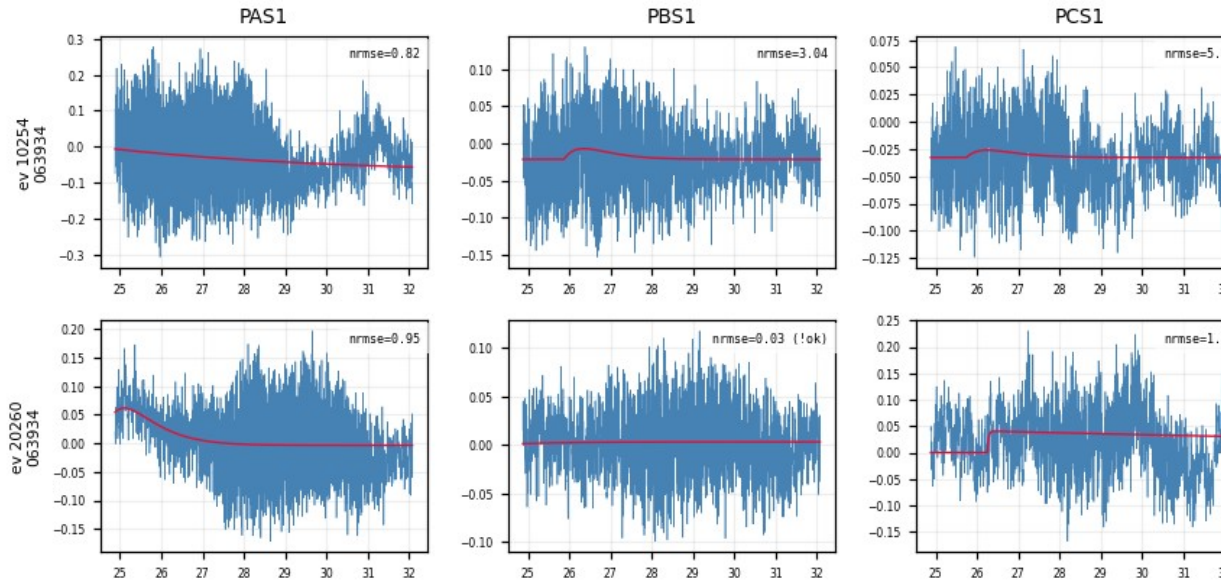
- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in **PTOFamps** (the red line in each panel below)
- A **PTOFamps** window bracketing the red line: position  $\times/\div 1.35$ , a rough eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata, saved as per-series pickle files (reusable for any later analysis)
- Events in the window per zip: Z1 10.1k Z4 26.1k Z6 17.0k Z7 2.2k Z9 2.6k Z10 7.6k Z13 9.1k Z15 4.2k Z16 1.6k Z18 22.1k Z19 23.3k Z22 25.0k Z24 22.9k (total 174k)
- **Focus: Z7, the best detector; examples use its PBS1 channel; other detectors in the backup slides**



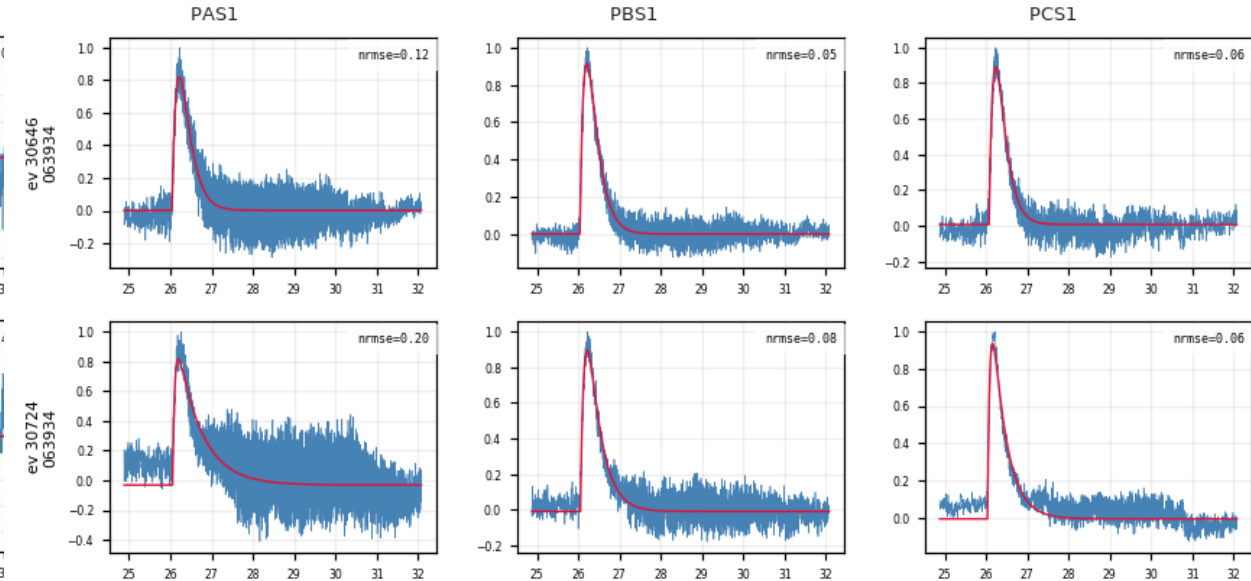
# Fit examples

- Real events and noise triggers can be told apart by visual comparison (no cut applied at this stage)

## NOISE



## REAL PULSE





# Per-trace algorithm: low-pass → fit → align

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1. **Low-pass**: 100 kHz Butterworth
2. **Normalize**: subtract the baseline (median of the pre-pulse samples), scale the trace to unit peak
3. **Fit**: two-exponential model:

$$y(t) = A[ e^{-(t - t_0)/\tau_{\text{fall}}} - e^{-(t - t_0)/\tau_{\text{rise}}} ] + b$$

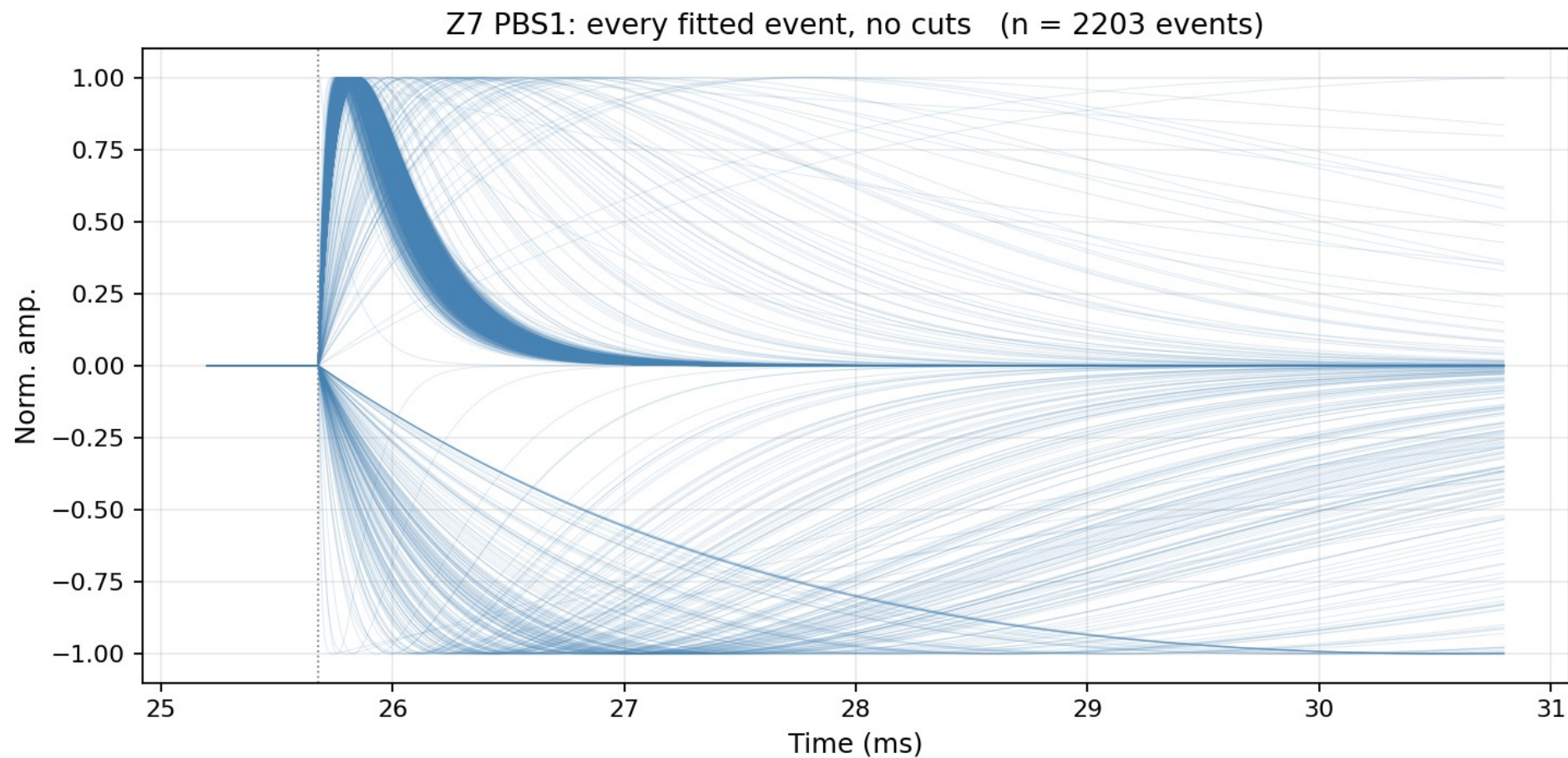
$t_0$  = pretrigger, free within  $16050 \pm 3000$  samples ( $\tau_{\text{rise}}$ ,  $\tau_{\text{fall}}$ : rise / fall time)

*Fit parameters: A amplitude ·  $\tau_{\text{rise}}$  rise time constant ·  $\tau_{\text{fall}}$  fall time constant ·  $t_0$  pulse onset (pretrigger) · b baseline*

4. **Two quality checks per trace** (calculated and recorded here, not cut yet):

- **fit\_ok**: amplitude is positive, and the pulse rises faster than it falls
- **NRMSE**: RMS of the fit residual ÷ pulse peak (smaller = better fit)

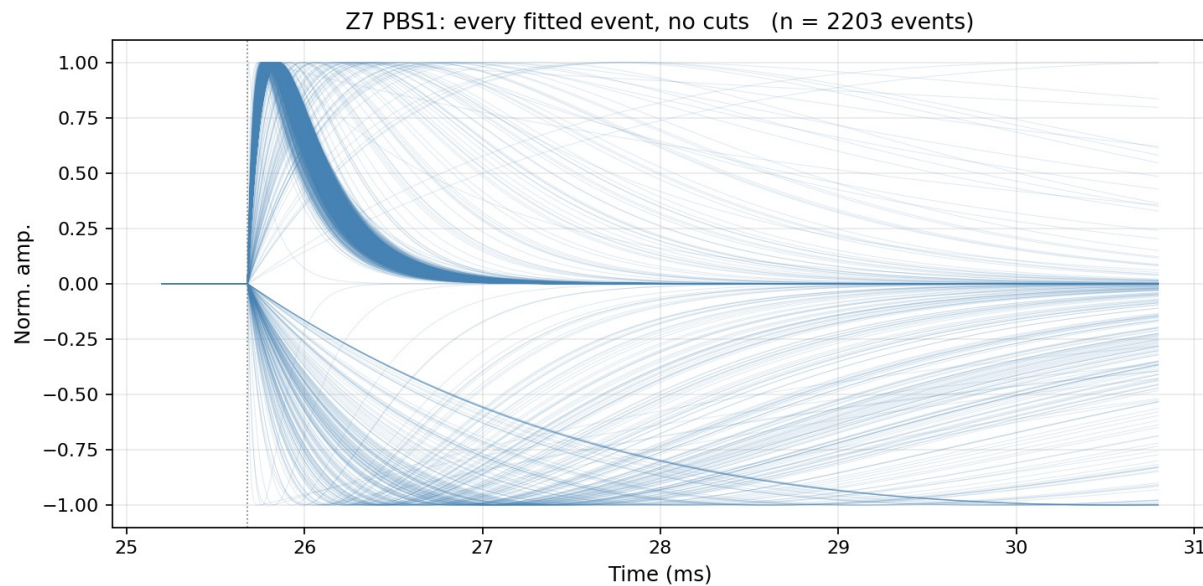
## 5. Align



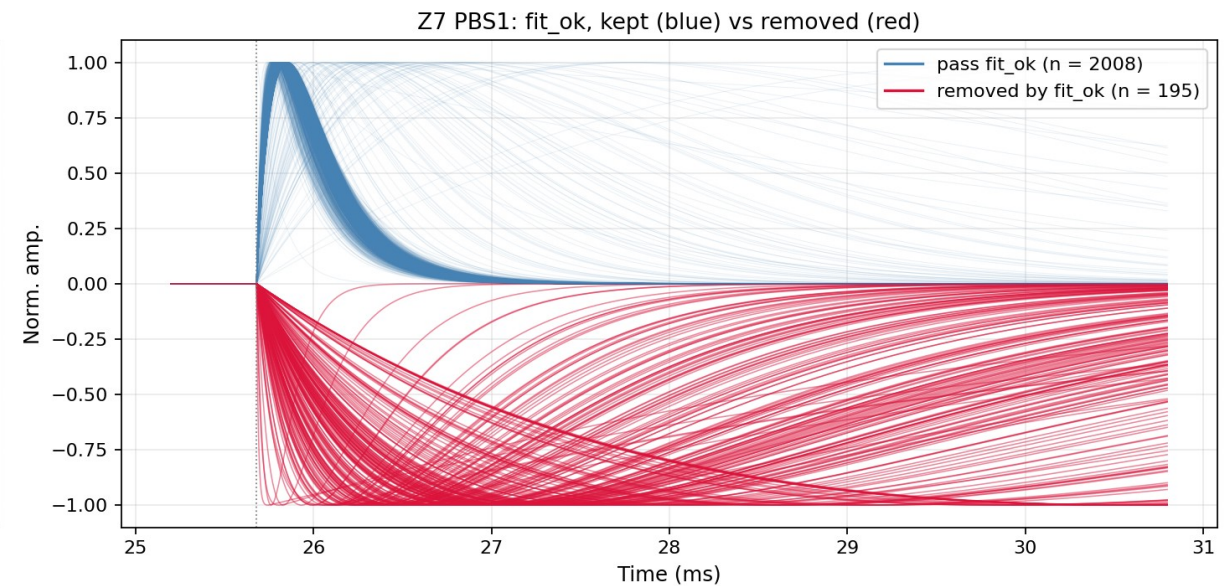
*Z7 PBS1: every fitted event drawn at the common pretrigger, no cuts of any kind (n = 2203, inverted fits included)*

# Quality cut 1: fit\_ok

- fit\_ok: amplitude > 0 AND rise faster than fall (the fit is a physical pulse)
- Z7 PBS1: 2008 of 2206 events pass; 195 removed (red, inverted or negative shapes), 3 fits did not converge



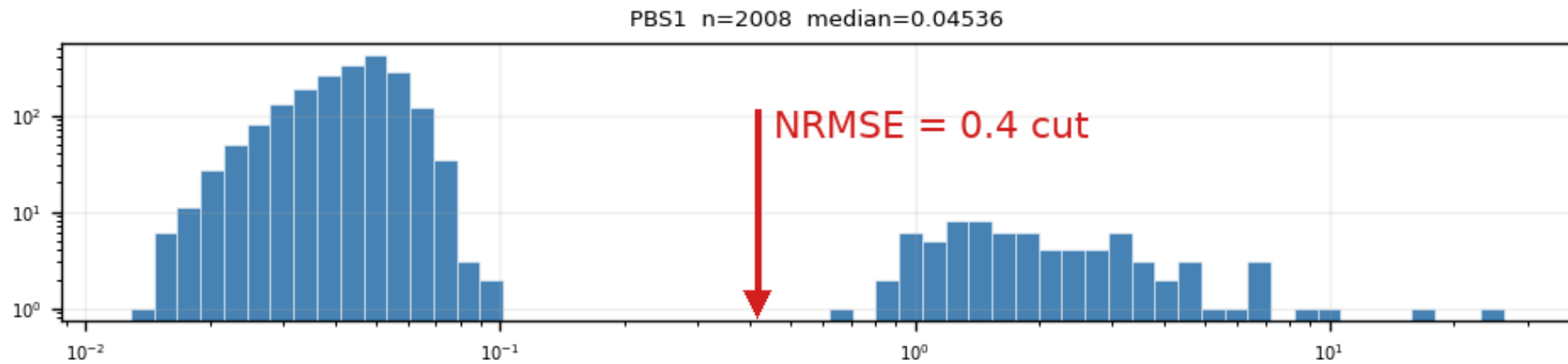
*All fitted events, no cuts (n = 2203 drawn, inverted fits included)*



*fit\_ok: kept (blue) vs removed (red, drawn at |peak| normalization)*

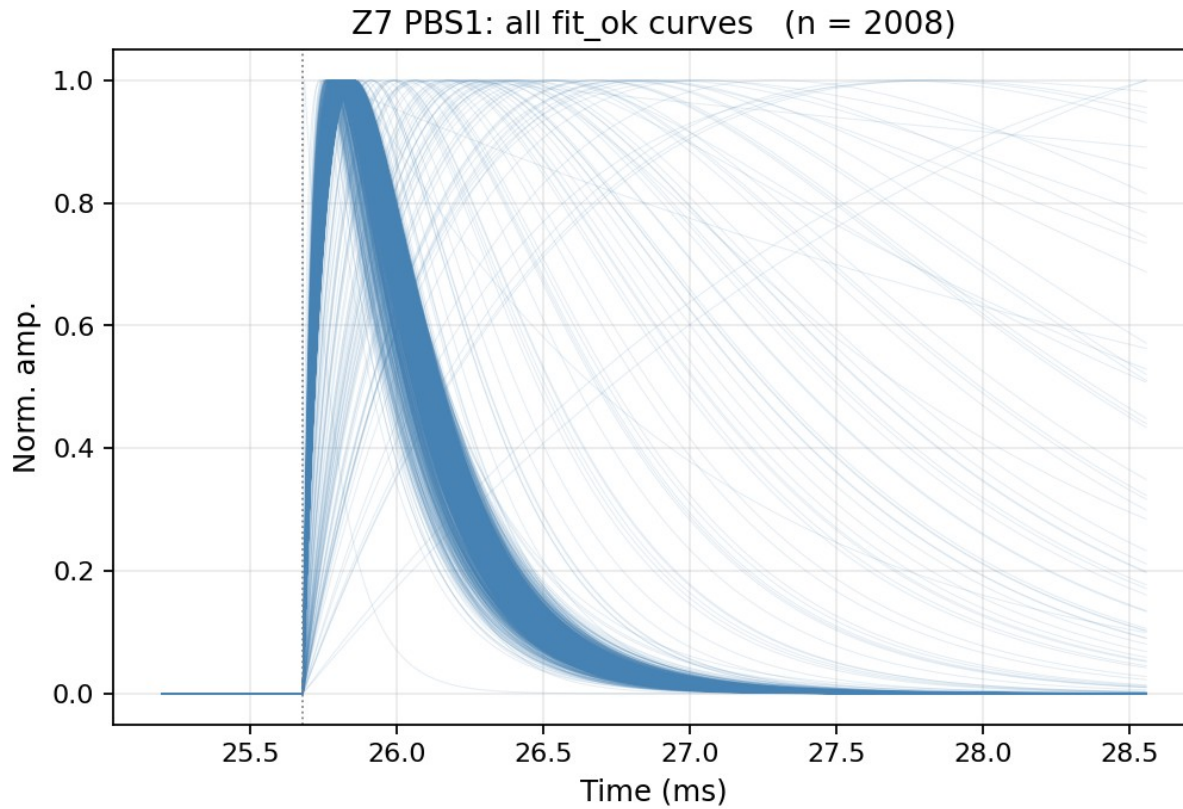
## Quality cut 2: $\text{NRMSE} \leq 0.4$ — where the number comes from

- NRMSE of fit\_ok events is bimodal: good fits (median  $\approx 0.05$ – $0.1$ ) vs noise triggers ( $\approx 1$ – $2$ ), valley at  $\approx 0.4$ – $0.5$
- The cut sits in the valley: eyeball read off the distribution
- Worse detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates

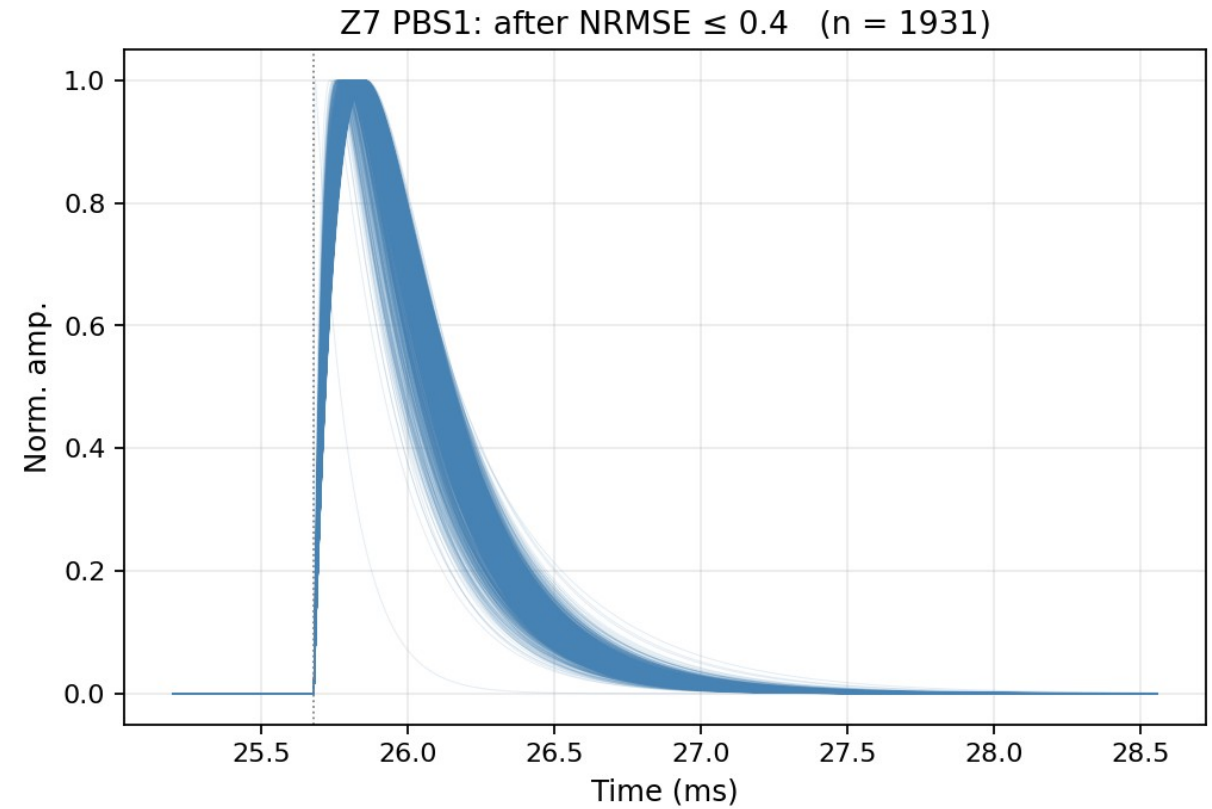


Z7 PBS1: two clean populations, log-log axes

# Aligned fitted curves: before and after the NRMSE cut



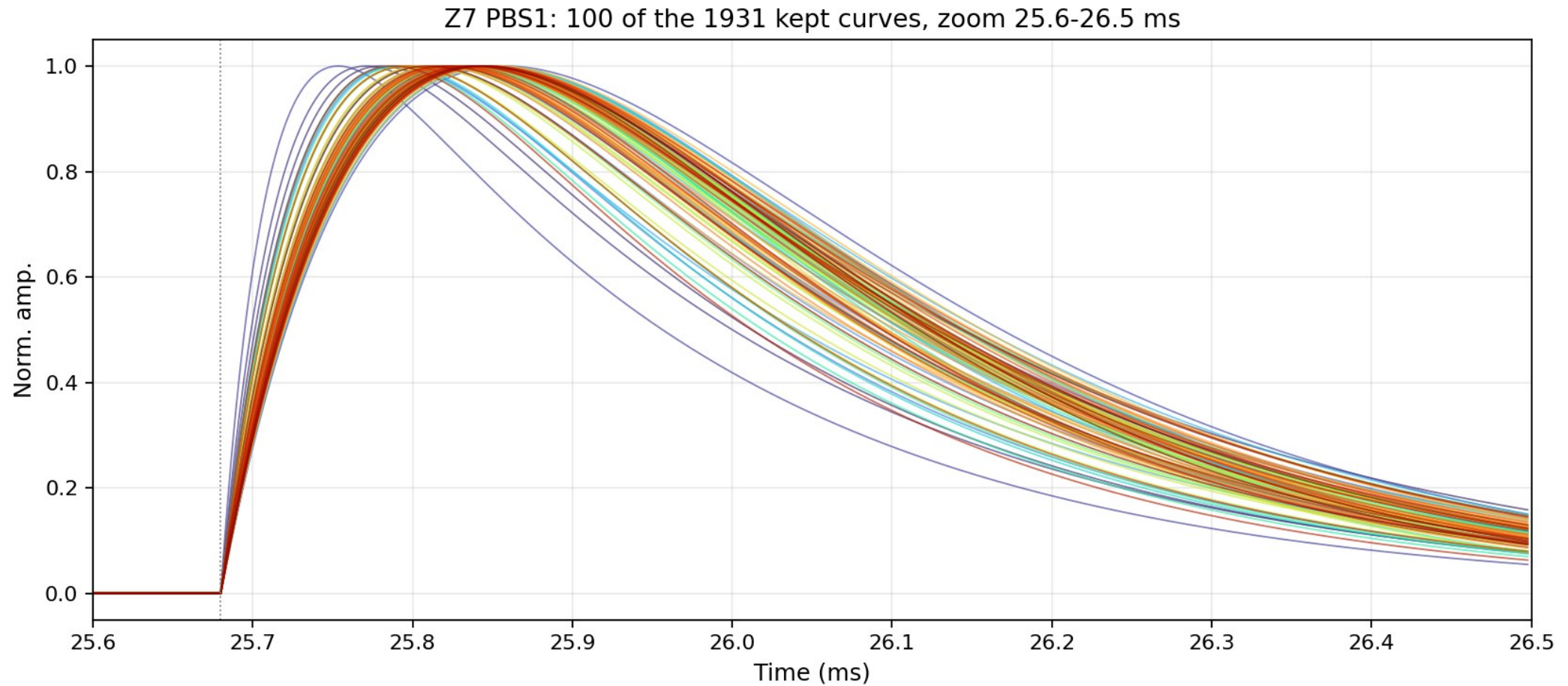
*All fit\_ok fitted curves (Z7 PBS1, n = 2008, every curve drawn)*



*After NRMSE  $\leq 0.4$  (n = 1931): one tight shape family*



## After $\text{NRMSE} \leq 0.4$ : a closer look

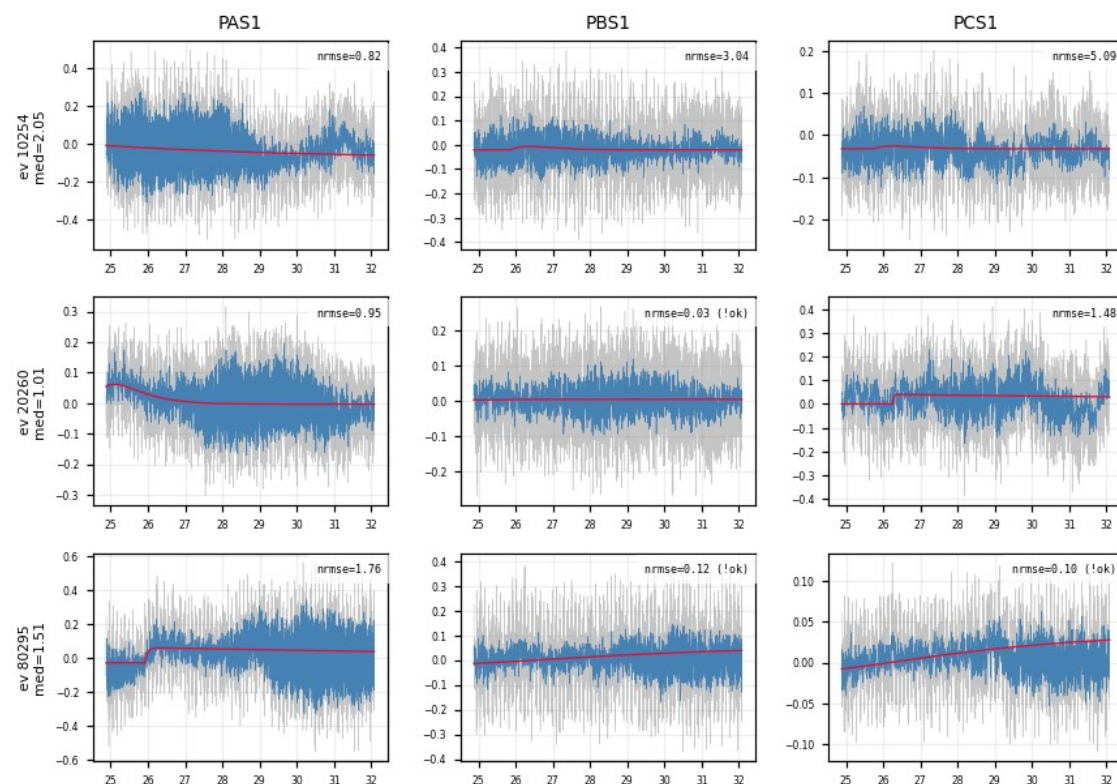


*Z7 PBS1: 100 randomly drawn curves of the 1931 kept, zoomed to 25.6–26.5 ms*

# The rejected population is noise — verified in the raw traces

- Example of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse
- the cut removes noise triggers, not physics

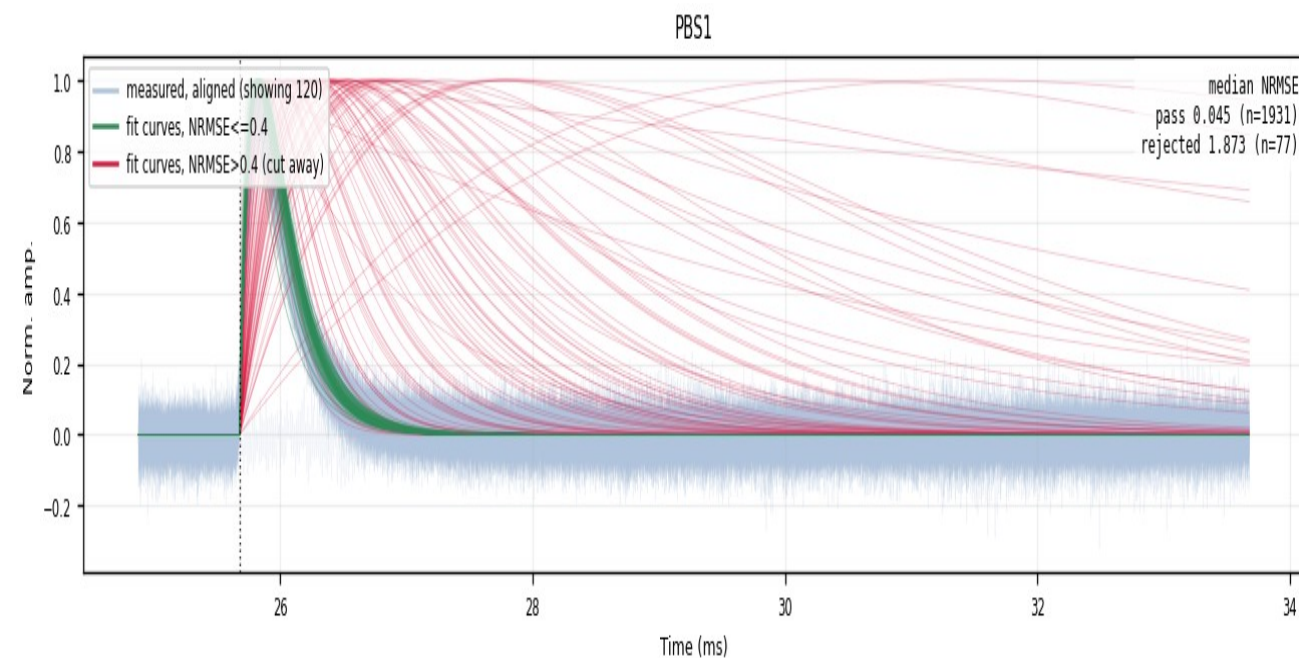
## Rejected events: raw vs fit



Z7: NRMSE-rejected events — raw traces are noise

## Aligned curves: kept vs rejected

median NRMSE: **kept 0.045 (n = 1931)** · **rejected 1.873 (n = 77)**

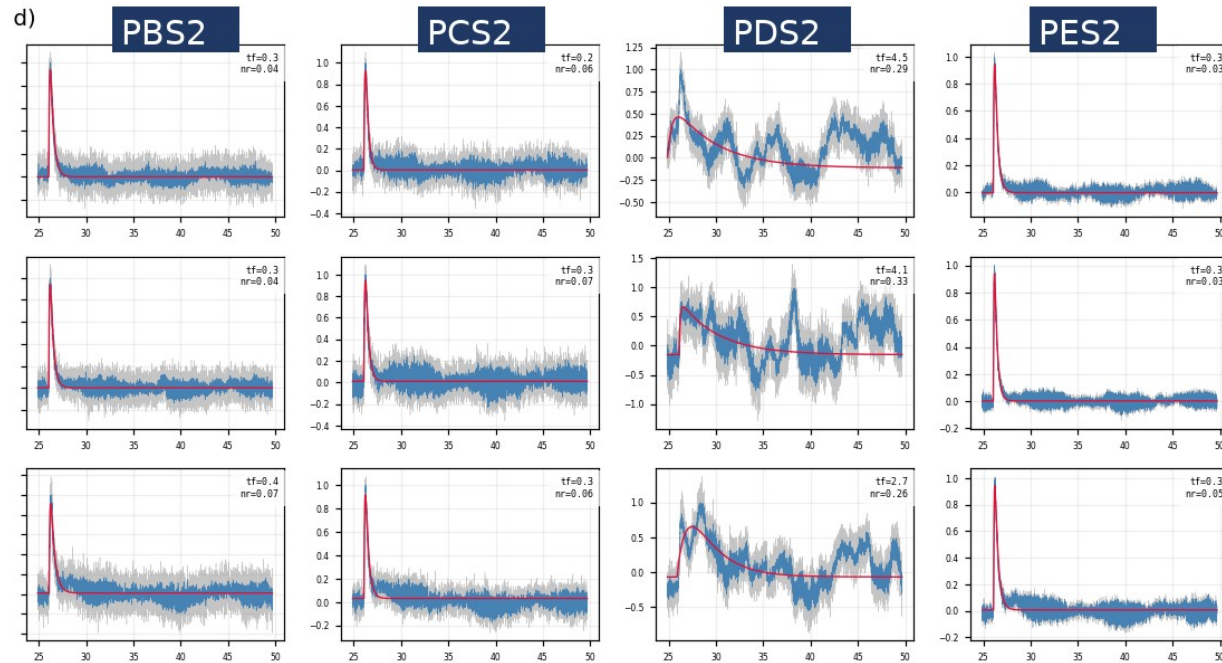


Z7 PBS1 — **passes the cut (kept)** — **cut away = rejected**  
faint blue = measured traces 1931 kept / 77 cut (~3.8%)



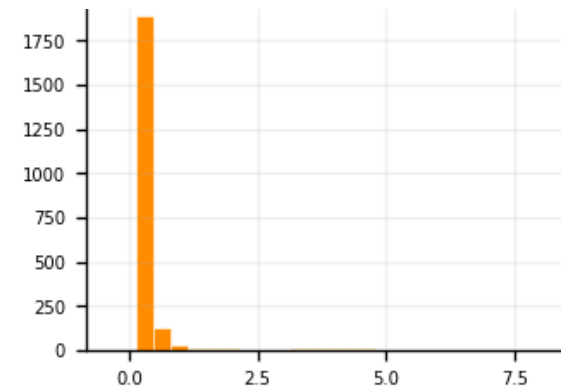
# Slow-fall tail in one particular channel

- PDS2 alone: can not make useful templates; hard to fix in analysis/software, the noise itself needs to be fixed
- Temporary solution: use the PDS1 template in place of PDS2 (applied in the ROOT files)

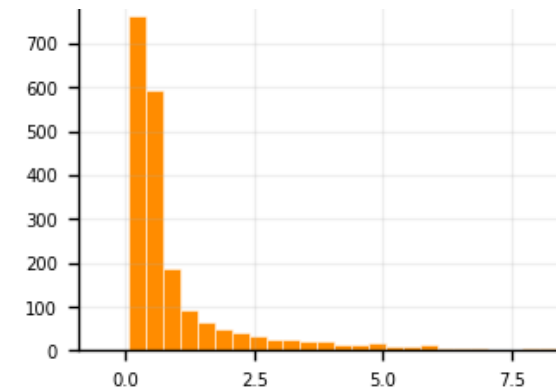


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

## Fitted $\tau_{\text{fall}}$ distribution, same 0–7 ms axis:



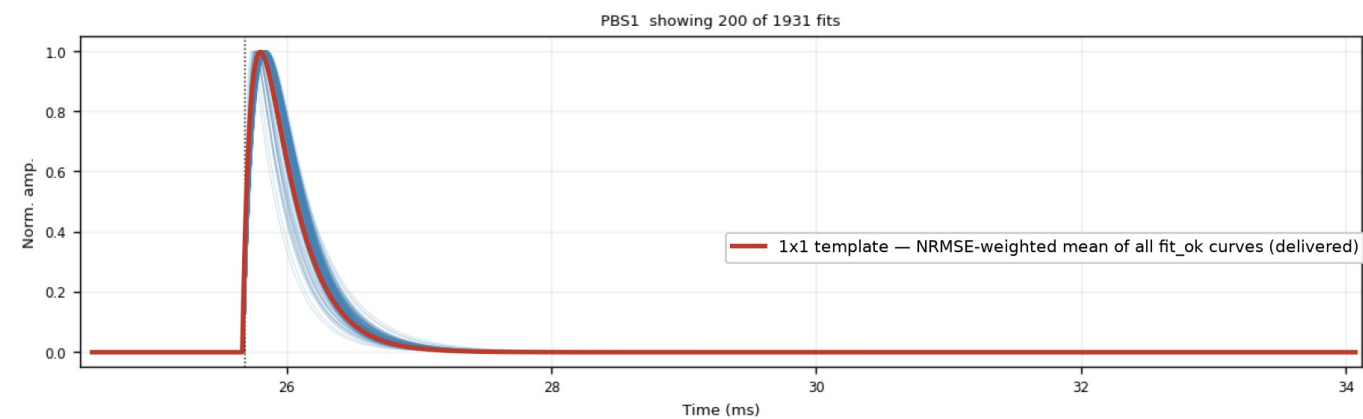
Z7 PAS1 — normal channel: narrow  
(median 0.25 ms)



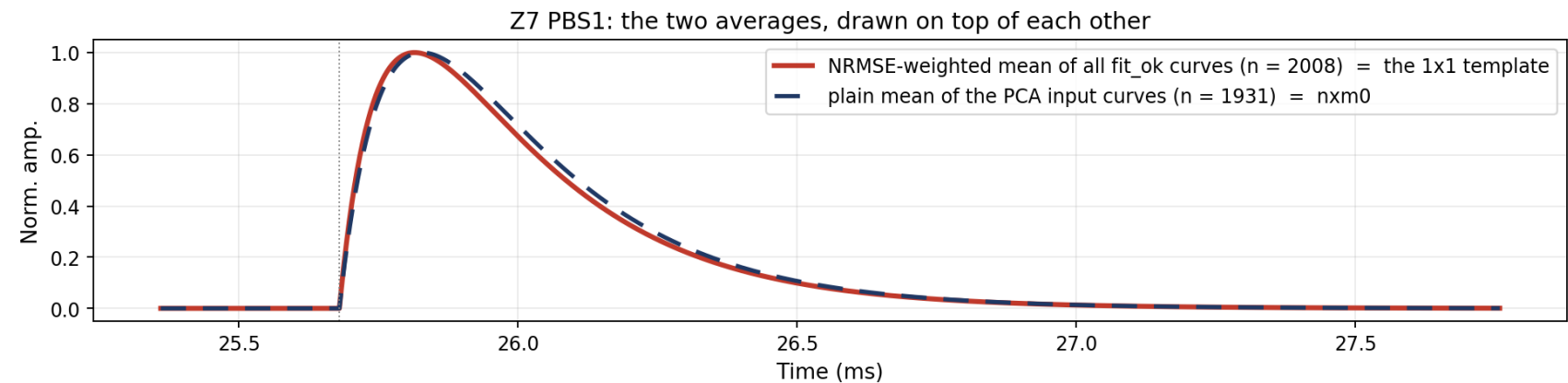
Z7 PDS2 — bad channel: broad tail  
(median 0.51 ms)

# Results: 1x1 Templates

- NRMSE-weighted mean of the fit\_ok 2-exp curves
- Smooth & noise-free by construction



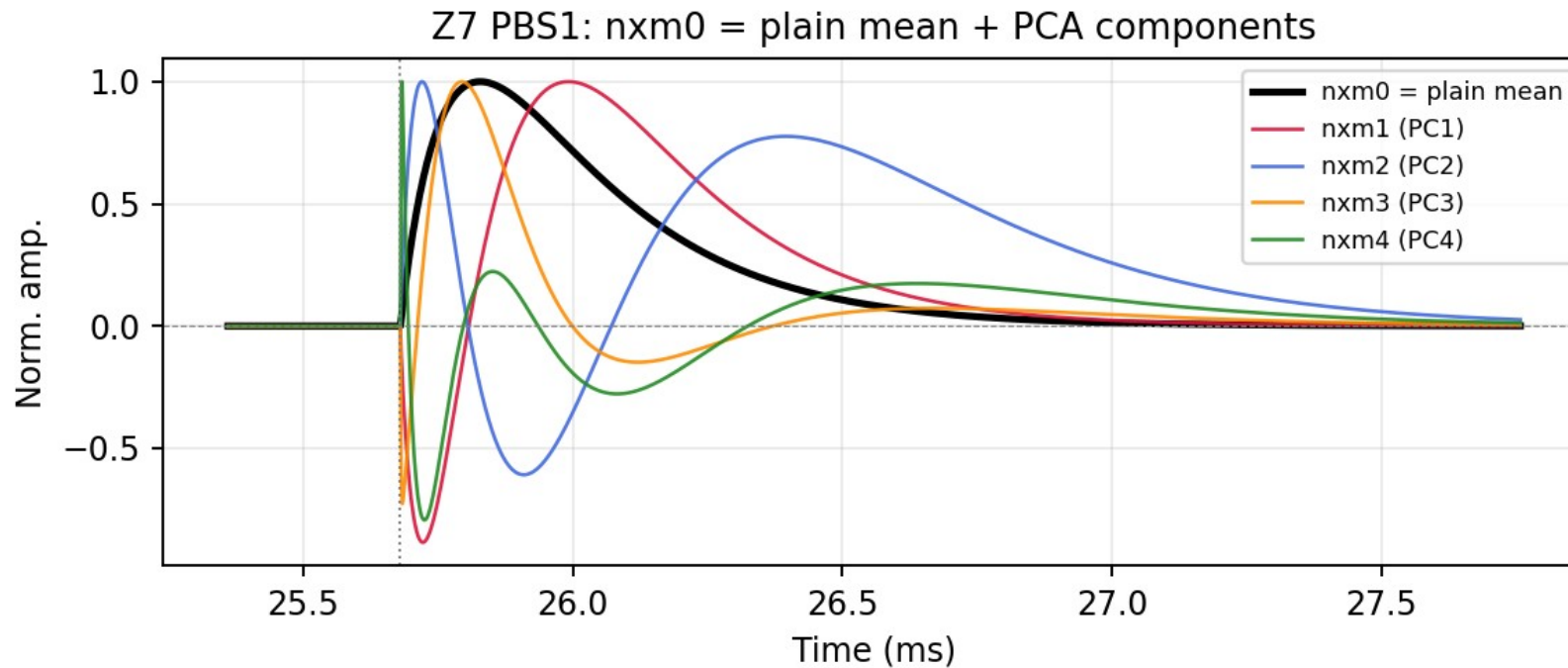
Z7 PBS1, blue: the fitted curves (200 of 1931 drawn); red: the 1x1 template



Solid red: NRMSE-weighted mean (= the 1x1 template); dashed navy: plain mean of the PCA input curves (= nxm0)

# Results: NxM Templates

- channel-specific PCA over the fitted curves ( $\text{fit\_ok} + \text{NRMSE} \leq 0.4$ )
- $\text{nxm0}$  = mean shape;  $\text{nxm1}$ –4 = principal components;  $\text{real pulse} = \sum_i \text{amp}_i \cdot \text{nxm}_i$
- $\text{PC1} + \text{PC2} \approx 96\text{--}98\%$  of the shape variance



Z7 PBS1:  $\text{nxm0}$  (plain mean over the  $\text{fit\_ok} + \text{NRMSE}$ -cut curves) +  $\text{nxm1}$ –4, zoomed to the pulse

# Template file

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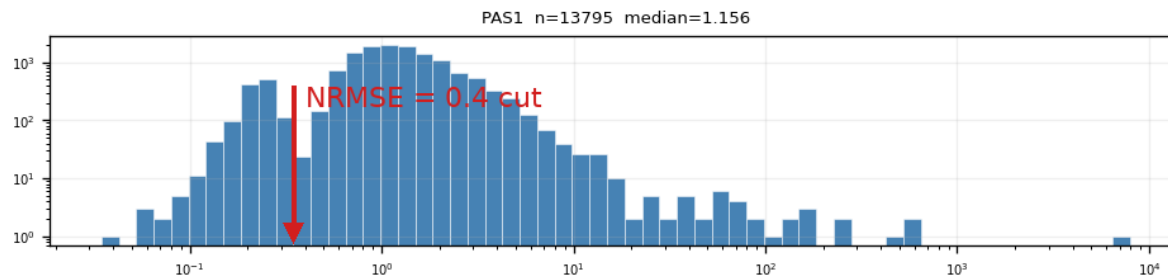
- Two template sets for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- In cdmsbats\_config feature branch, ready for merge

## Future Steps

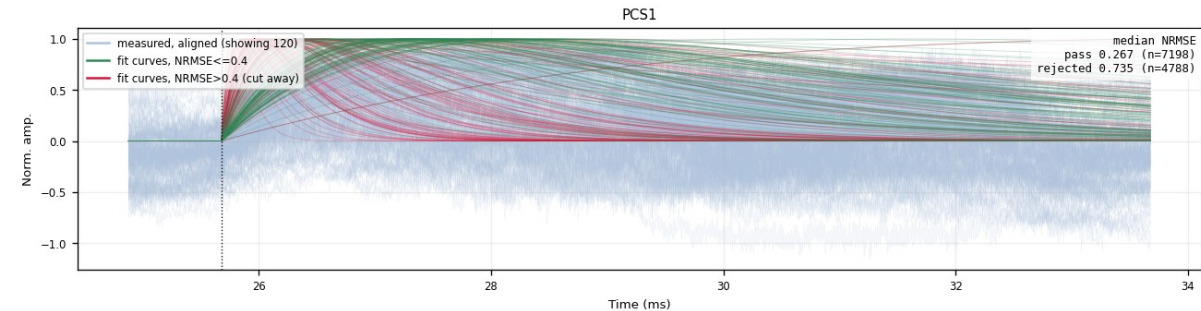
- Three-, four-exponential fits
- Exercise NxM processing
- Other detectors: worse noise, a few more cuts added, in the backup slides

# Backup · weak detectors (example: Z22)

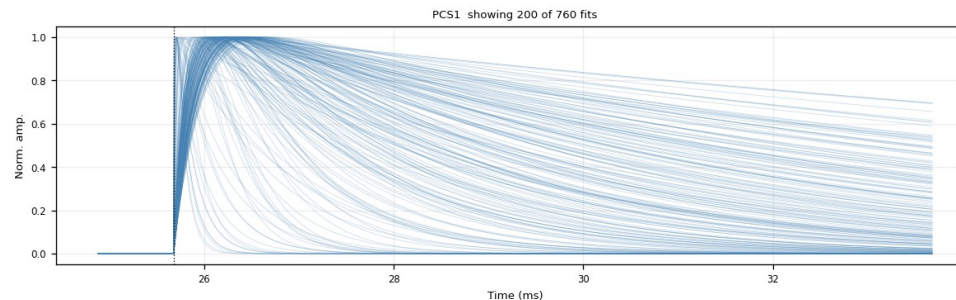
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line inside the noise population, the window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominant; the 0.4 cut digs the real pulses out (Z22 PCS1: 7198 kept / 4788 cut, ~40%)
- Kept bundle broader; steep red curves = fits latching onto noise spikes, no real pulses lost
- $\tau_{\text{rise}} \leq 0.3$  ms (after NRMSE): removes 55–74% on weak zips (Z22: 71%) vs 1.6% on Z7; removes residual slow drift



Z22 PAS1 NRMSE histogram: noise dominates



Z22 PCS1: kept (green) vs cut (red)

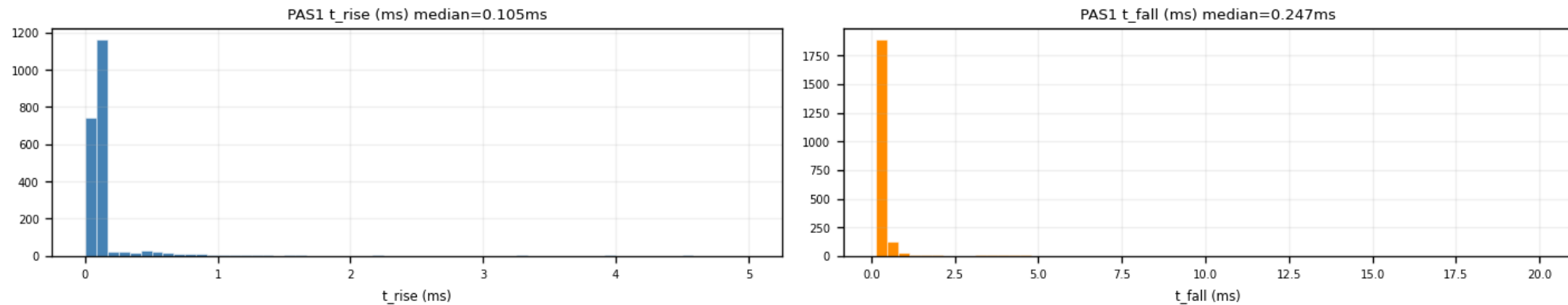


Z22 PCS1 after  $\text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3$  ms: final shape family (760 fits)



# Backup · $\tau_{\text{rise}} \leq 0.3$ ms ceiling (PCA input)

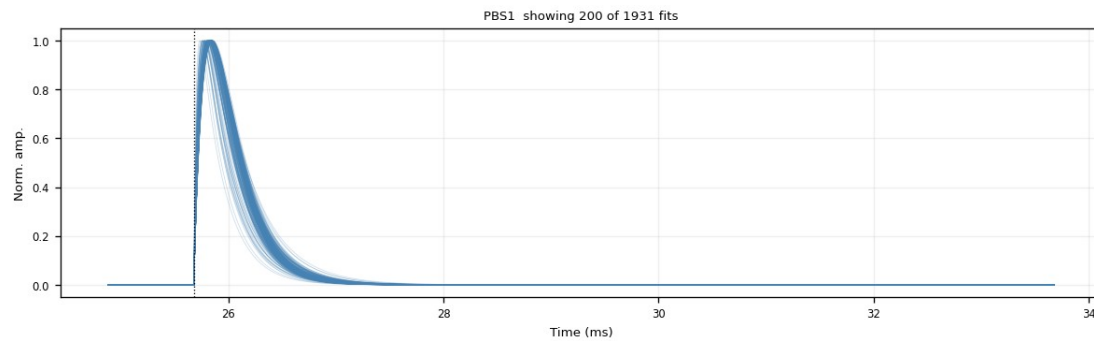
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at  $\tau_{\text{rise}} \approx 0.1$  ms (p90  $\approx 0.15$  ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%), weak 55–74% (Z22 71%), all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



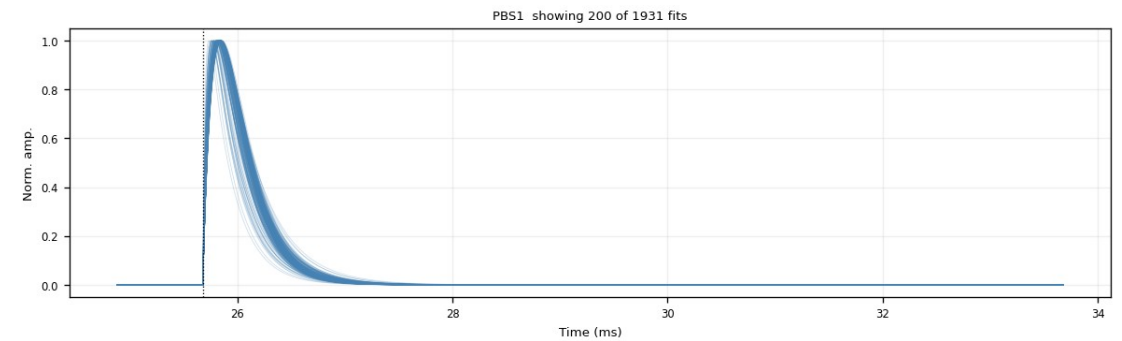
*Z7 PAS1: fitted  $\tau_{\text{rise}}$  (median 0.105 ms) and  $\tau_{\text{fall}}$  distributions*

# Backup · $\tau_{\text{rise}}$ cut: good vs bad channel (Z7)

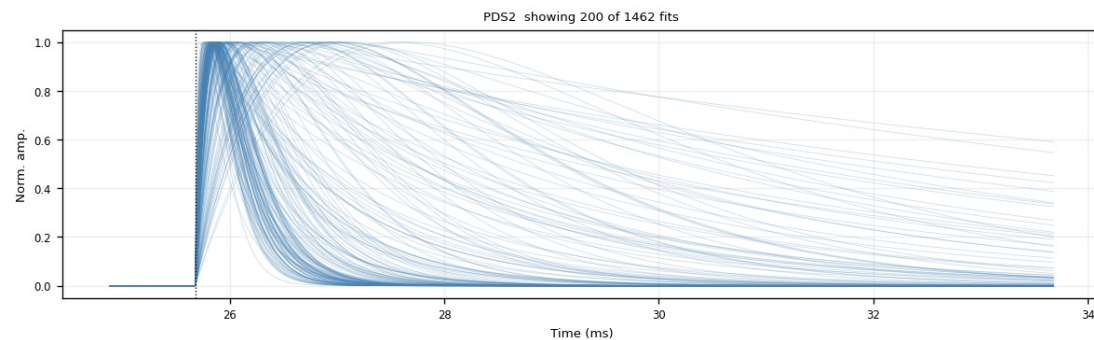
- Good channel PBS1: tight clean bundle; the cut removes ~0% (nearly free on quiet channels)
- Bad channel PDS2: broad messy fan, the channel itself is bad; the cut trims slow risers, 1462  $\rightarrow$  1154 (21%)



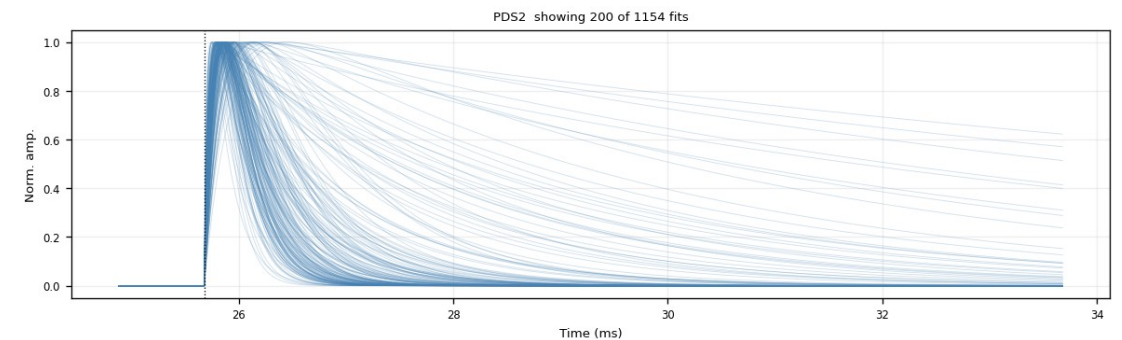
*PBS1 (good):  $\text{NRMSE} \leq 0.4$*



*PBS1 (good):  $+\tau_{\text{rise}} \leq 0.3 \text{ ms}$  (unchanged)*



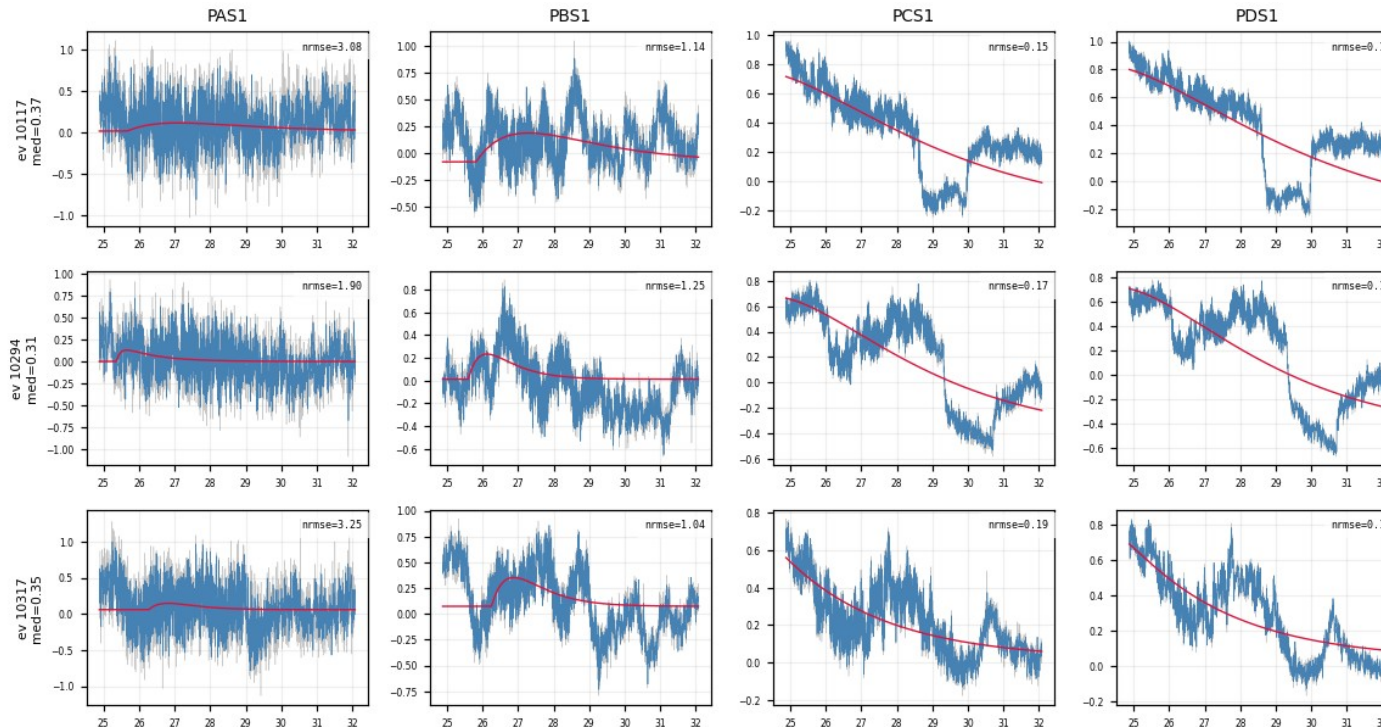
*PDS2 (bad):  $\text{NRMSE} \leq 0.4$*



*PDS2 (bad):  $+\tau_{\text{rise}} \leq 0.3 \text{ ms}$  (slow risers trimmed)*

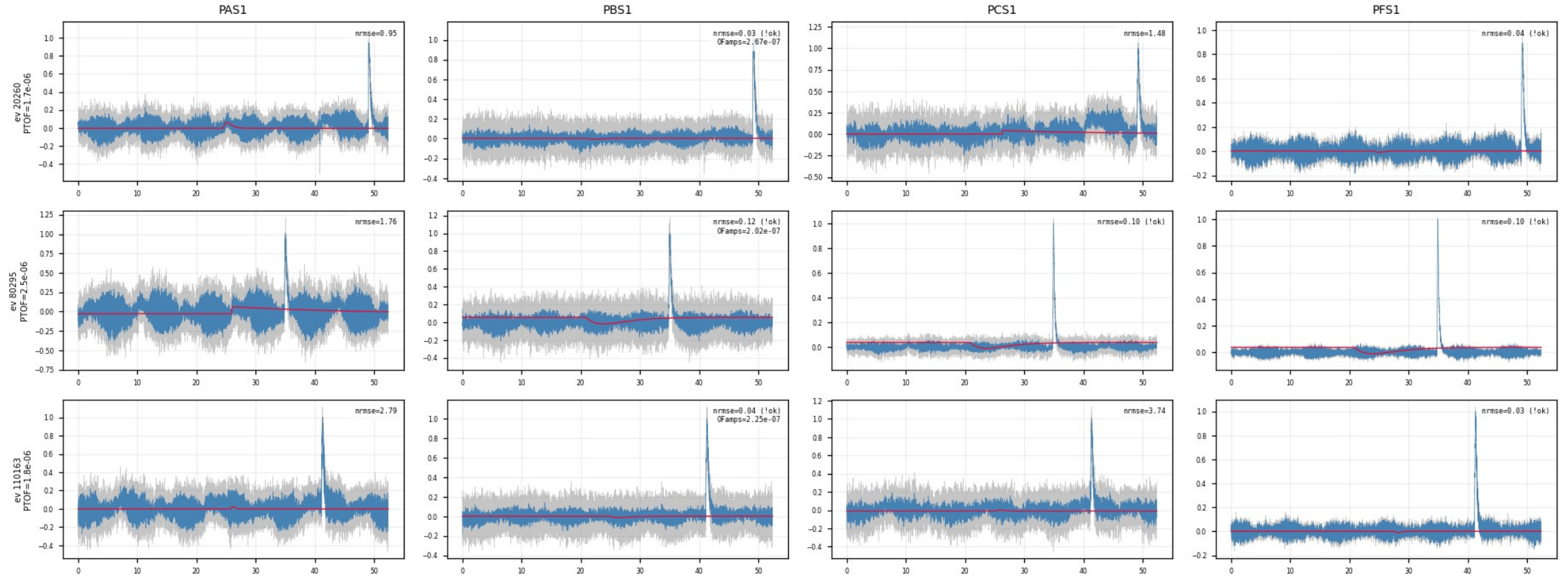
# Backup · what the $\tau_{\text{rise}}$ cut removes (Z22)

- Events that PASS  $\text{NRMSE} \leq 0.4$  but are REMOVED by  $\tau_{\text{rise}} \leq 0.3$  ms: the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at  $\text{NRMSE} \approx 0.15$  (a slow 2-exp hugs the drift); the raw trace has no clean fast pulse
- Trade-off: also trims genuine slow-rise pulses; decision pending



Z22, 3 events  $\times$  PAS1/PBS1/PCS1/PDS1: pass  $\text{NRMSE} \leq 0.4$ , removed by  $\tau_{\text{rise}} \leq 0.3$  ms; PCS1/PDS1 = pure slow drift

# Backup · fit\_ok-rejected events: normal amplitude, late pulse



Z7, 3 fit\_ok-rejected events × PAS1/PBS1/PCS1/PFS1, full 0–52 ms trace: the real pulse sits far right of the fit window; red (!ok) fits are the negative solutions

# Backup · Z7 full results gallery

every diagnostic and result figure of the Z7 pipeline, in processing order

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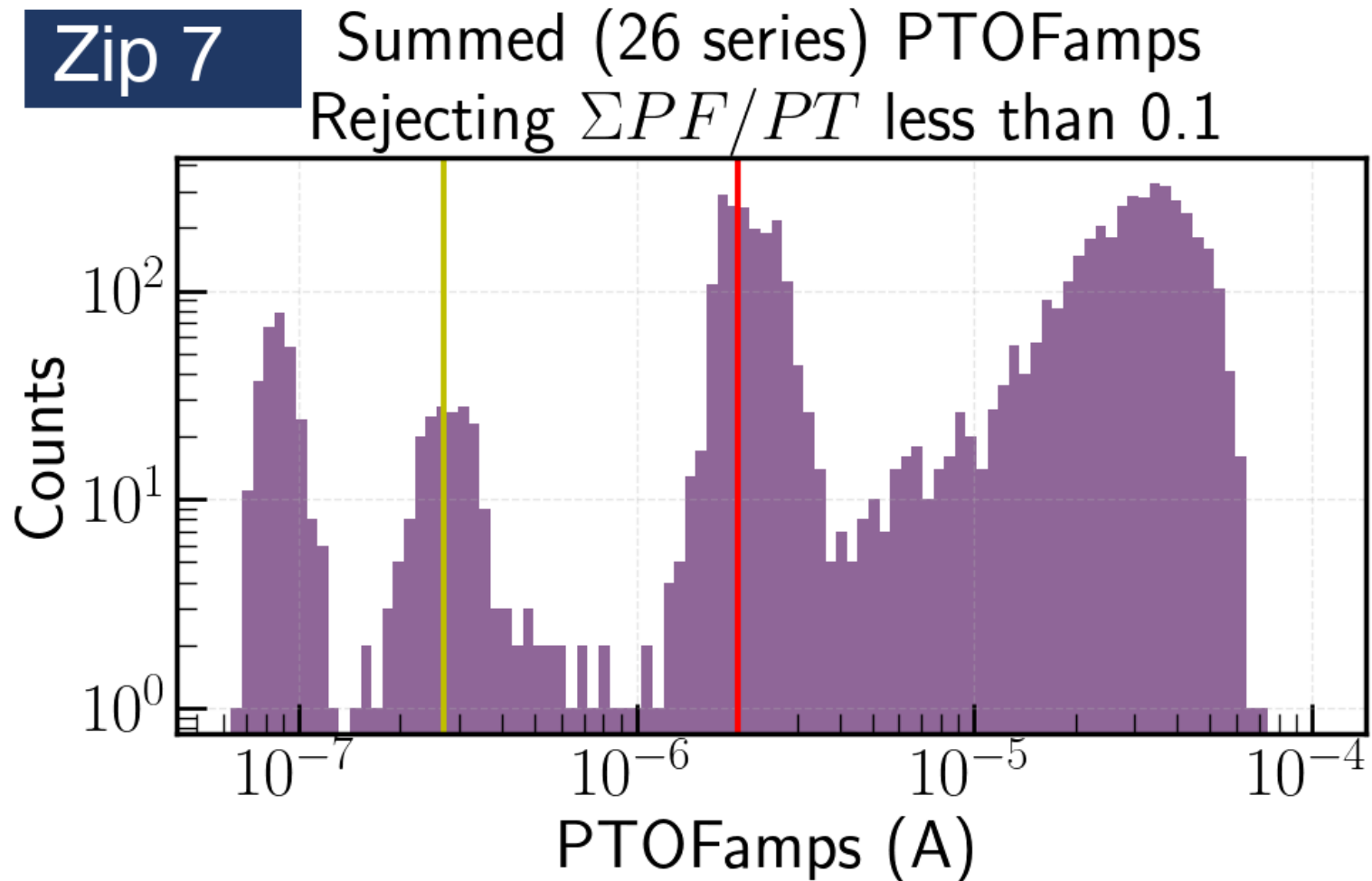
- Dataset: the K-line selection window
- Example grids: LP + fit and raw vs fit, event × channel
- Fit diagnostics: NRMSE, fitted pretrigger,  $\tau_{\text{rise}}$  /  $\tau_{\text{fall}}$  distributions
- Aligned fans: no cut  $\rightarrow$   $\text{NRMSE} \leq 0.4 \rightarrow + \tau_{\text{rise}} \leq 0.3$  ms; kept vs rejected; measured traces
- Special populations: rejected events, well-fit slow-rise events, PDS2 slow-fall study
- Templates: PCA nxm0–4 per channel, summed PT / PS1 / PS2
- Tall figures are cut at white space into full-width panels, two per page; grids into 3-event × half-channel blocks



# Backup · Z7 · K-line selection window

figure: kline\_zip7.png

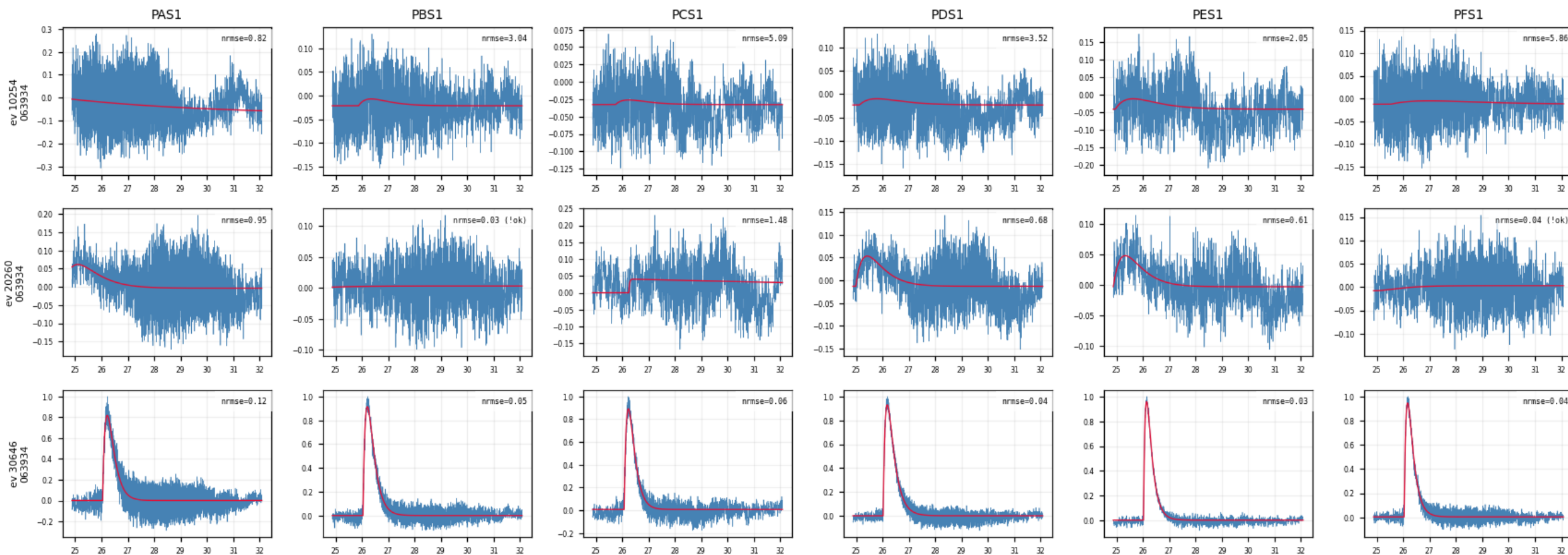
Summed PTOFamps spectrum of Z7; red line = Prof. Saab's 10.37 keV K-line position, shaded band = the  $\times/\div 1.35$  selection window; everything inside is cached raw.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (1/10)

figure: zip7\_fit\_examples.png, event rows 1–3 of 15, left half of the channels

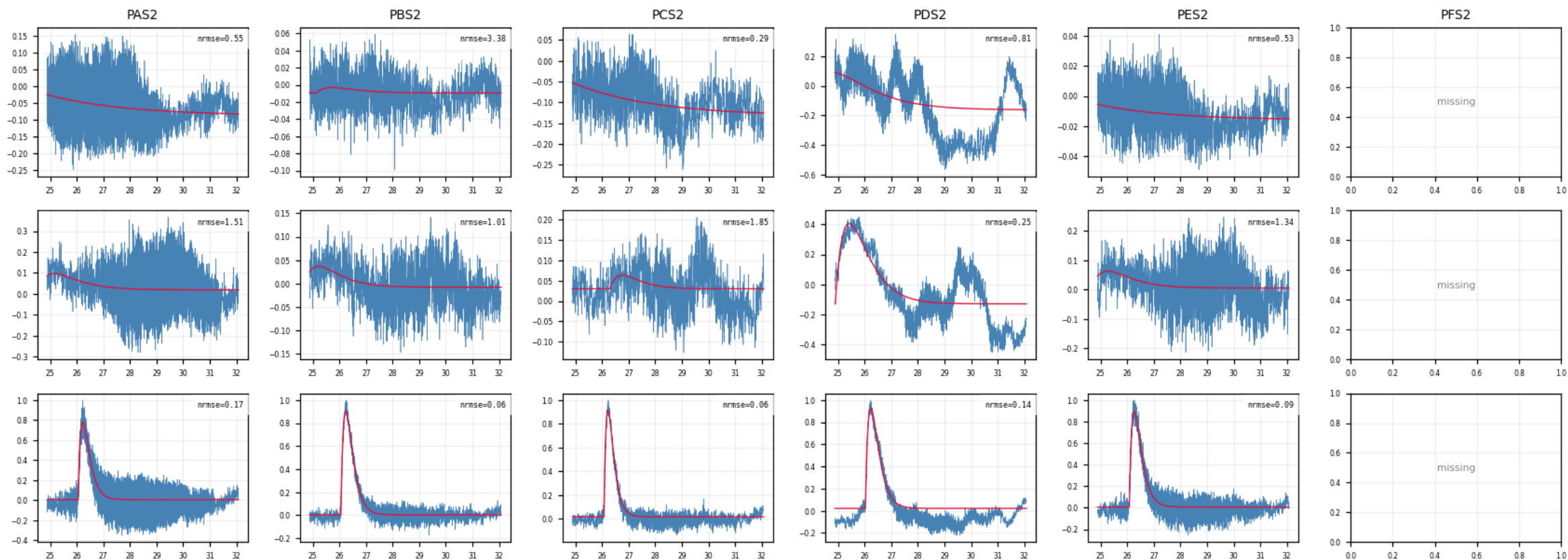
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (2/10)

figure: zip7\_fit\_examples.png, event rows 1–3 of 15, right half of the channels

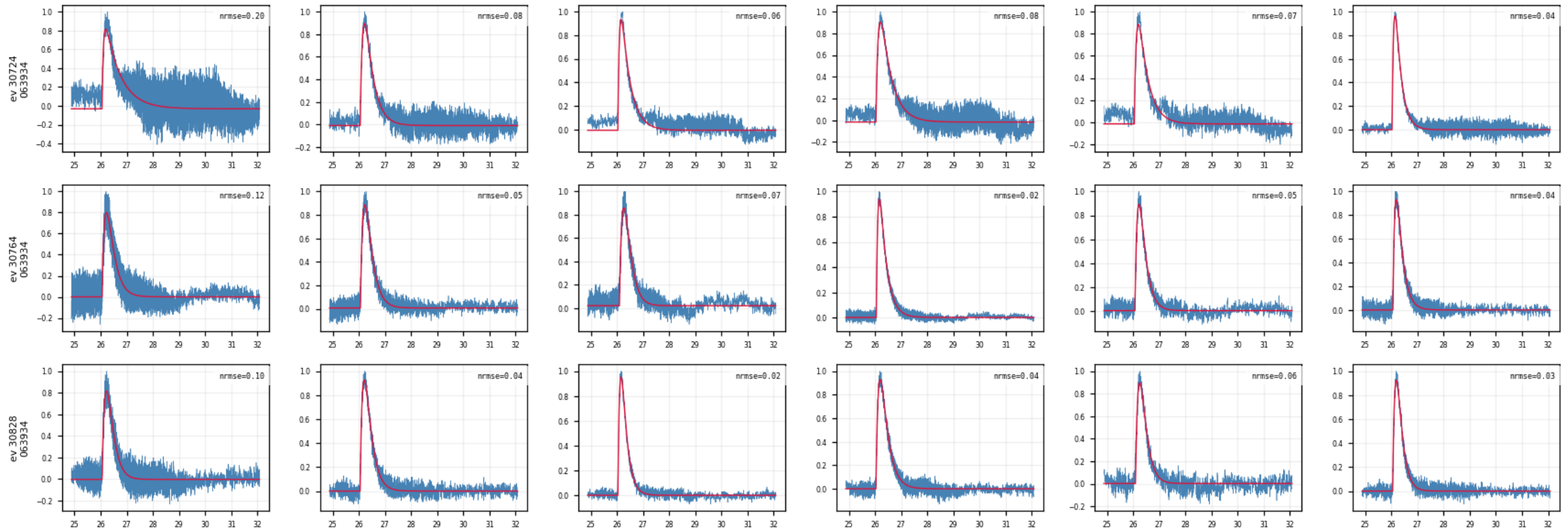
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (3/10)

figure: zip7\_fit\_examples.png, event rows 4–6 of 15, left half of the channels

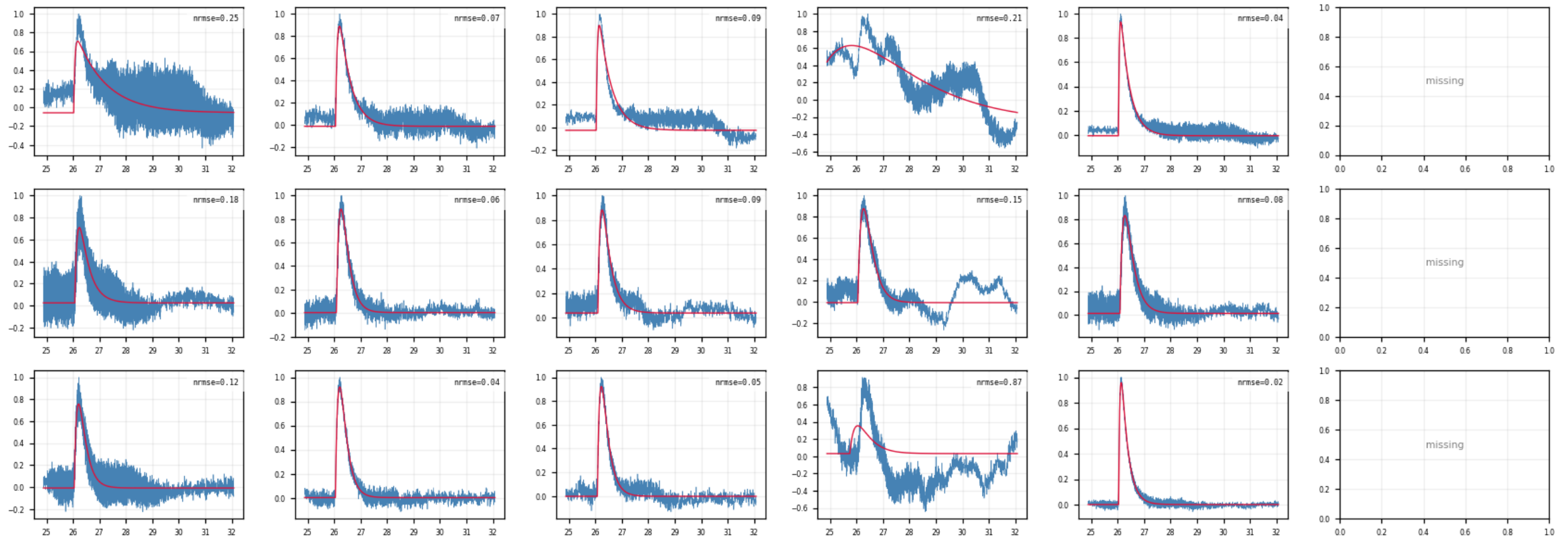
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (4/10)

figure: zip7\_fit\_examples.png, event rows 4–6 of 15, right half of the channels

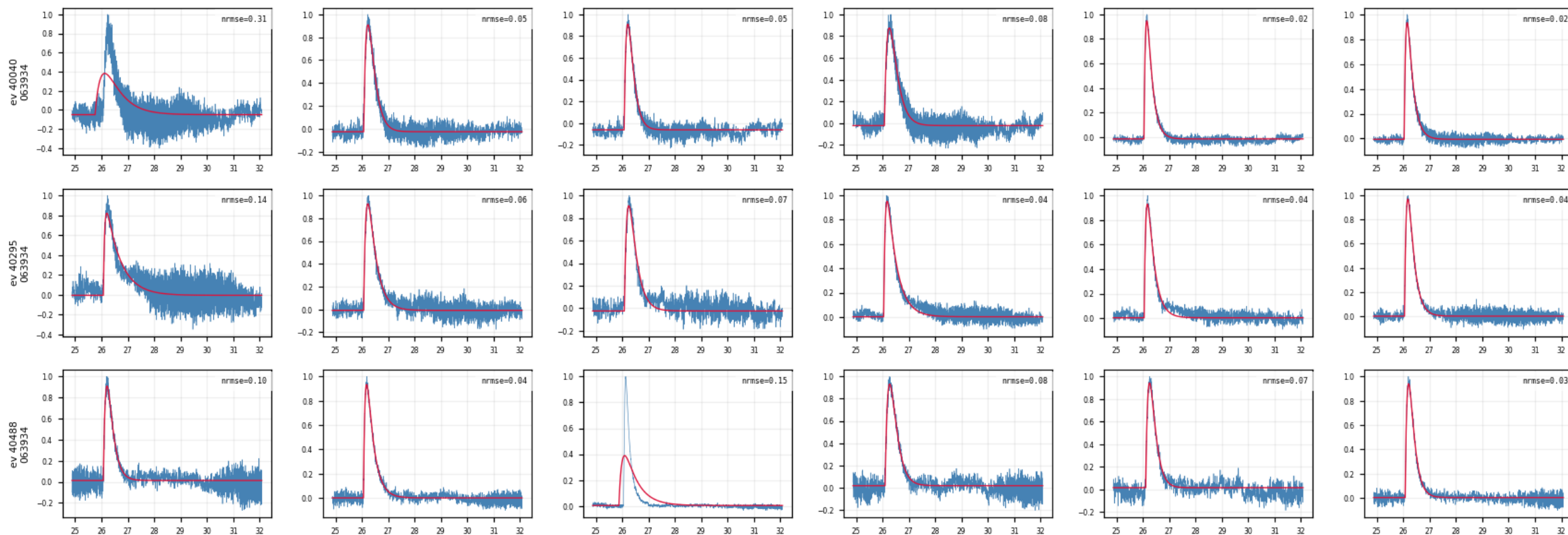
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (5/10)

figure: zip7\_fit\_examples.png, event rows 7–9 of 15, left half of the channels

First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.

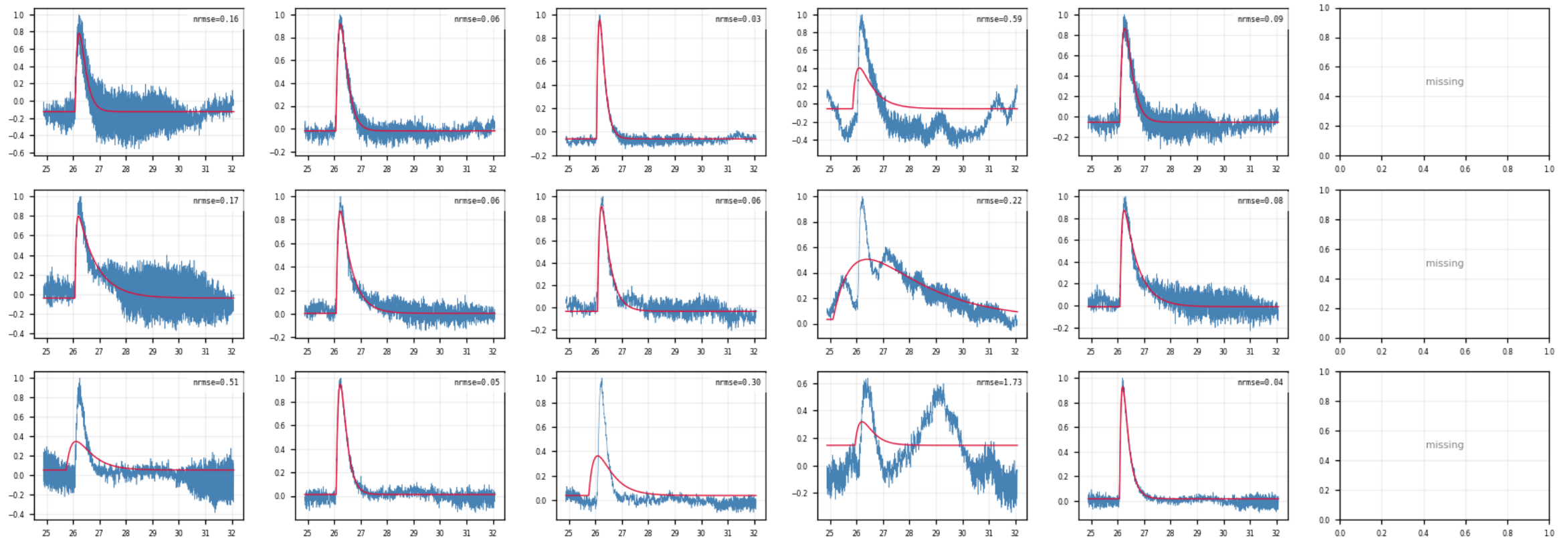




# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (6/10)

figure: zip7\_fit\_examples.png, event rows 7–9 of 15, right half of the channels

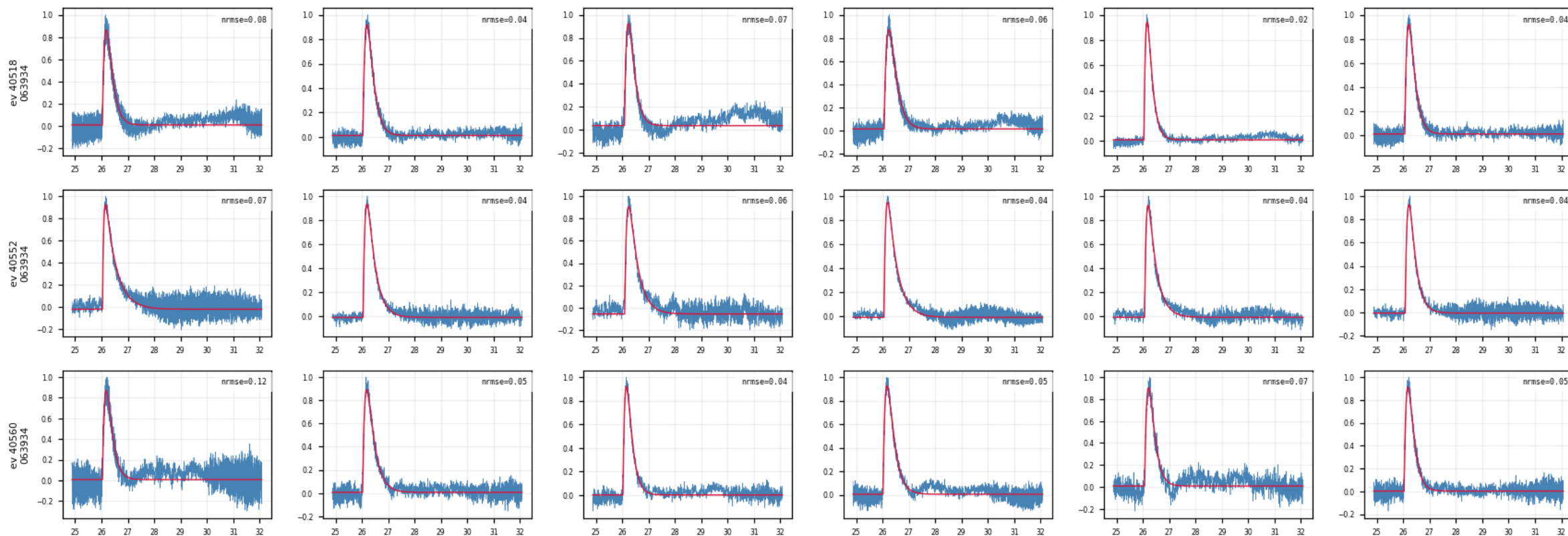
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (7/10)

figure: zip7\_fit\_examples.png, event rows 10–12 of 15, left half of the channels

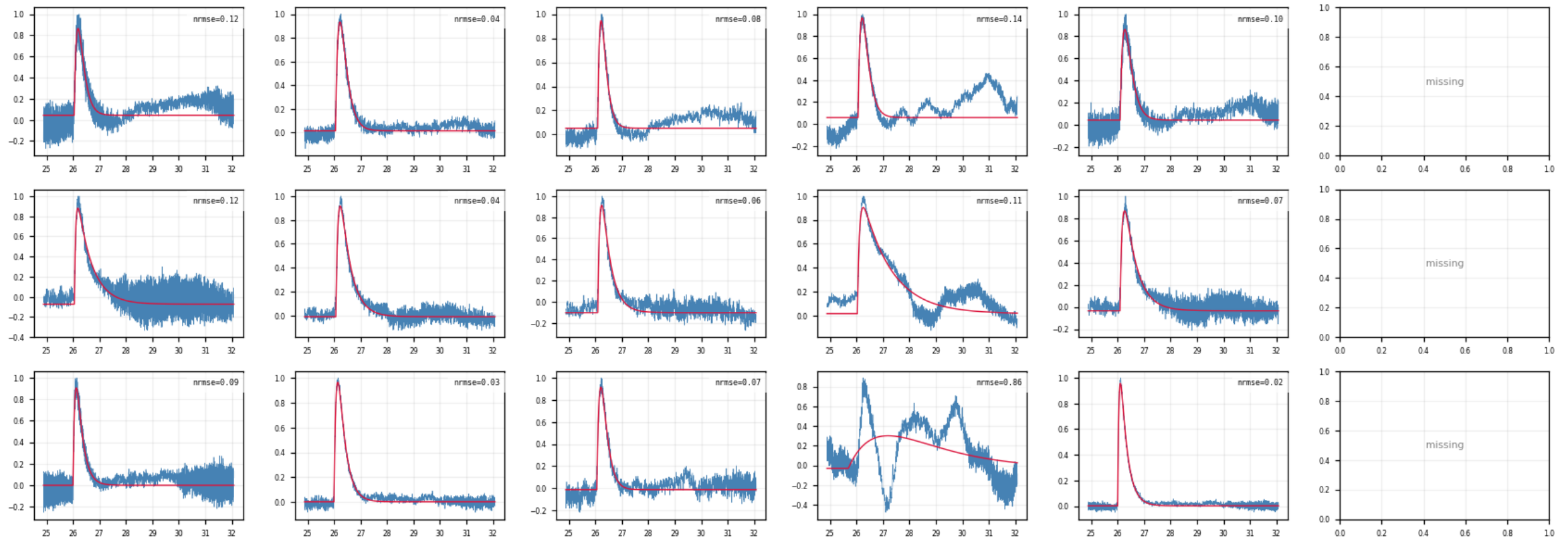
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (8/10)

figure: zip7\_fit\_examples.png, event rows 10–12 of 15, right half of the channels

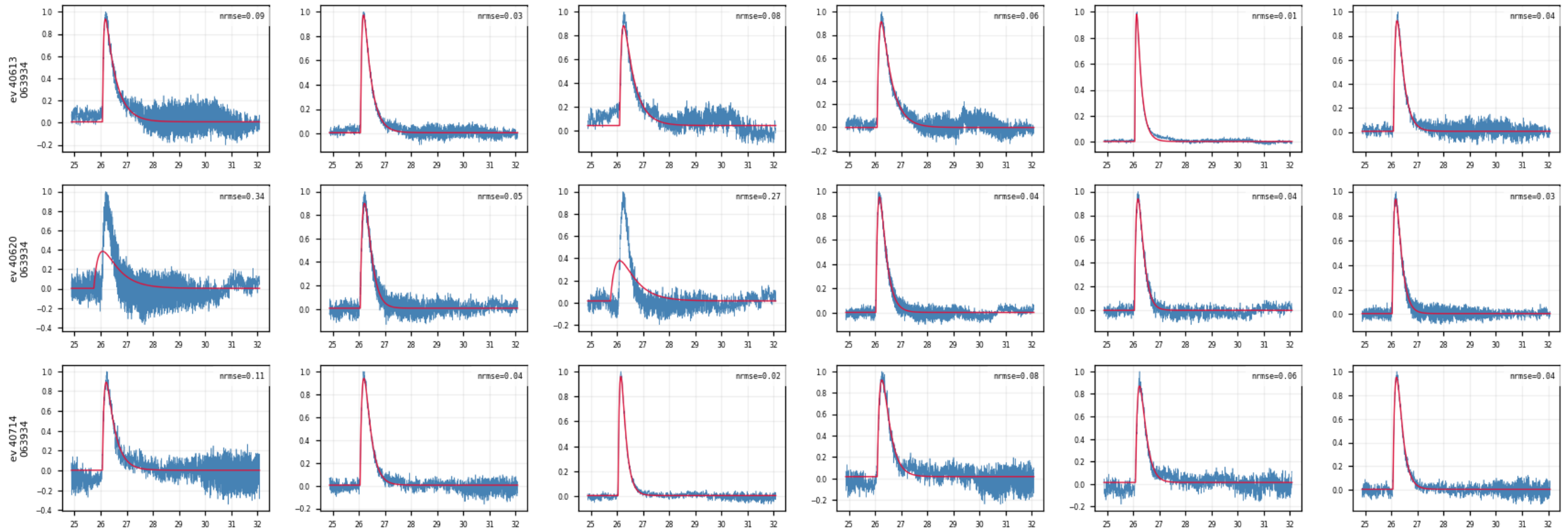
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (9/10)

figure: zip7\_fit\_examples.png, event rows 13–15 of 15, left half of the channels

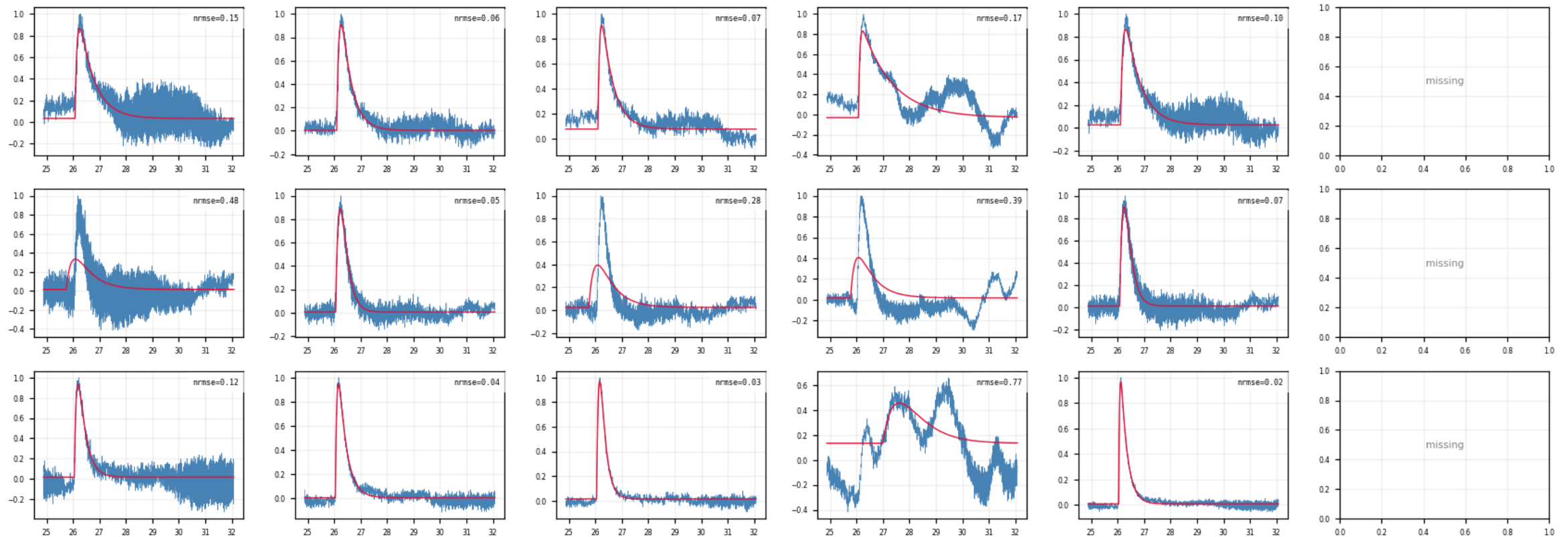
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



# Backup · Z7 · fit\_examples: LP trace vs 2-exp fit (10/10)

figure: zip7\_fit\_examples.png, event rows 13–15 of 15, right half of the channels

First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.

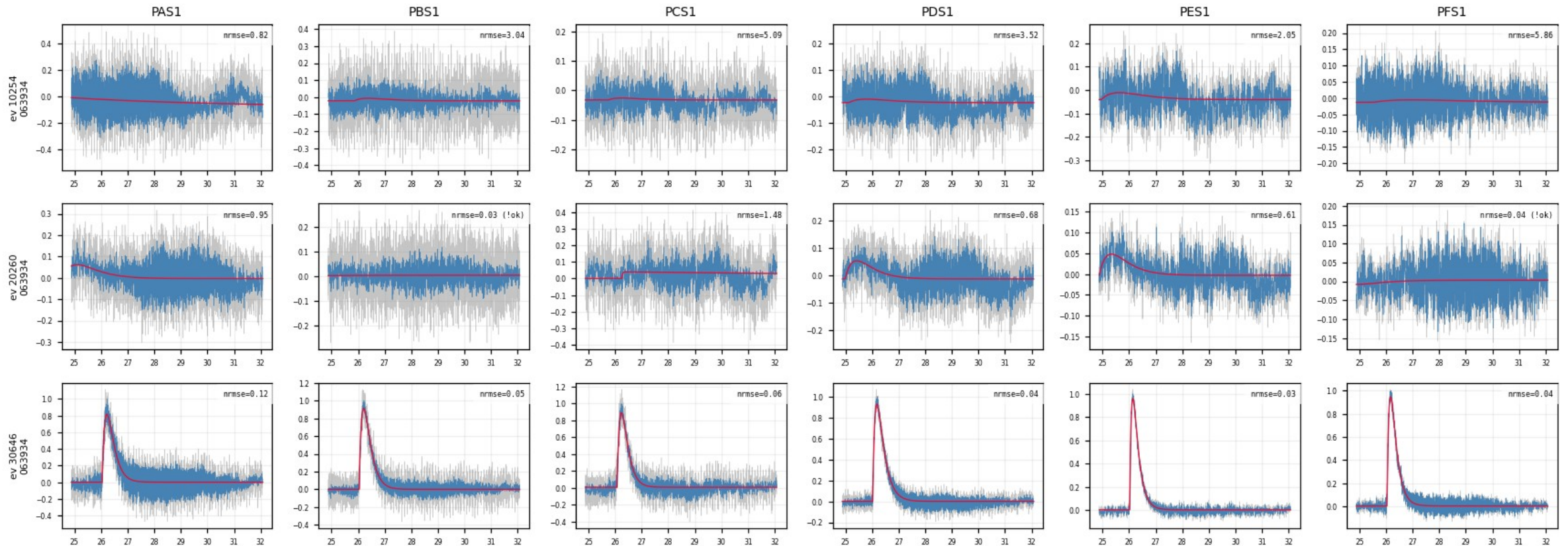




# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (1/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 1–3 of 15, left half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

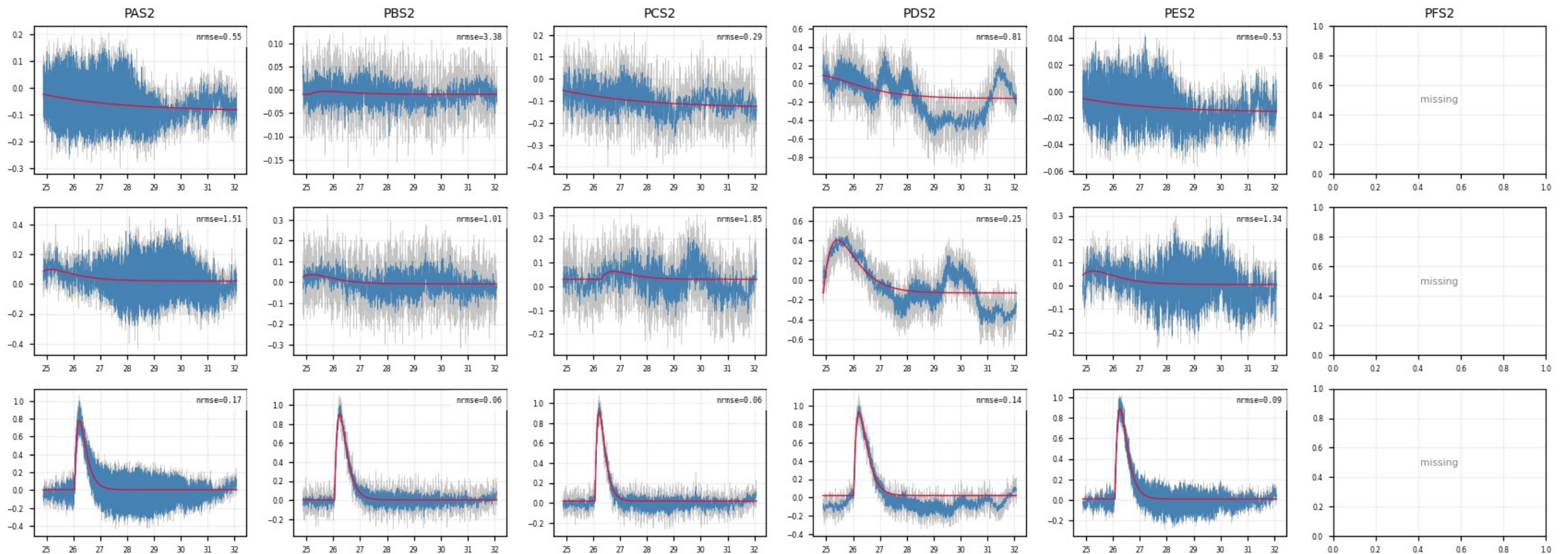




# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (2/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 1–3 of 15, right half of the channels

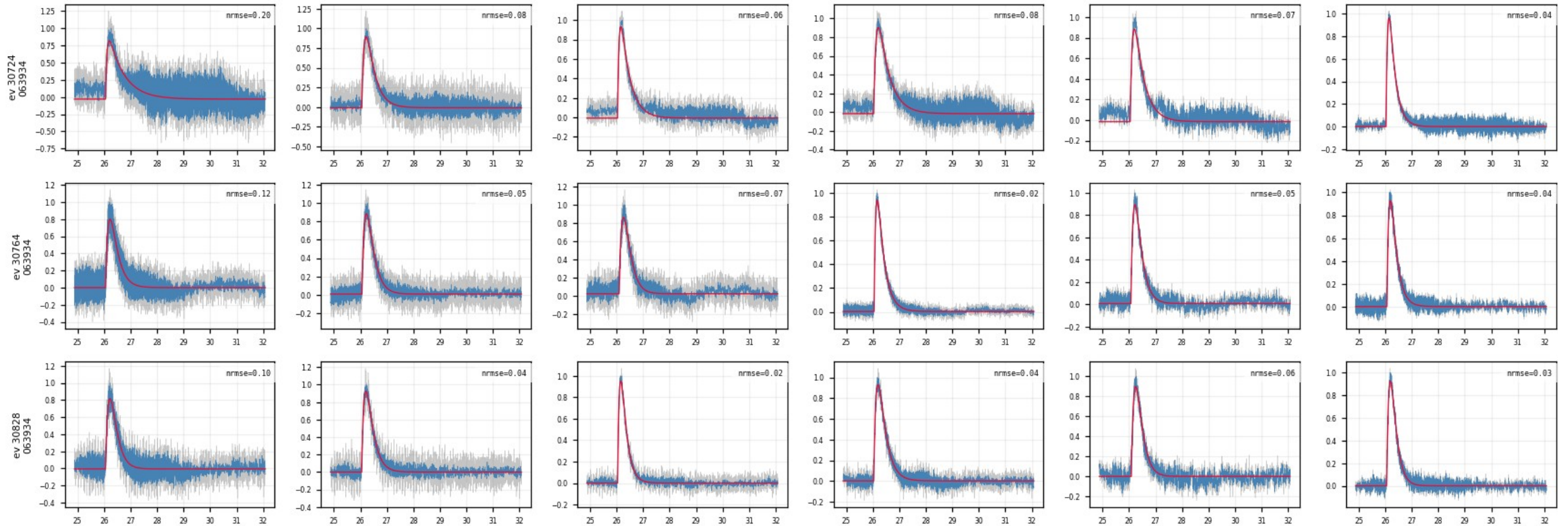
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (3/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 4–6 of 15, left half of the channels

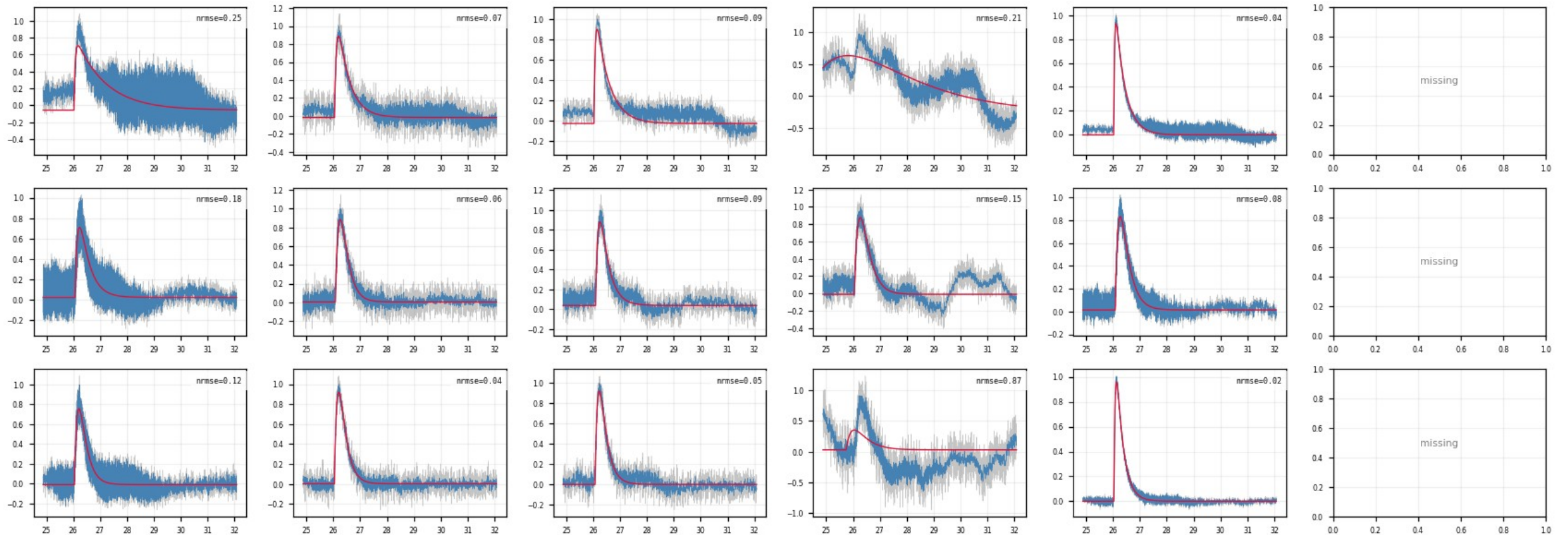
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (4/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 4–6 of 15, right half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

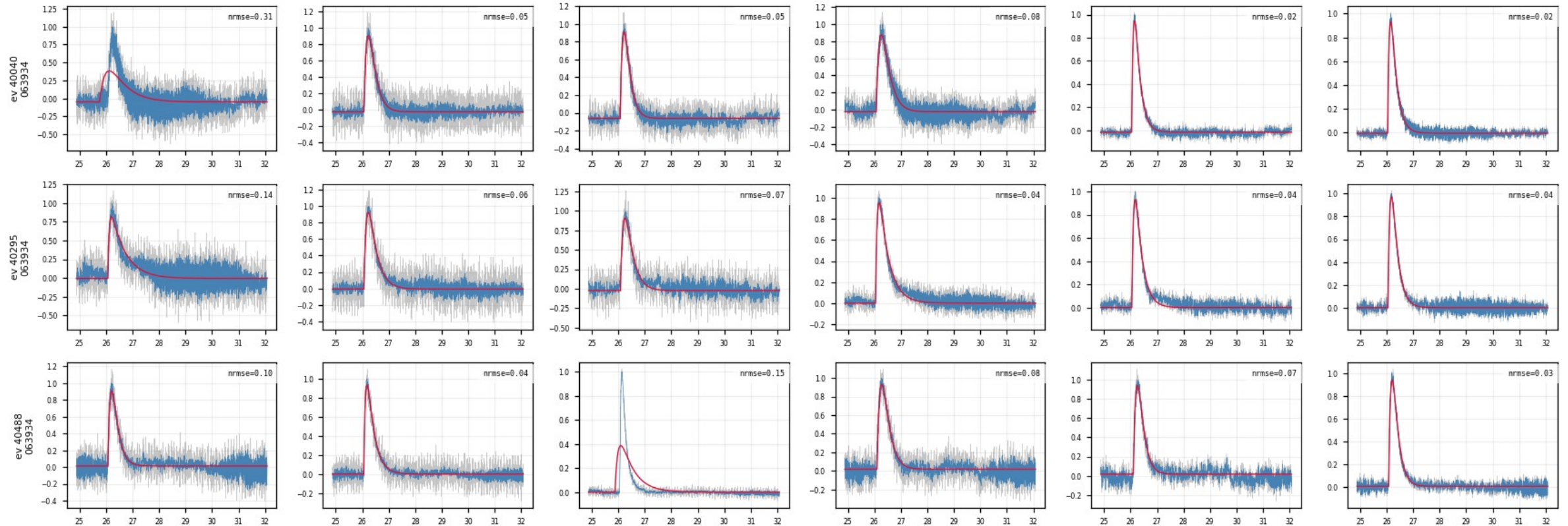




# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (5/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 7–9 of 15, left half of the channels

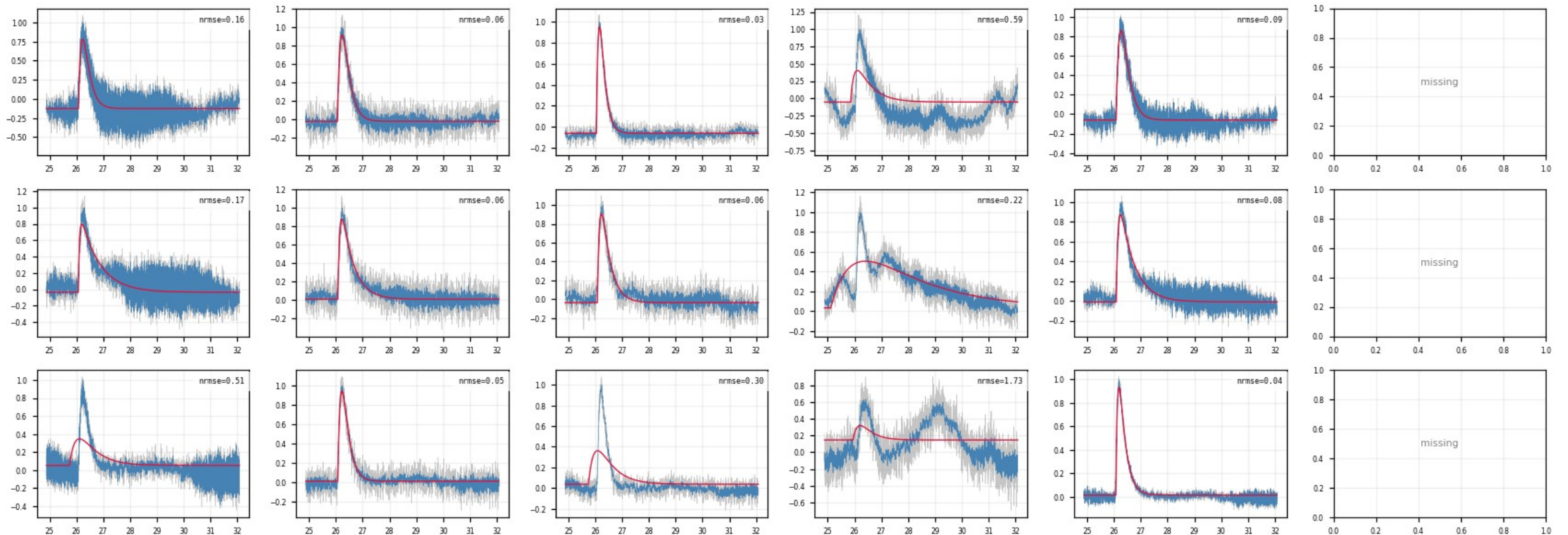
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (6/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 7–9 of 15, right half of the channels

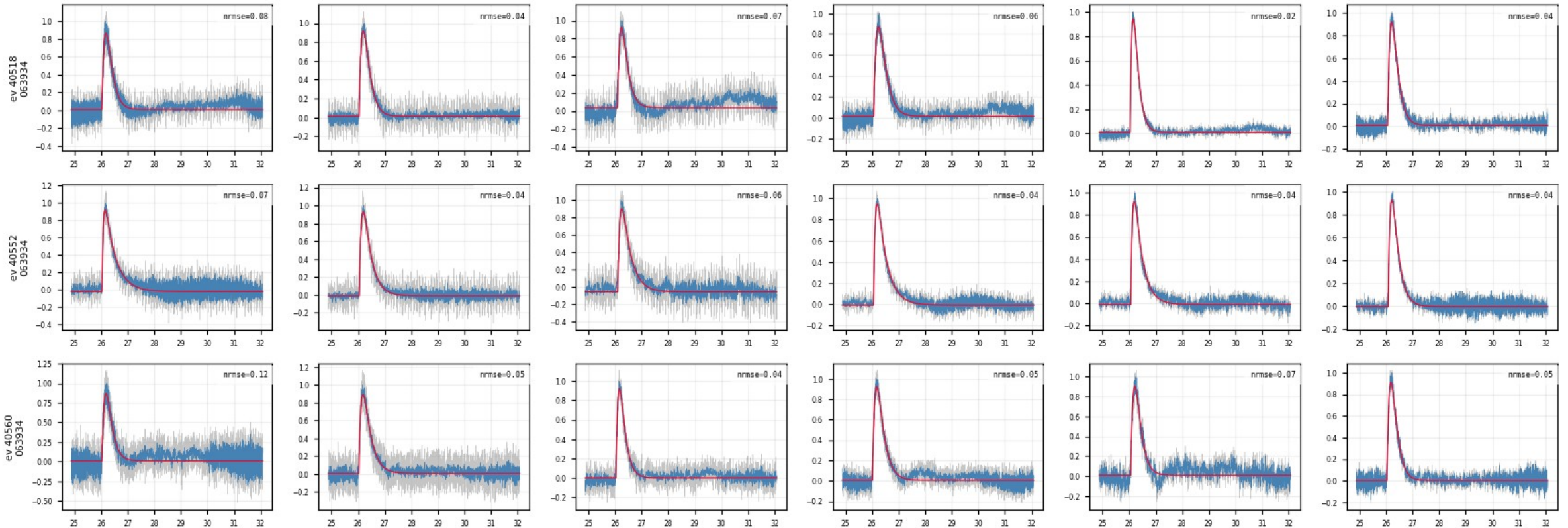
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (7/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 10–12 of 15, left half of the channels

Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.

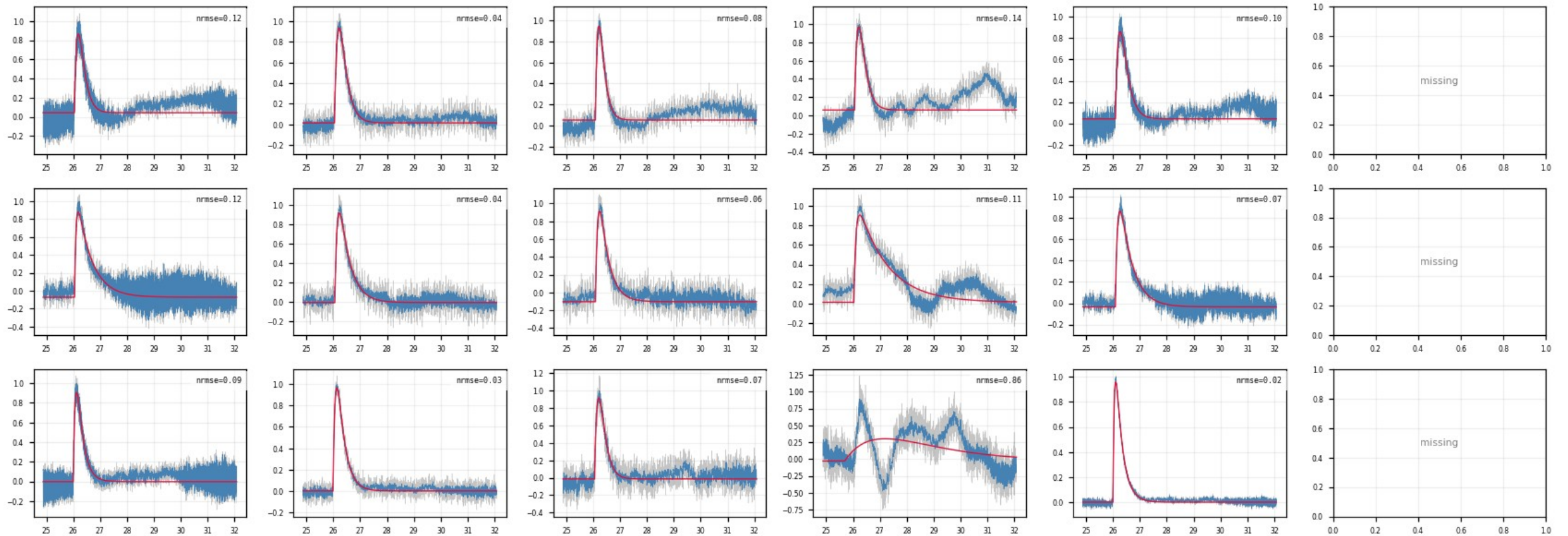




# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (8/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 10–12 of 15, right half of the channels

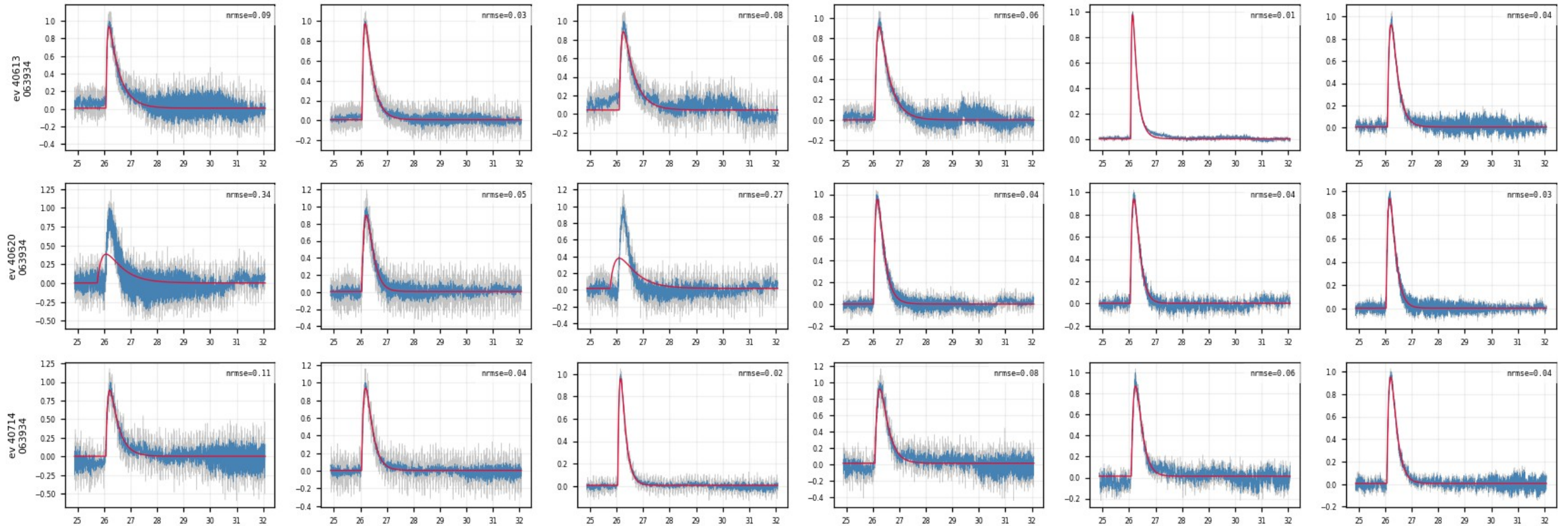
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (9/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 13–15 of 15, left half of the channels

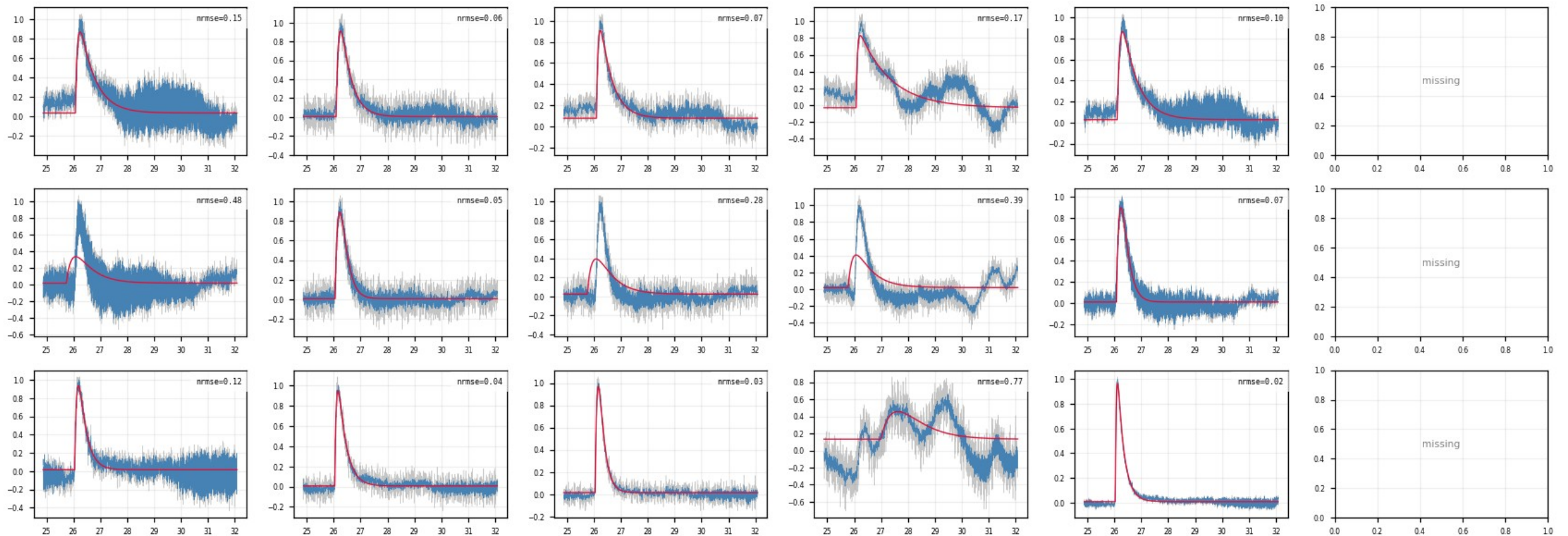
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Backup · Z7 · raw\_vs\_fit\_examples: raw trace vs fit (10/10)

figure: zip7\_raw\_vs\_fit\_examples.png, event rows 13–15 of 15, right half of the channels

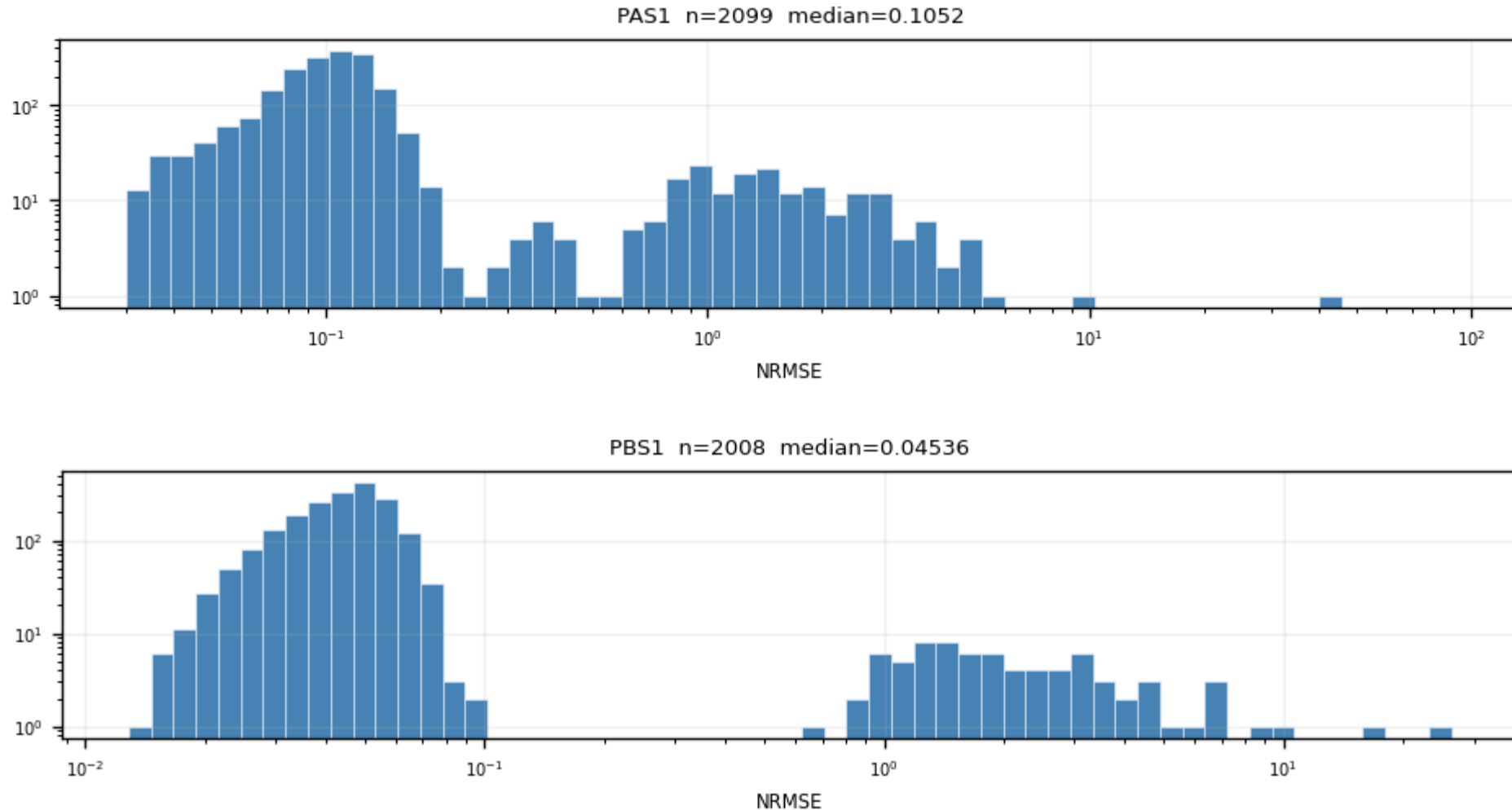
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



# Backup · Z7 · nrmse: NRMSE distribution per channel (1/6)

figure: zip7\_nrmse.png, channel panels 1–2 of 11, top to bottom

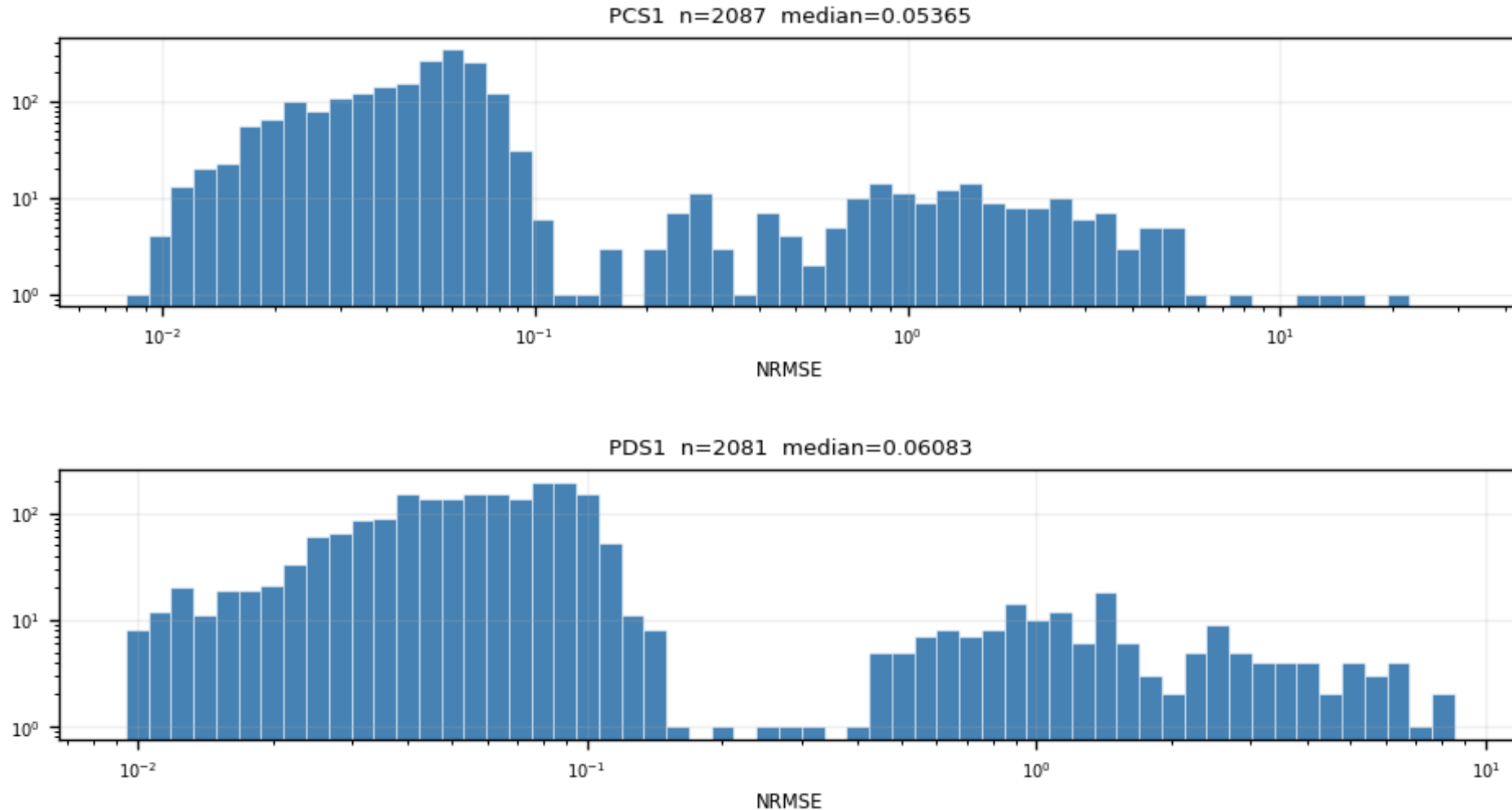
NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits ( $\sim 0.05\text{--}0.1$ ) vs noise triggers ( $\sim 1\text{--}2$ ); the valley at  $\approx 0.4$  sets quality cut 2.



# Backup · Z7 · nrmse: NRMSE distribution per channel (2/6)

figure: zip7\_nrmse.png, channel panels 3–4 of 11, top to bottom

NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits ( $\sim 0.05\text{--}0.1$ ) vs noise triggers ( $\sim 1\text{--}2$ ); the valley at  $\approx 0.4$  sets quality cut 2.

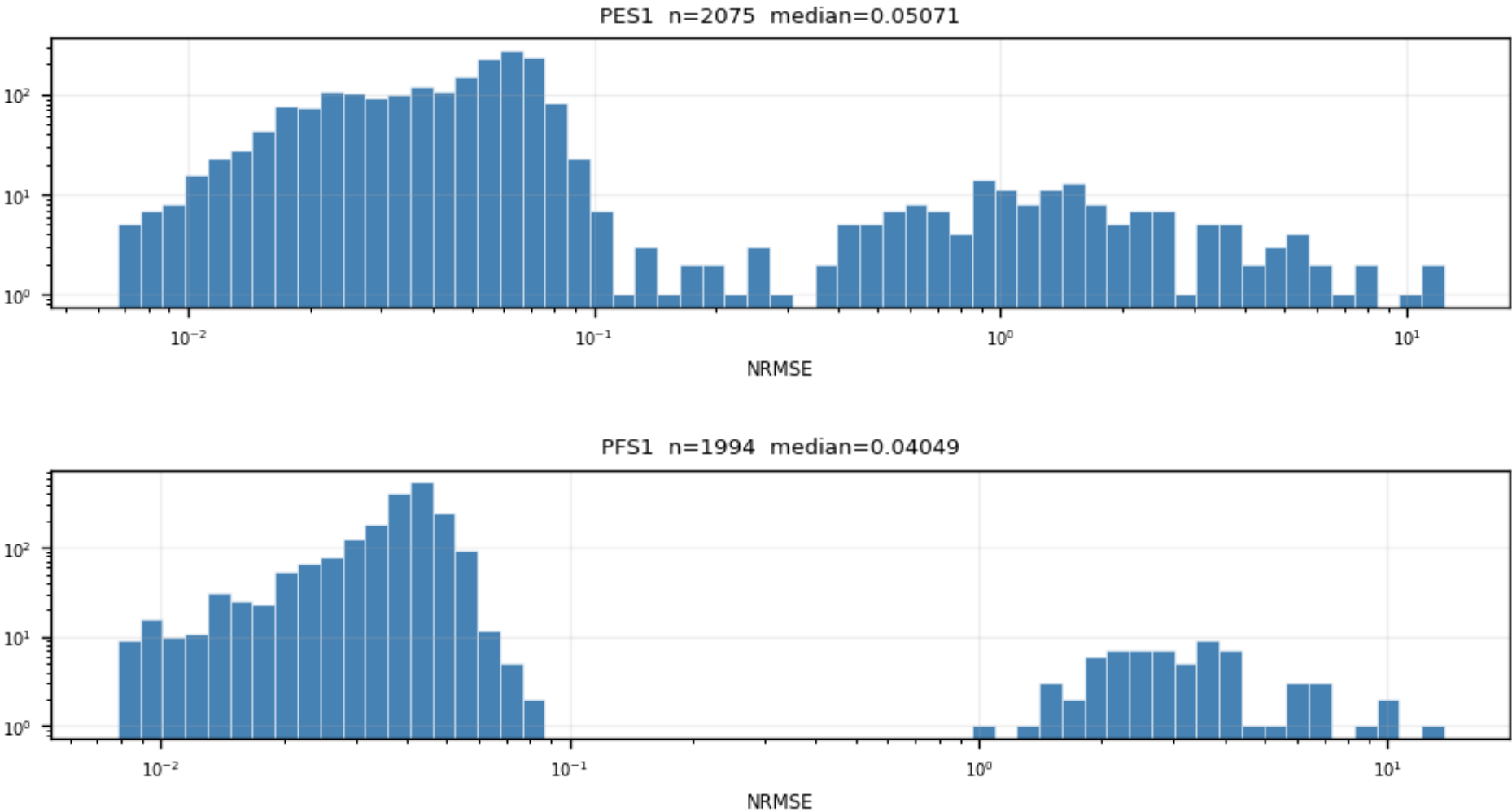




# Backup · Z7 · nrmse: NRMSE distribution per channel (3/6)

figure: zip7\_nrmse.png, channel panels 5–6 of 11, top to bottom

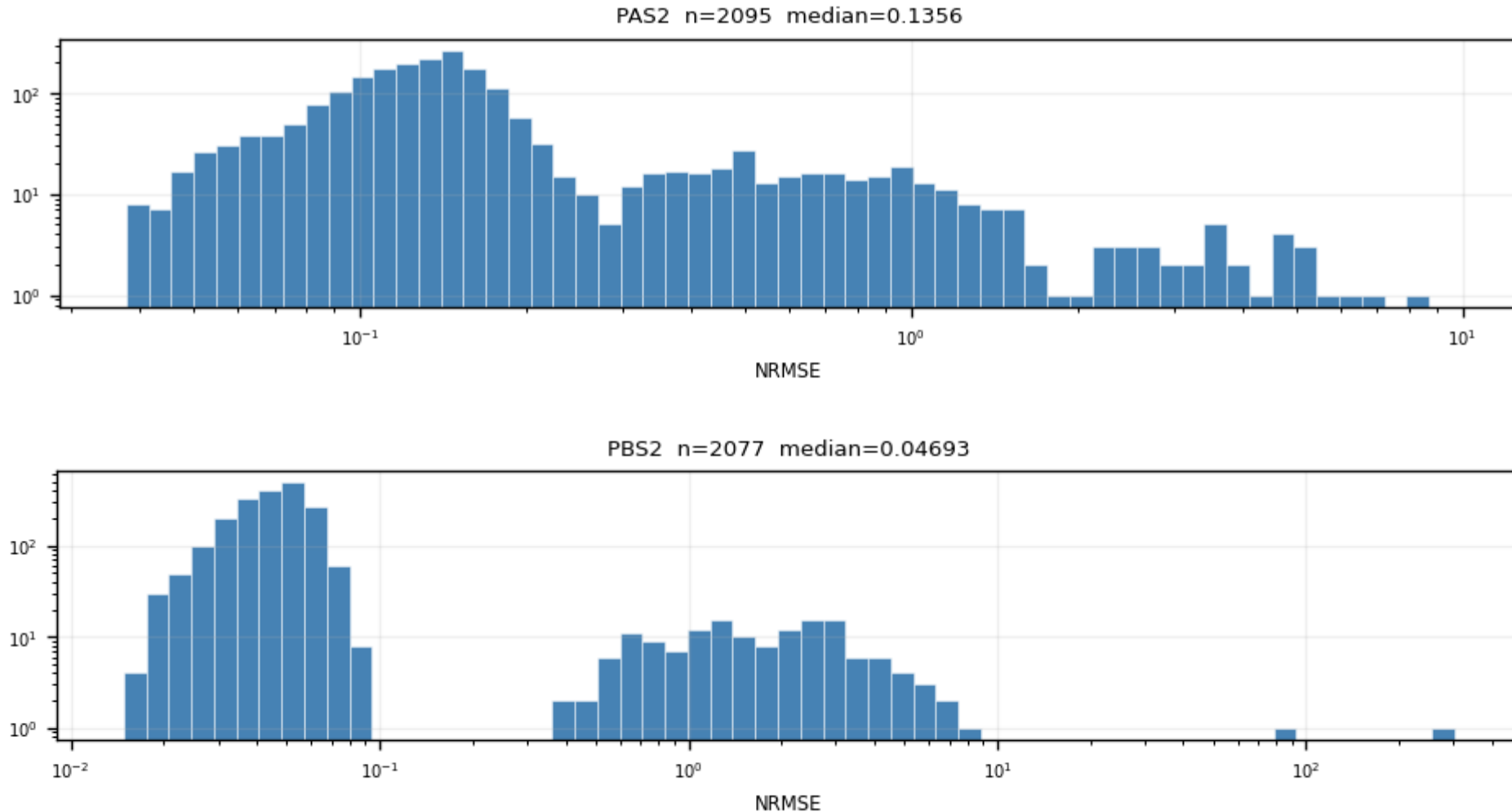
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.



# Backup · Z7 · nrmse: NRMSE distribution per channel (4/6)

figure: zip7\_nrmse.png, channel panels 7–8 of 11, top to bottom

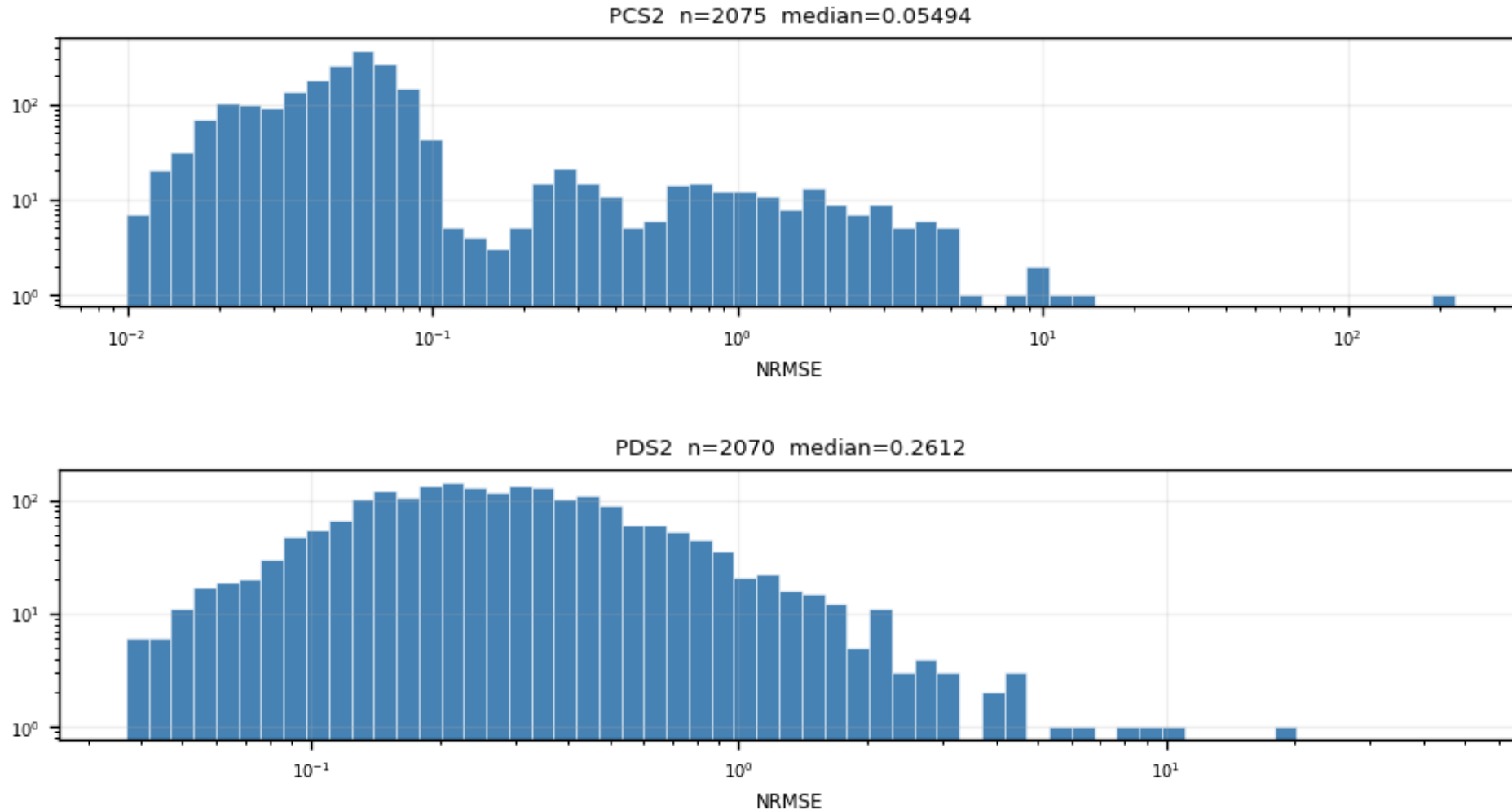
NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits ( $\sim 0.05\text{--}0.1$ ) vs noise triggers ( $\sim 1\text{--}2$ ); the valley at  $\approx 0.4$  sets quality cut 2.



# Backup · Z7 · nrmse: NRMSE distribution per channel (5/6)

figure: zip7\_nrmse.png, channel panels 9–10 of 11, top to bottom

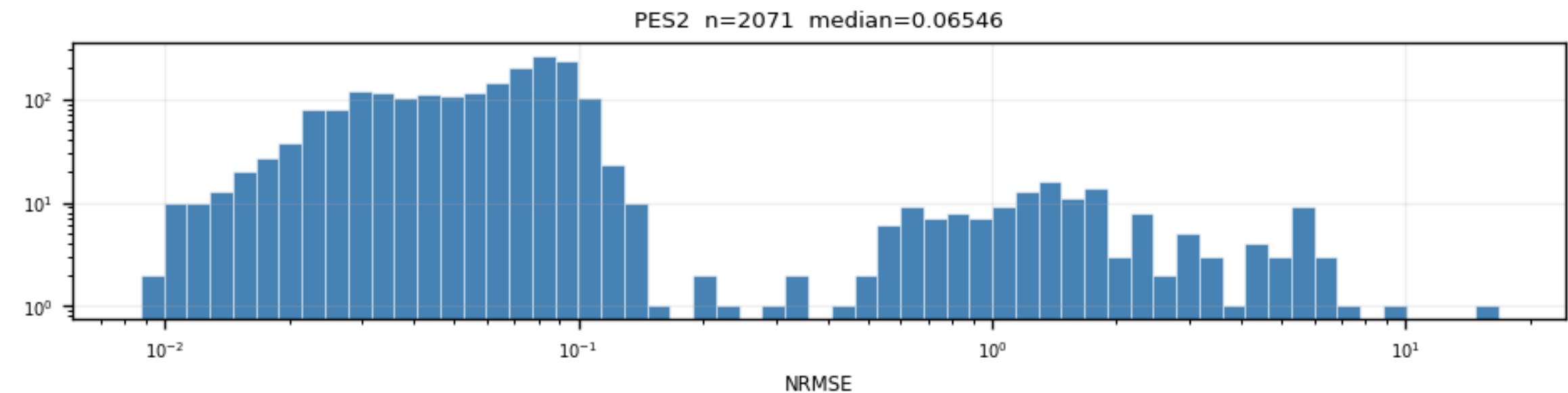
NRMSE =  $\text{RMS}(\text{fit residual}) / \text{fitted peak}$ , log-log. Bimodal: good fits ( $\sim 0.05\text{--}0.1$ ) vs noise triggers ( $\sim 1\text{--}2$ ); the valley at  $\approx 0.4$  sets quality cut 2.



# Backup · Z7 · nrmse: NRMSE distribution per channel (6/6)

figure: zip7\_nrmse.png, channel panels 11–11 of 11, top to bottom

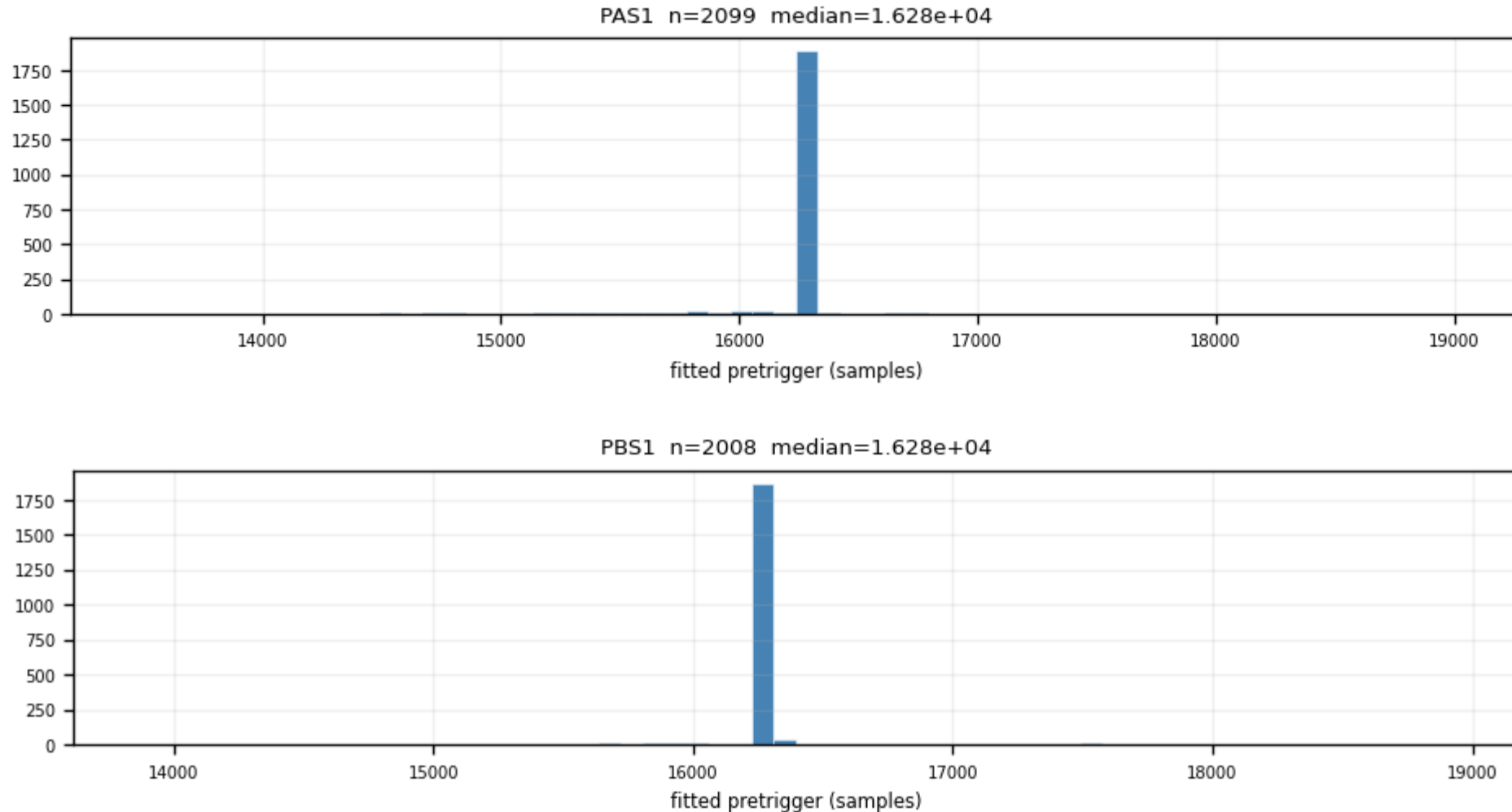
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits ( $\sim 0.05\text{--}0.1$ ) vs noise triggers ( $\sim 1\text{--}2$ ); the valley at  $\approx 0.4$  sets quality cut 2.



# Backup · Z7 · pretrigger: fitted onset per channel (1/6)

figure: zip7\_pretrigger.png, channel panels 1–2 of 11, top to bottom

The onset is a FREE fit parameter; fitted values cluster  $\approx 230$  samples after the nominal 16050; pinning it inside the fit would bias every other parameter.

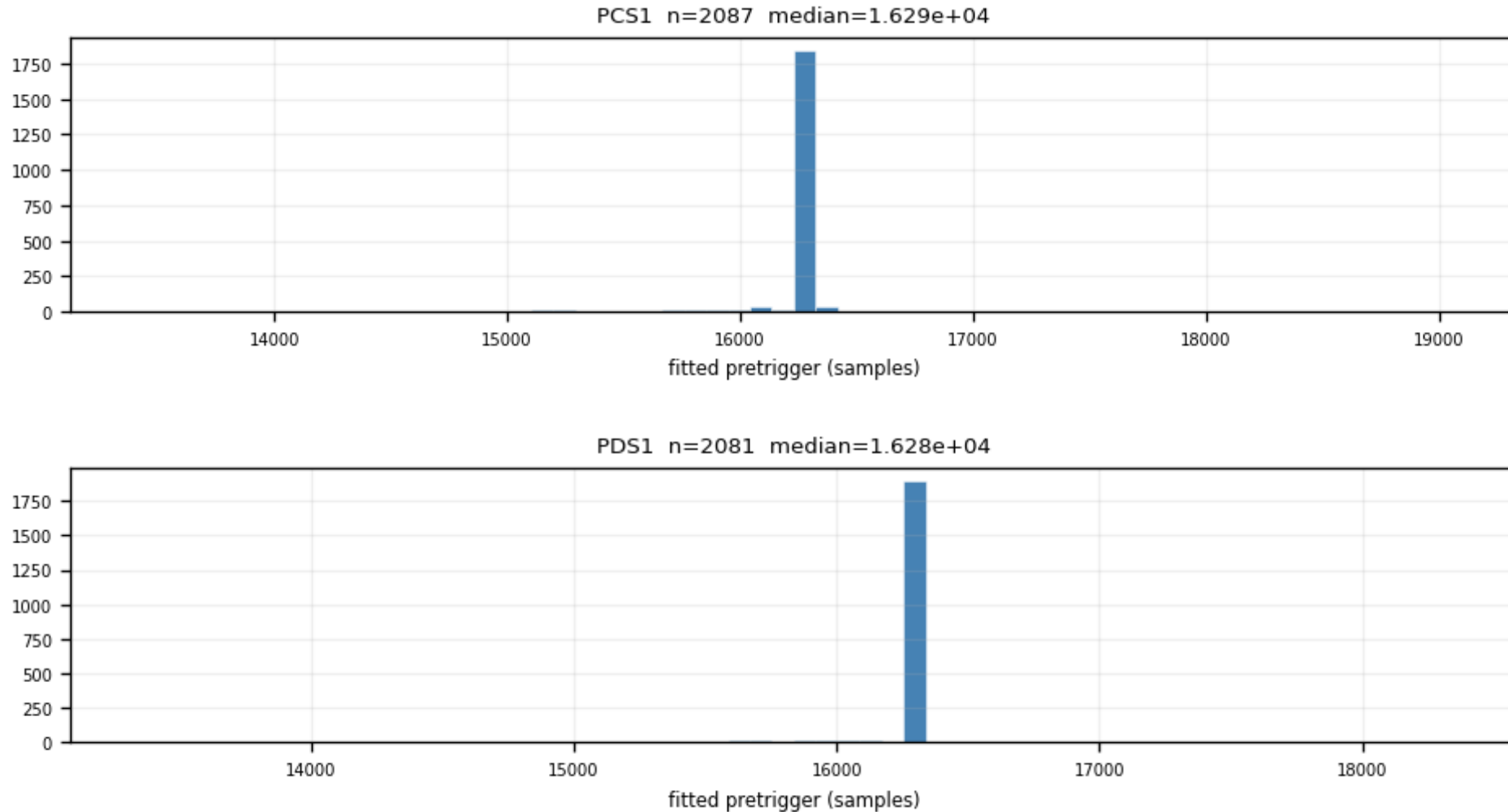




# Backup · Z7 · pretrigger: fitted onset per channel (2/6)

figure: zip7\_pretrigger.png, channel panels 3–4 of 11, top to bottom

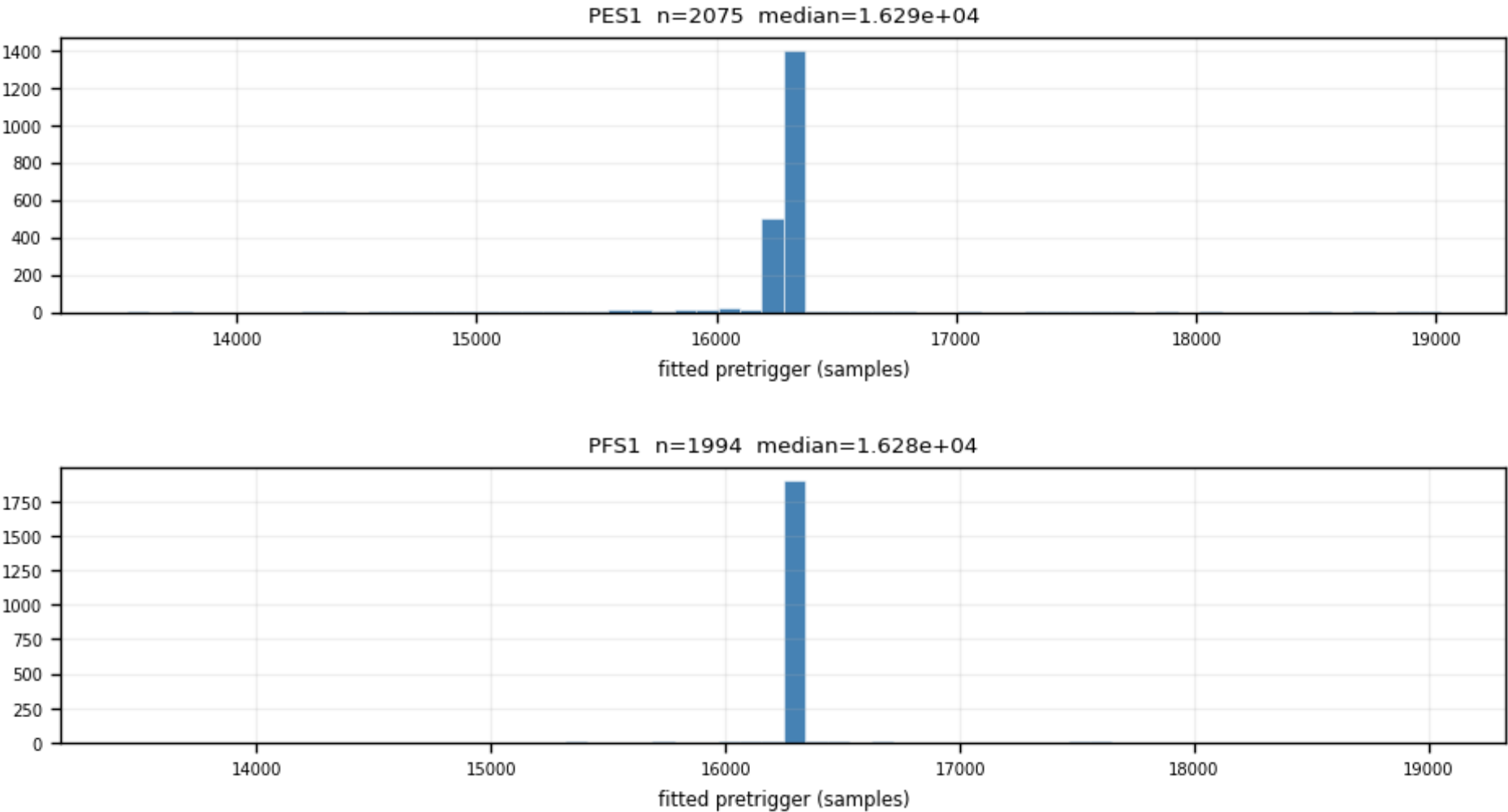
The onset is a FREE fit parameter; fitted values cluster  $\approx 230$  samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



# Backup · Z7 · pretrigger: fitted onset per channel (3/6)

figure: zip7\_pretrigger.png, channel panels 5–6 of 11, top to bottom

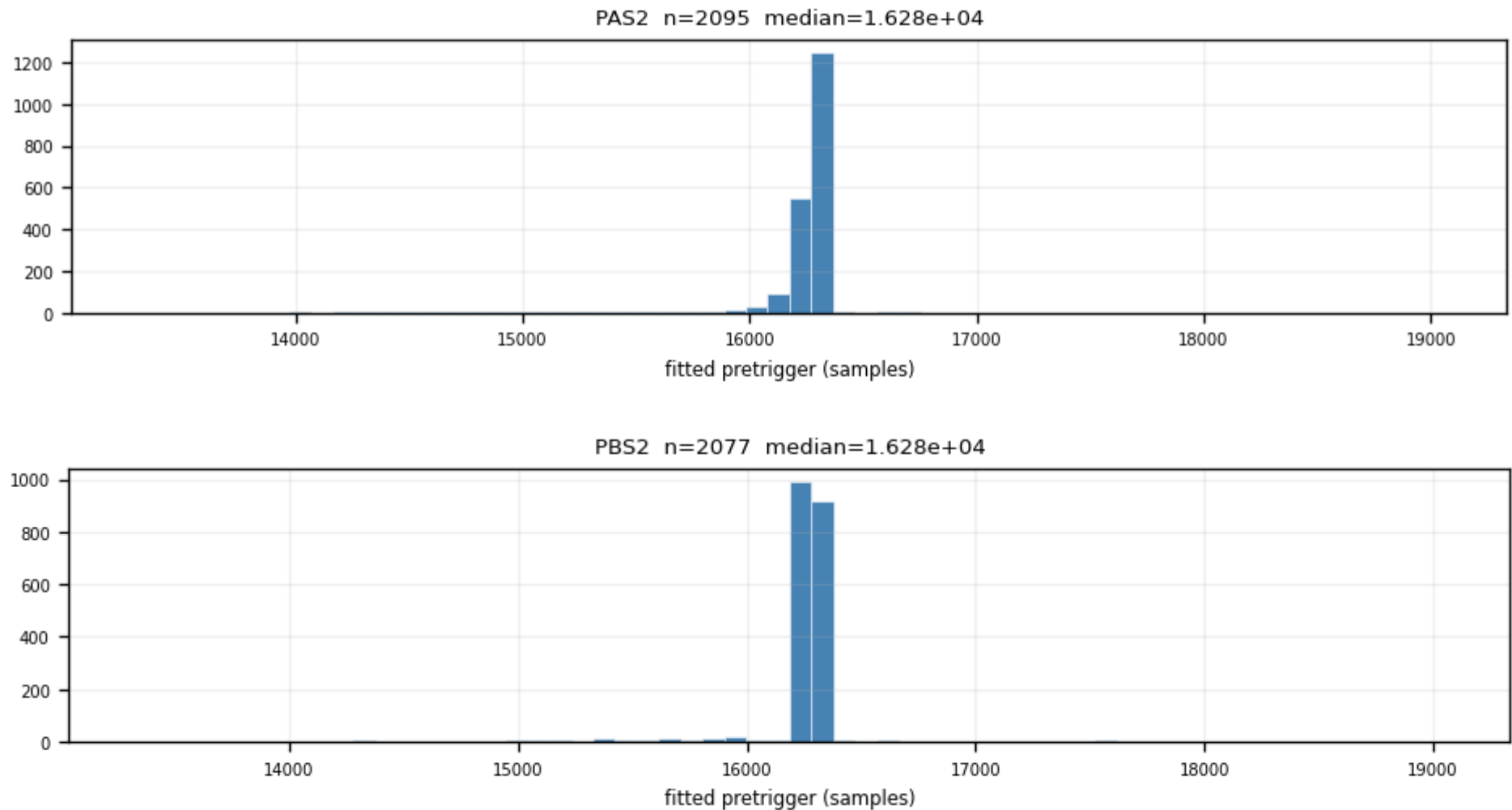
The onset is a FREE fit parameter; fitted values cluster  $\approx 230$  samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



# Backup · Z7 · pretrigger: fitted onset per channel (4/6)

figure: zip7\_pretrigger.png, channel panels 7–8 of 11, top to bottom

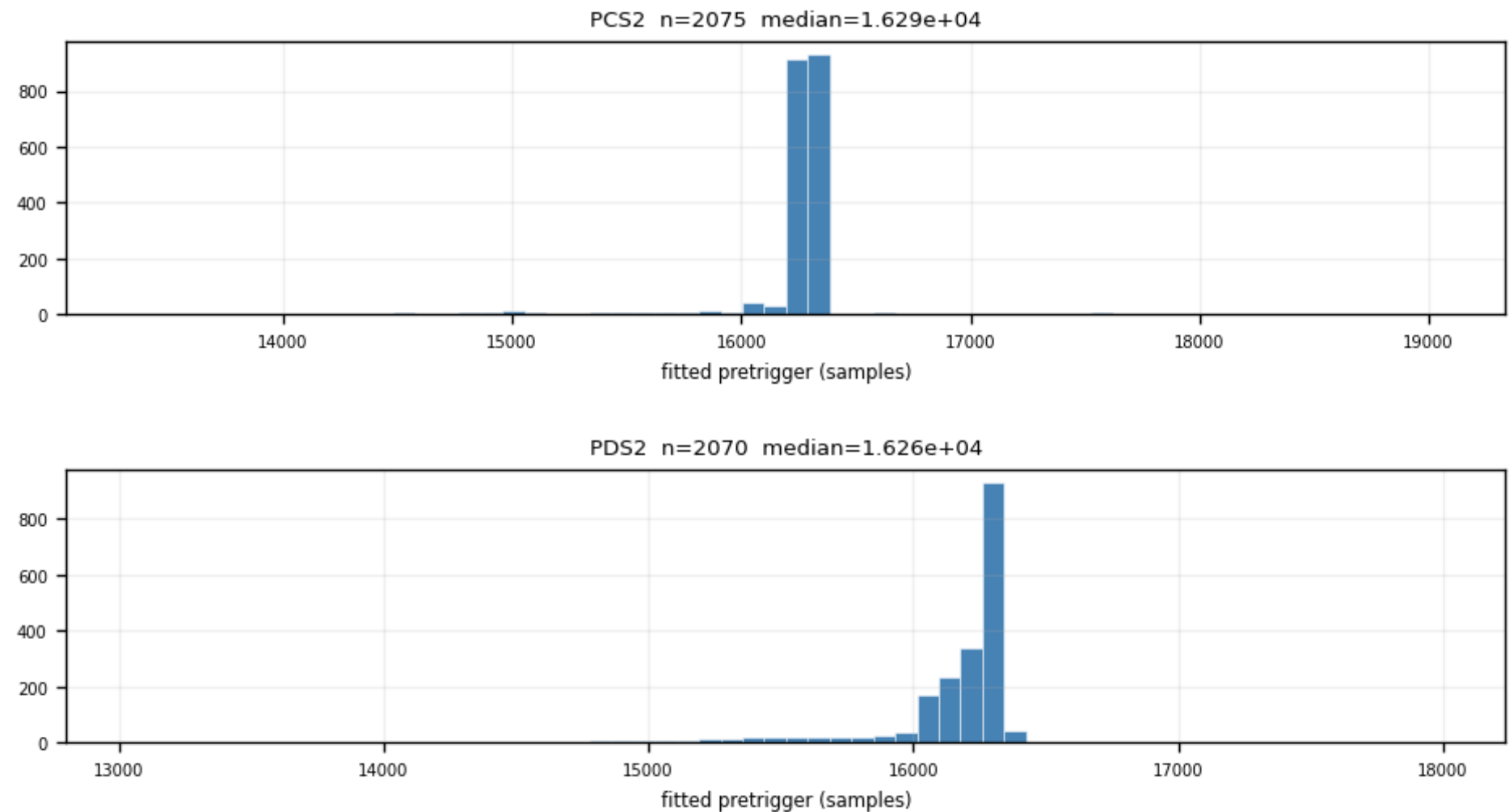
The onset is a FREE fit parameter; fitted values cluster  $\approx 230$  samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



# Backup · Z7 · pretrigger: fitted onset per channel (5/6)

figure: zip7\_pretrigger.png, channel panels 9–10 of 11, top to bottom

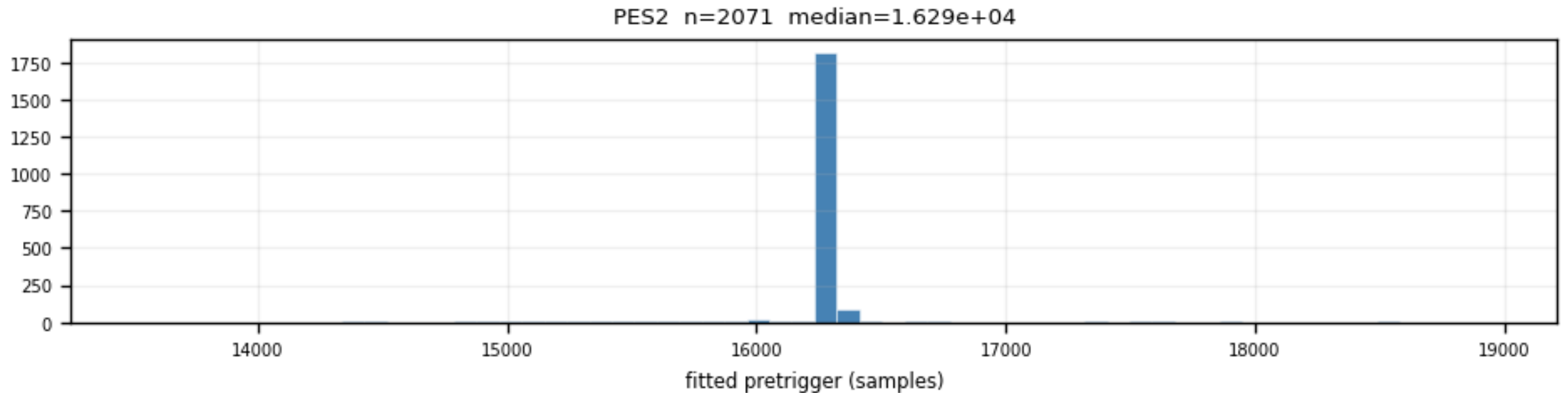
The onset is a FREE fit parameter; fitted values cluster  $\approx 230$  samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



# Backup · Z7 · pretrigger: fitted onset per channel (6/6)

figure: zip7\_pretrigger.png, channel panels 11–11 of 11, top to bottom

The onset is a FREE fit parameter; fitted values cluster  $\approx 230$  samples after the nominal 16050; pinning it inside the fit would bias every other parameter.

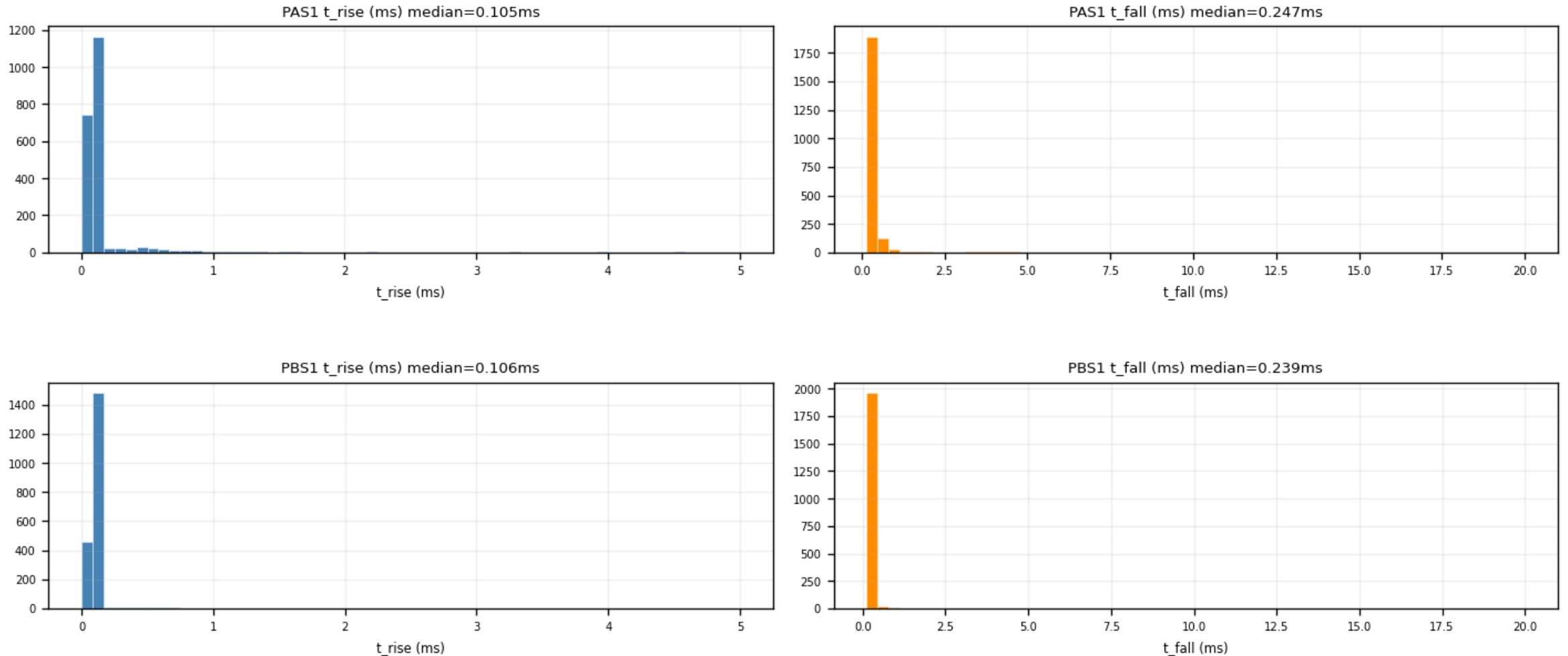




# Backup · Z7 · time\_constants: $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (1/6)

figure: zip7\_time\_constants.png, channel panels 1–2 of 11, top to bottom

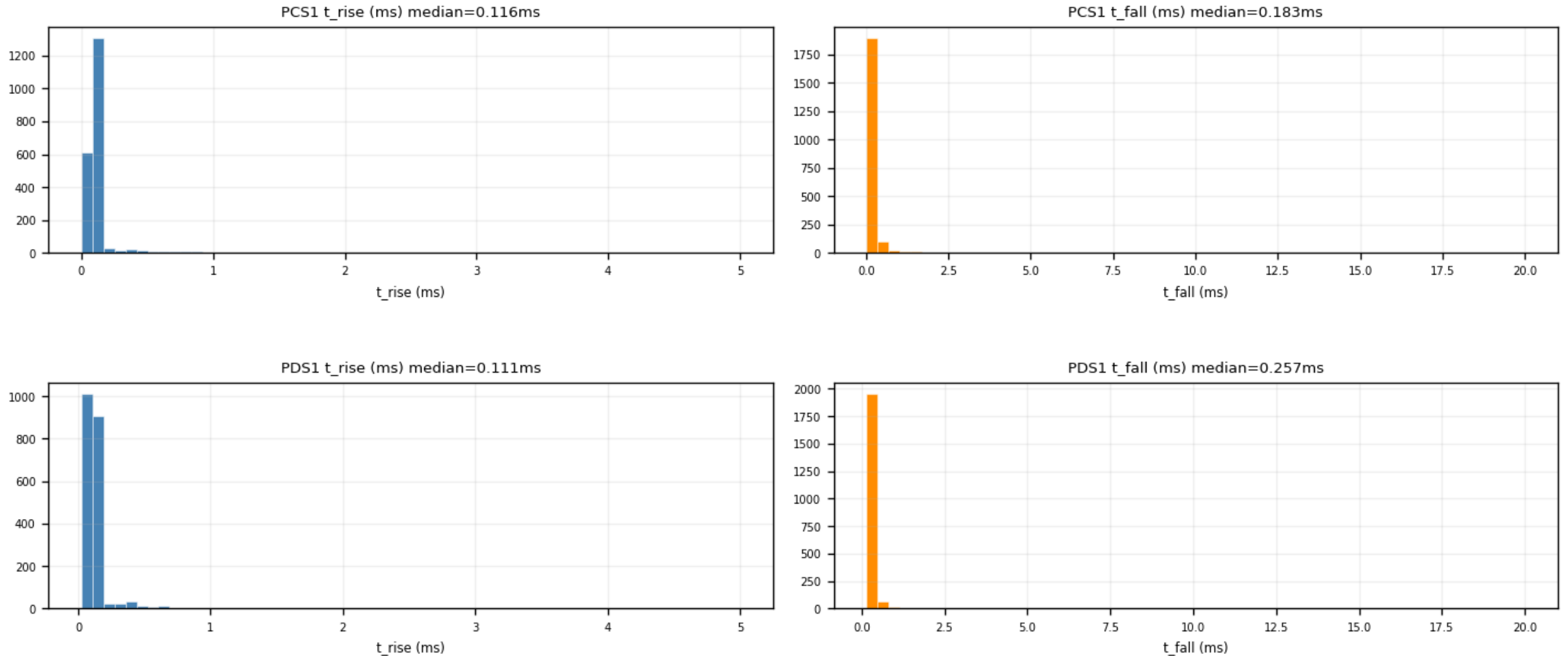
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



# Backup · Z7 · time\_constants: $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (2/6)

figure: zip7\_time\_constants.png, channel panels 3–4 of 11, top to bottom

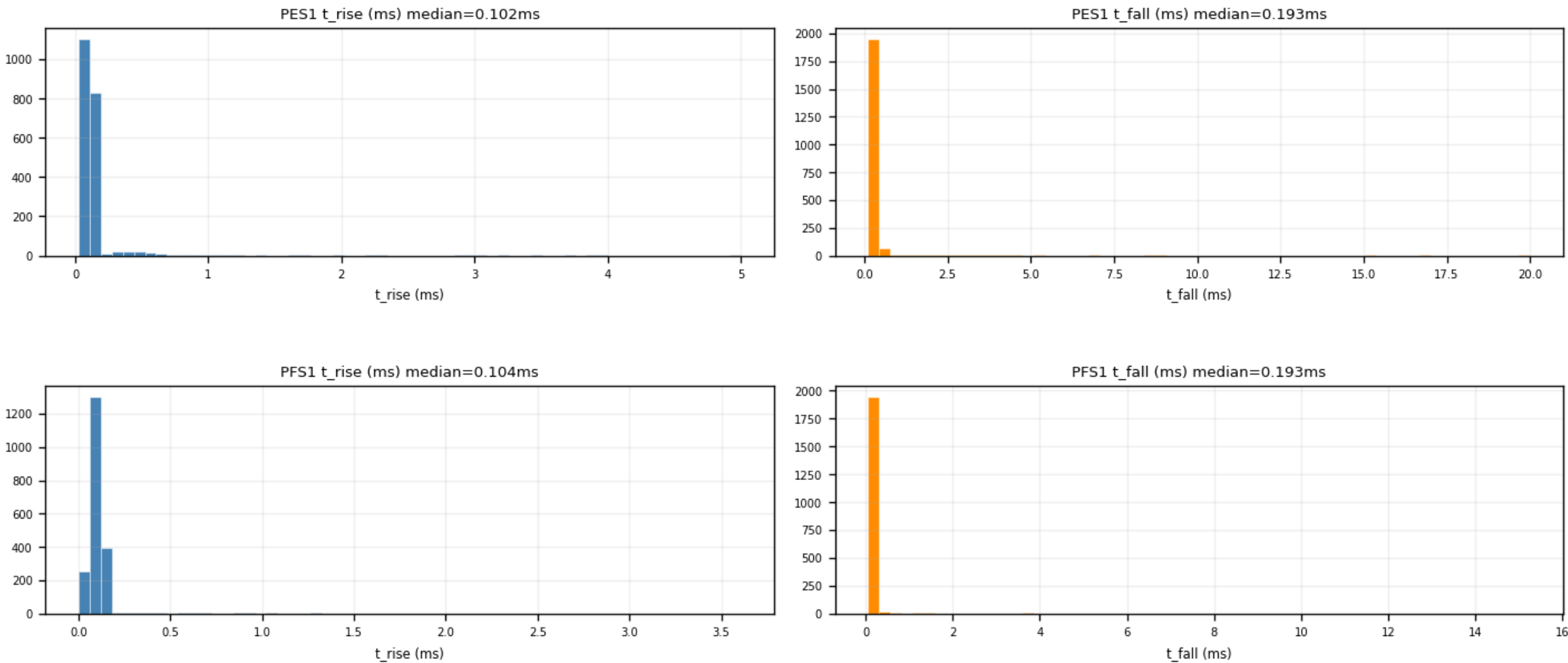
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



# Backup · Z7 · time\_constants: $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (3/6)

figure: zip7\_time\_constants.png, channel panels 5–6 of 11, top to bottom

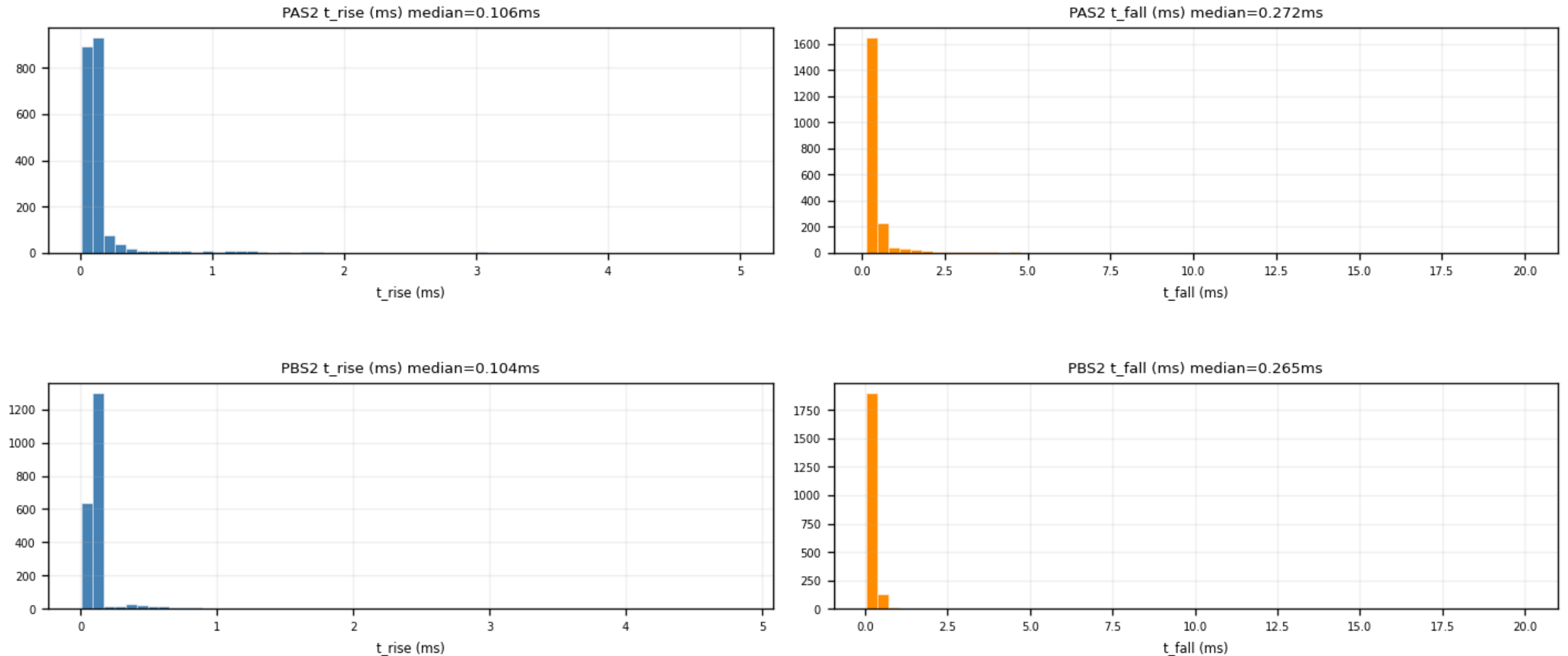
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



# Backup · Z7 · time\_constants: $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (4/6)

figure: zip7\_time\_constants.png, channel panels 7–8 of 11, top to bottom

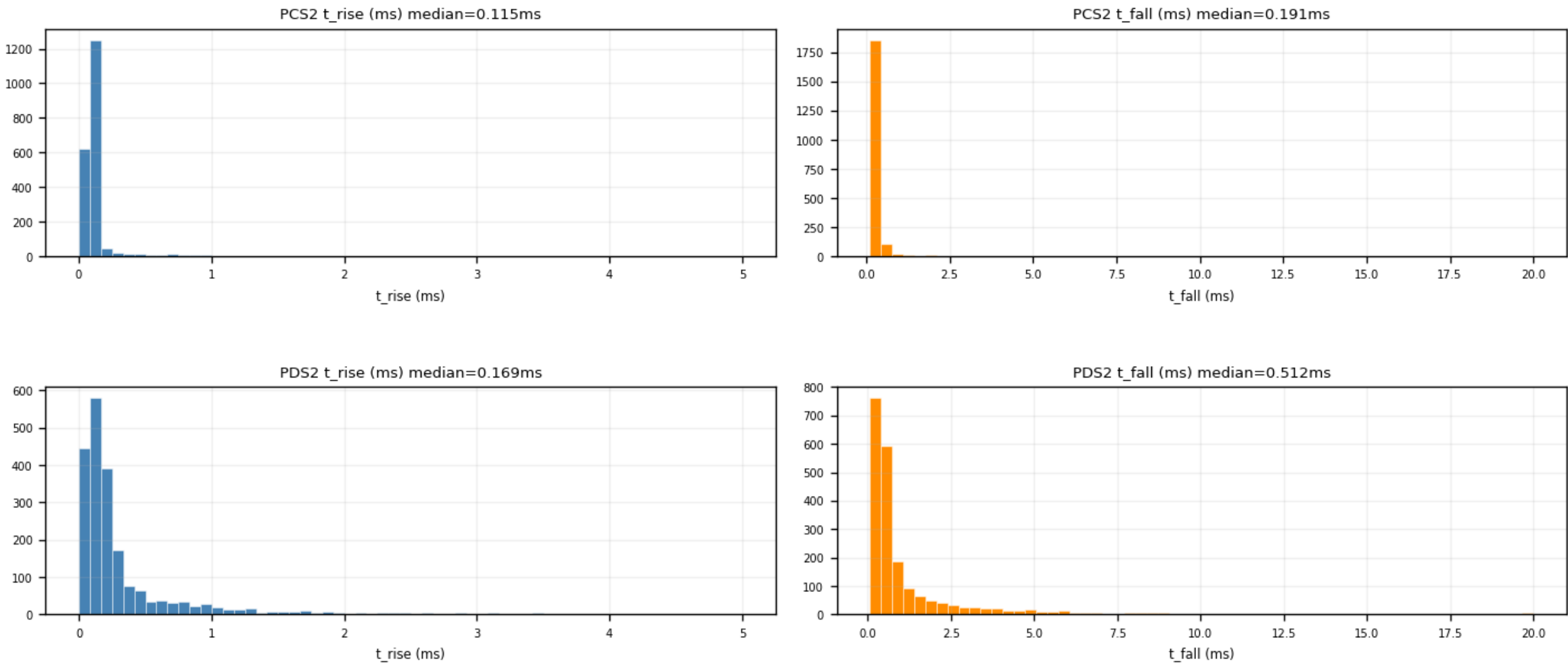
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



# Backup · Z7 · time\_constants: $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (5/6)

figure: zip7\_time\_constants.png, channel panels 9–10 of 11, top to bottom

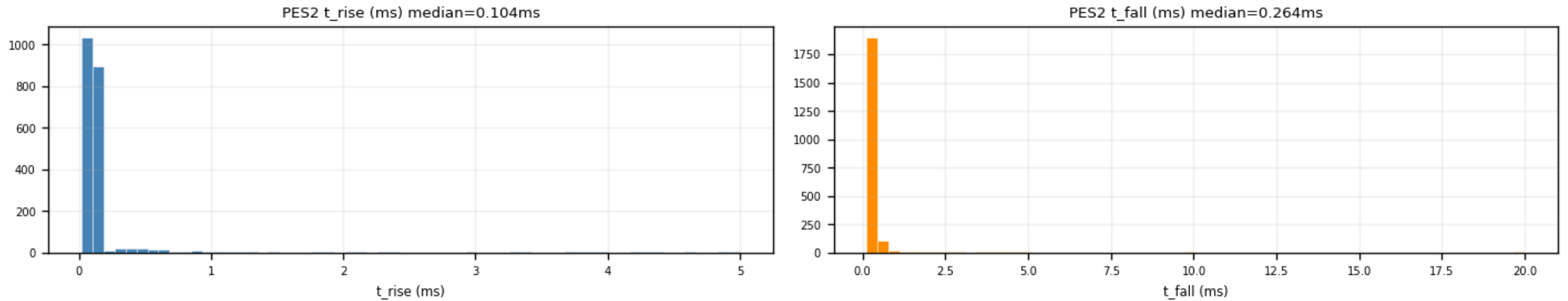
Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



# Backup · Z7 · time\_constants: $\tau_{\text{rise}}$ / $\tau_{\text{fall}}$ per channel (6/6)

figure: zip7\_time\_constants.png, channel panels 11–11 of 11, top to bottom

Fast-pulse population at  $\tau_{\text{rise}} \approx 0.1$  ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.

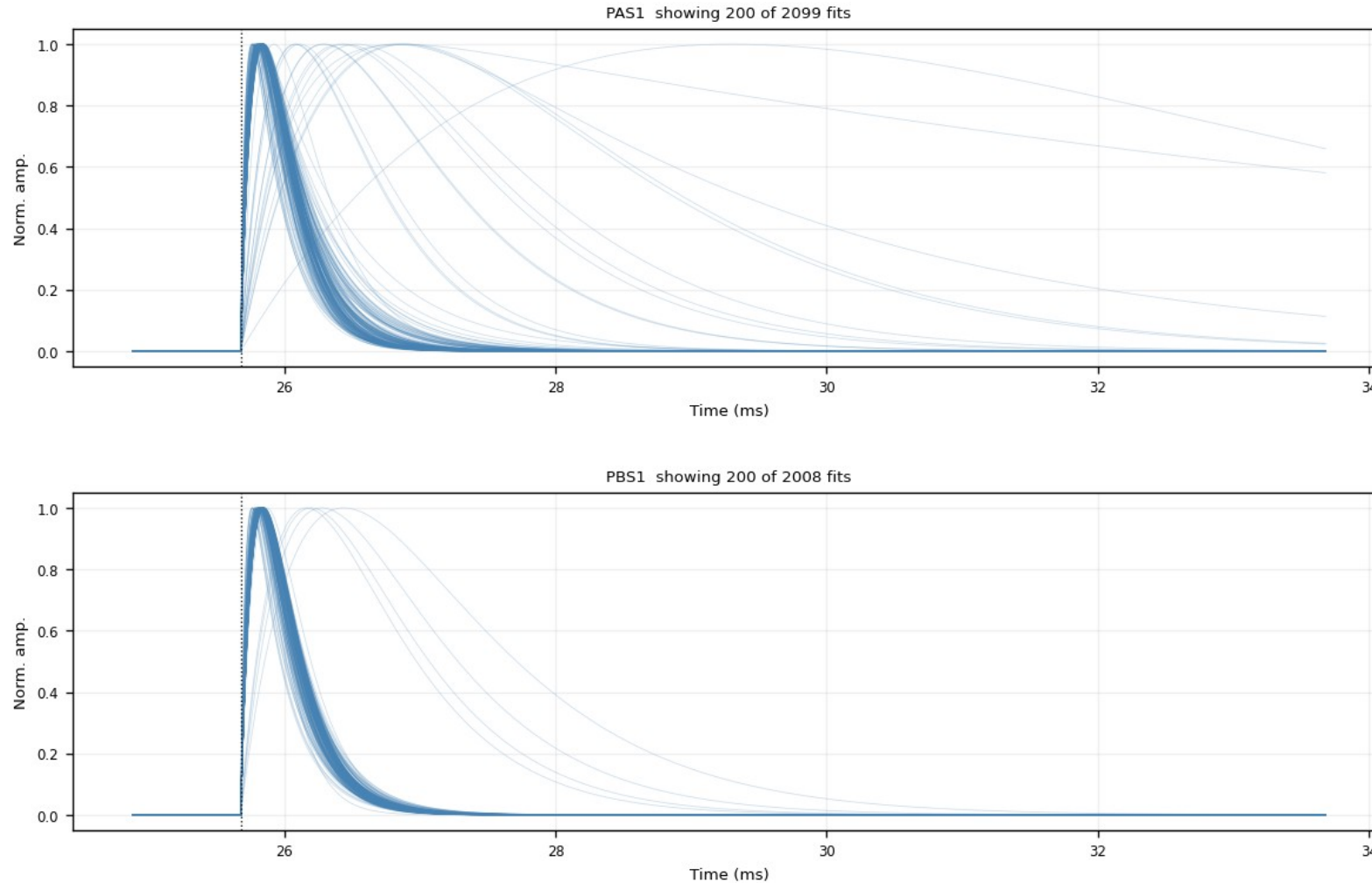




# Backup · Z7 · fitted\_curves\_overlay: fan, no cut (1/6)

figure: zip7\_fitted\_curves\_overlay.png, channel panels 1–2 of 11, top to bottom

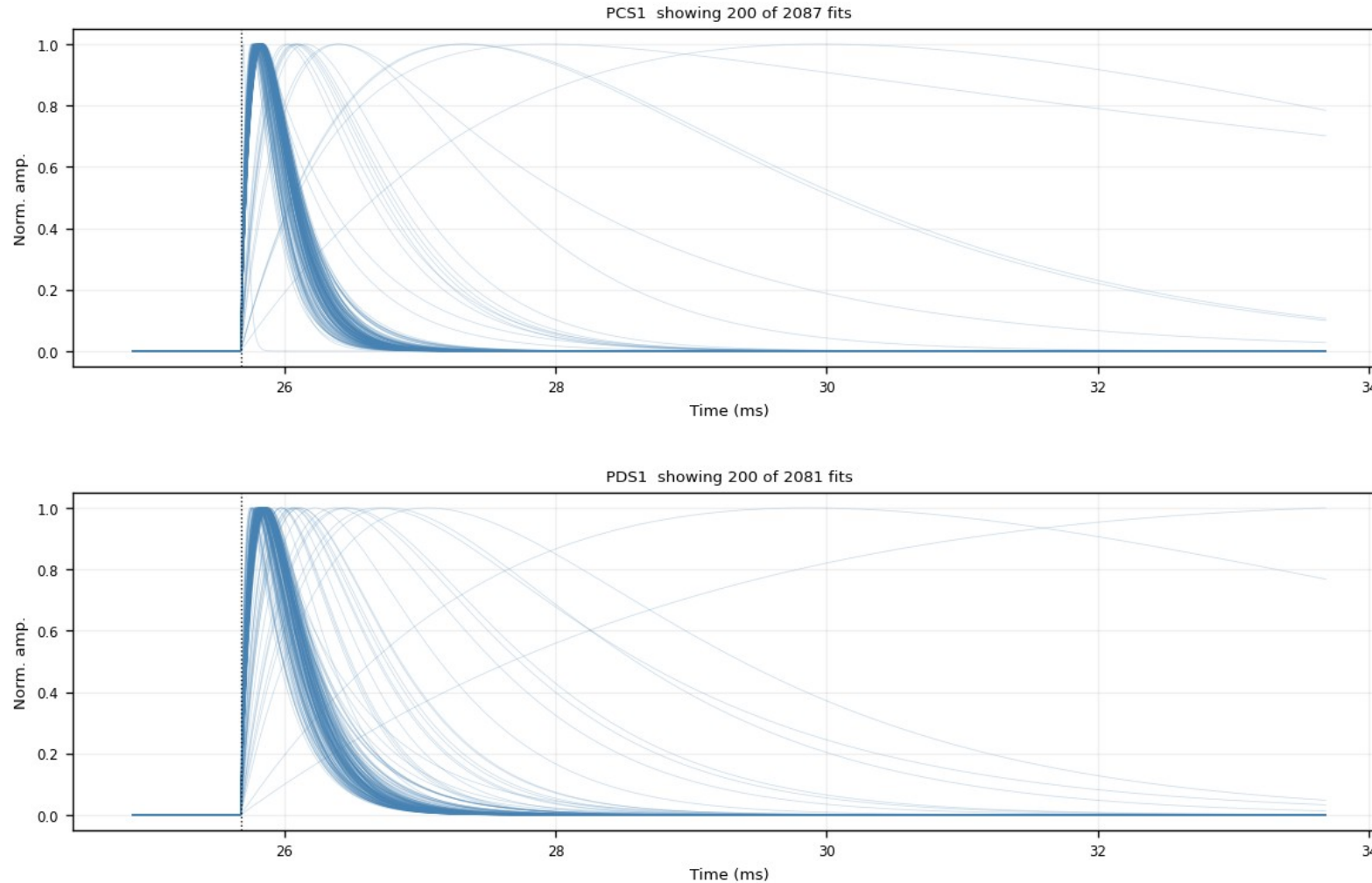
Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



# Backup · Z7 · fitted\_curves\_overlay: fan, no cut (2/6)

figure: zip7\_fitted\_curves\_overlay.png, channel panels 3–4 of 11, top to bottom

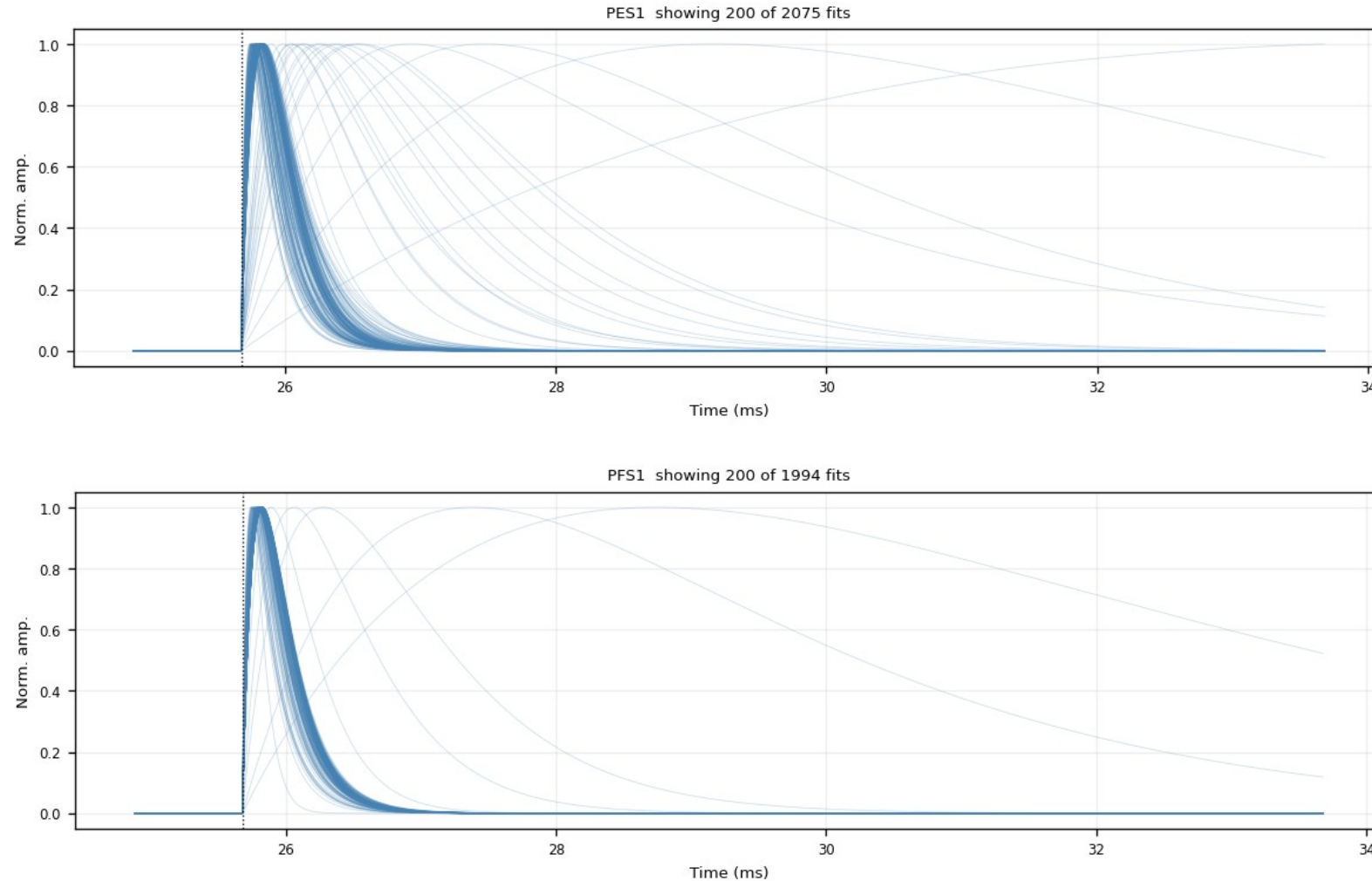
Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



# Backup · Z7 · fitted\_curves\_overlay: fan, no cut (3/6)

figure: zip7\_fitted\_curves\_overlay.png, channel panels 5–6 of 11, top to bottom

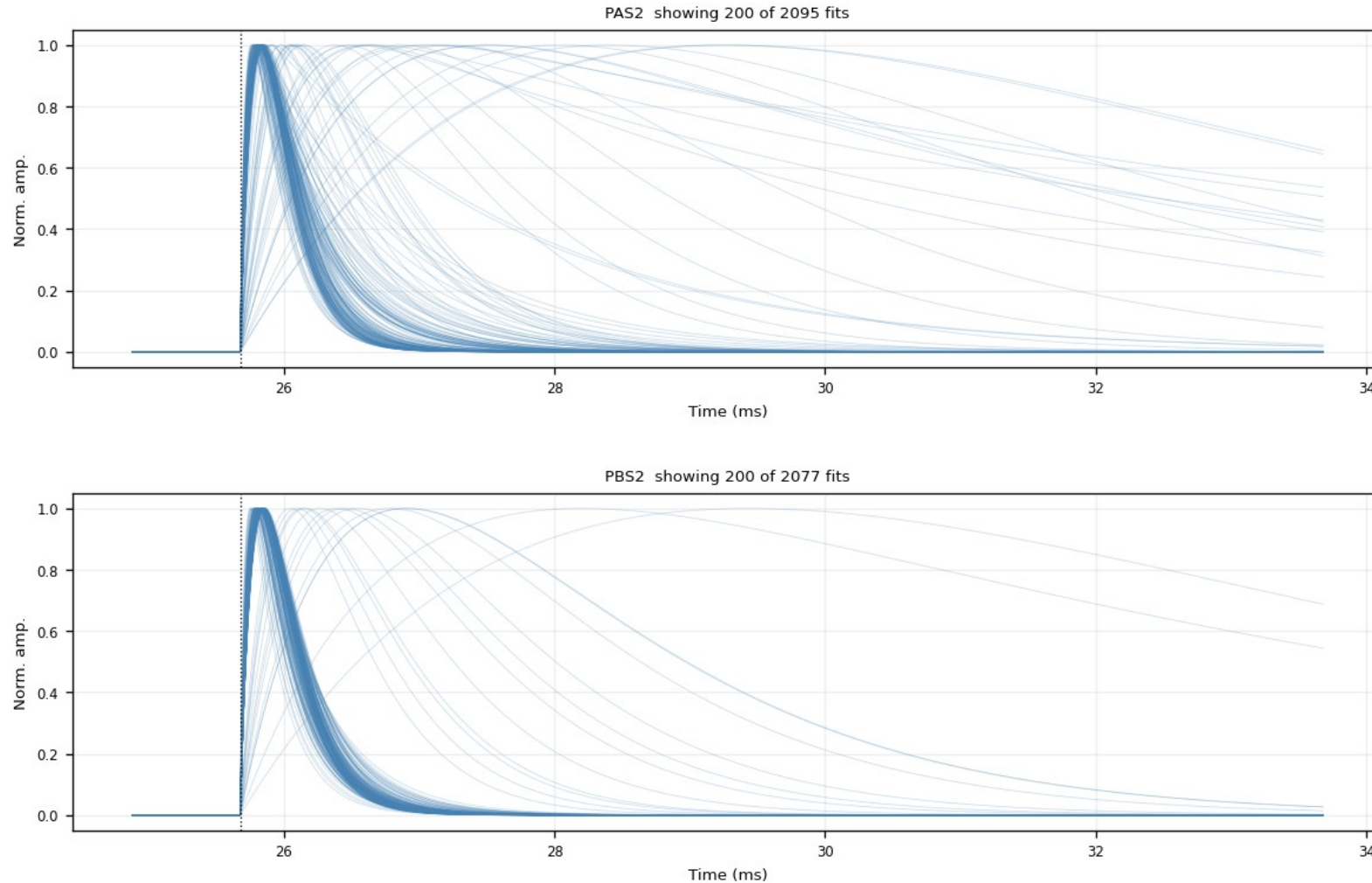
Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



# Backup · Z7 · fitted\_curves\_overlay: fan, no cut (4/6)

figure: zip7\_fitted\_curves\_overlay.png, channel panels 7–8 of 11, top to bottom

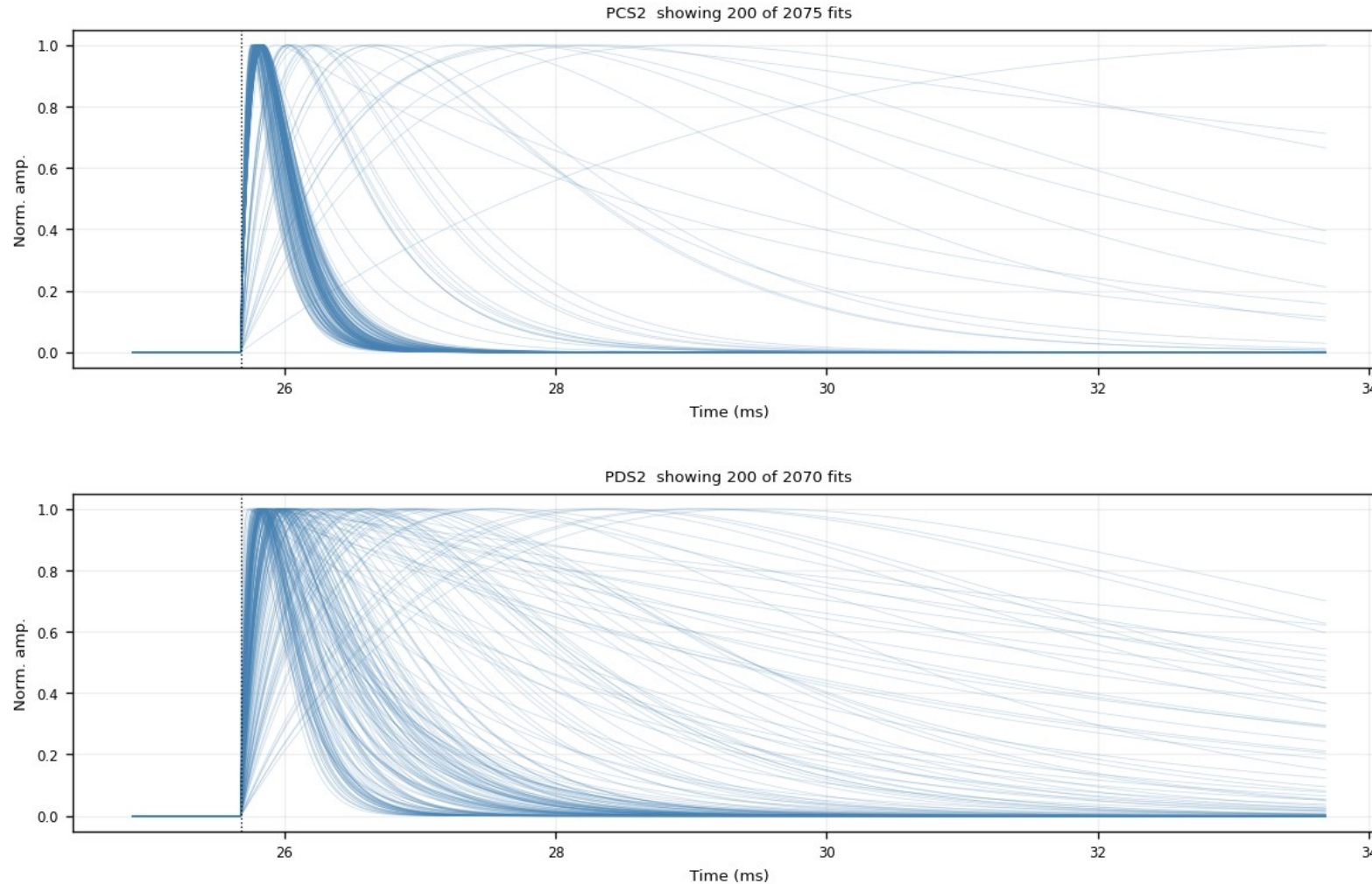
Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



# Backup · Z7 · fitted\_curves\_overlay: fan, no cut (5/6)

figure: zip7\_fitted\_curves\_overlay.png, channel panels 9–10 of 11, top to bottom

Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

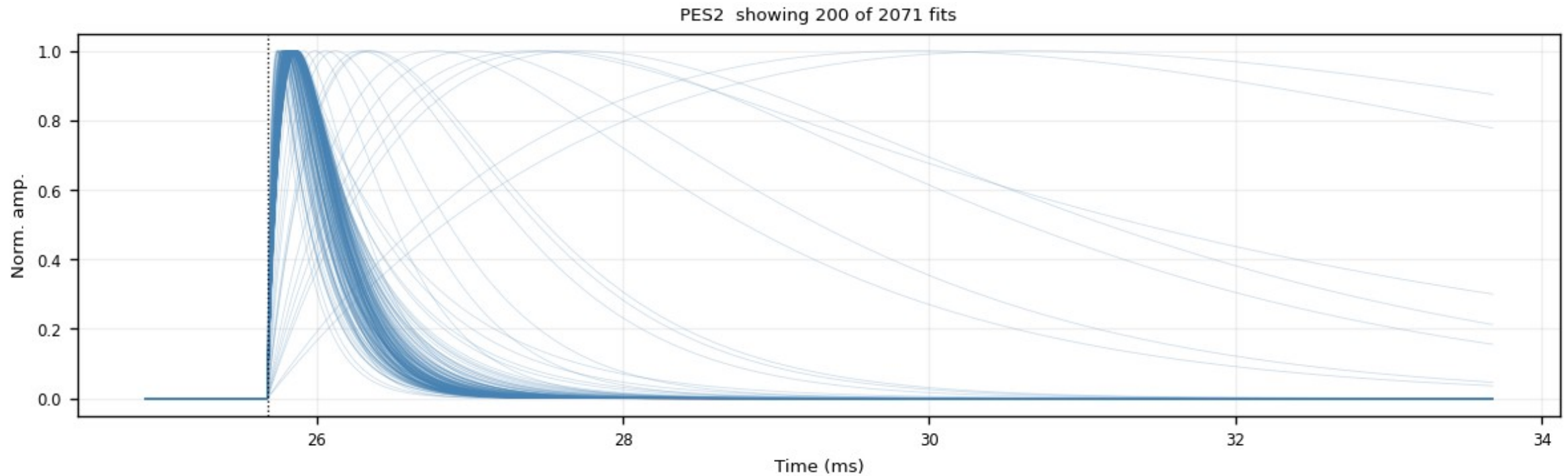




## Backup · Z7 · fitted\_curves\_overlay: fan, no cut (6/6)

figure: zip7\_fitted\_curves\_overlay.png, channel panels 11–11 of 11, top to bottom

Every fit\_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

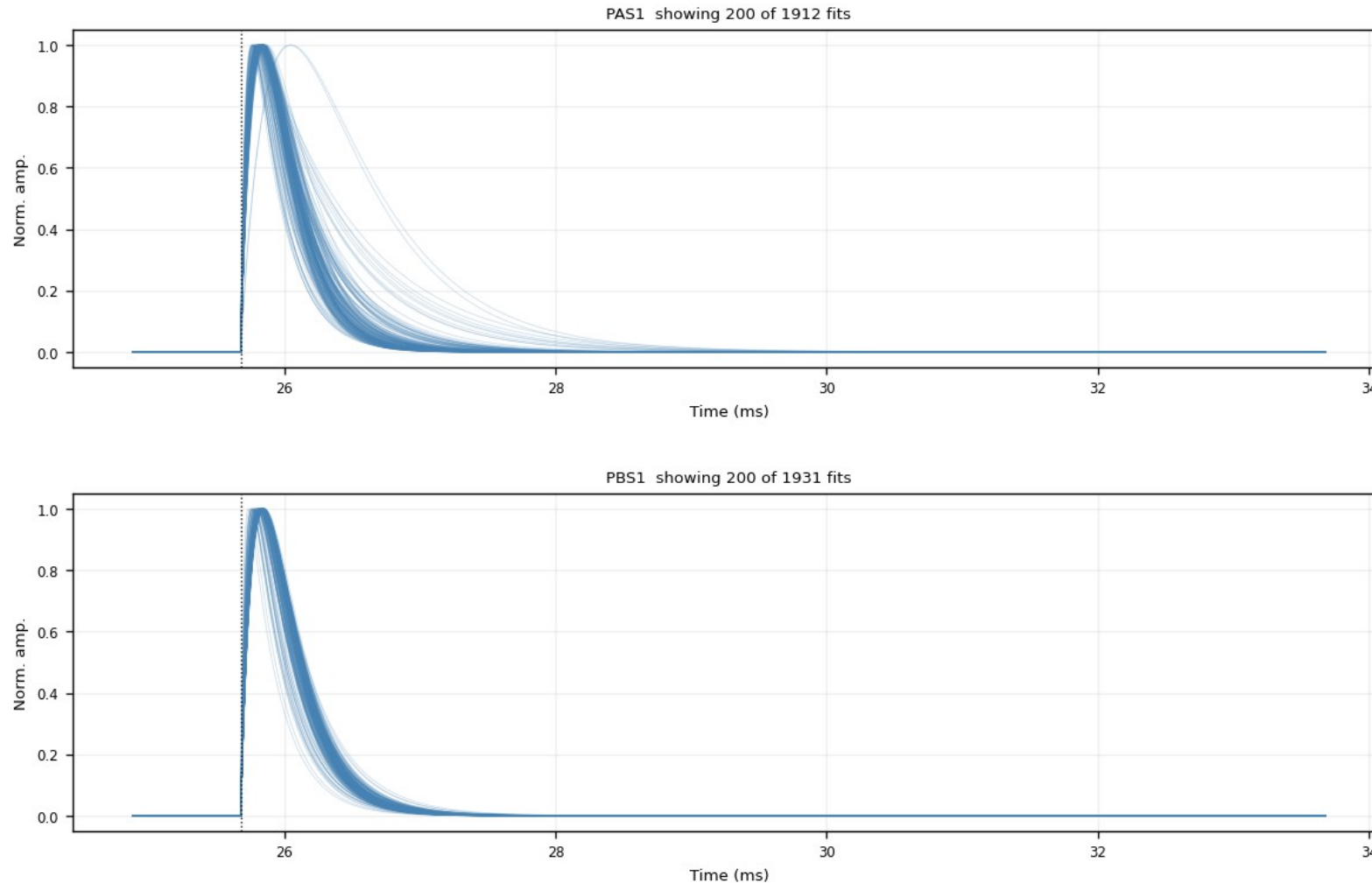




# Backup · Z7 · fitted\_curves\_overlay: after $\text{NRMSE} \leq 0.4$ (1/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

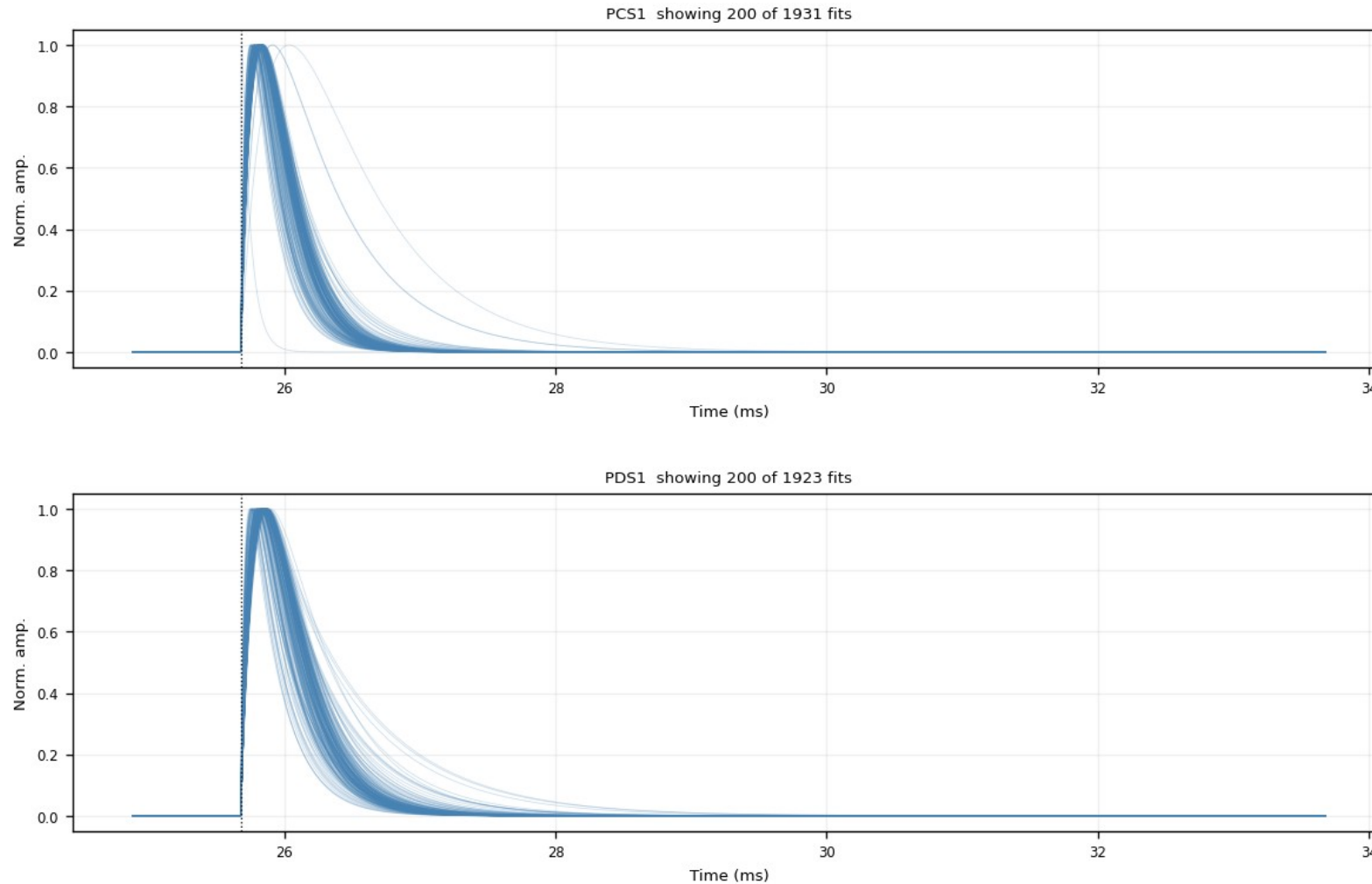
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



# Backup · Z7 · fitted\_curves\_overlay: after $\text{NRMSE} \leq 0.4$ (2/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

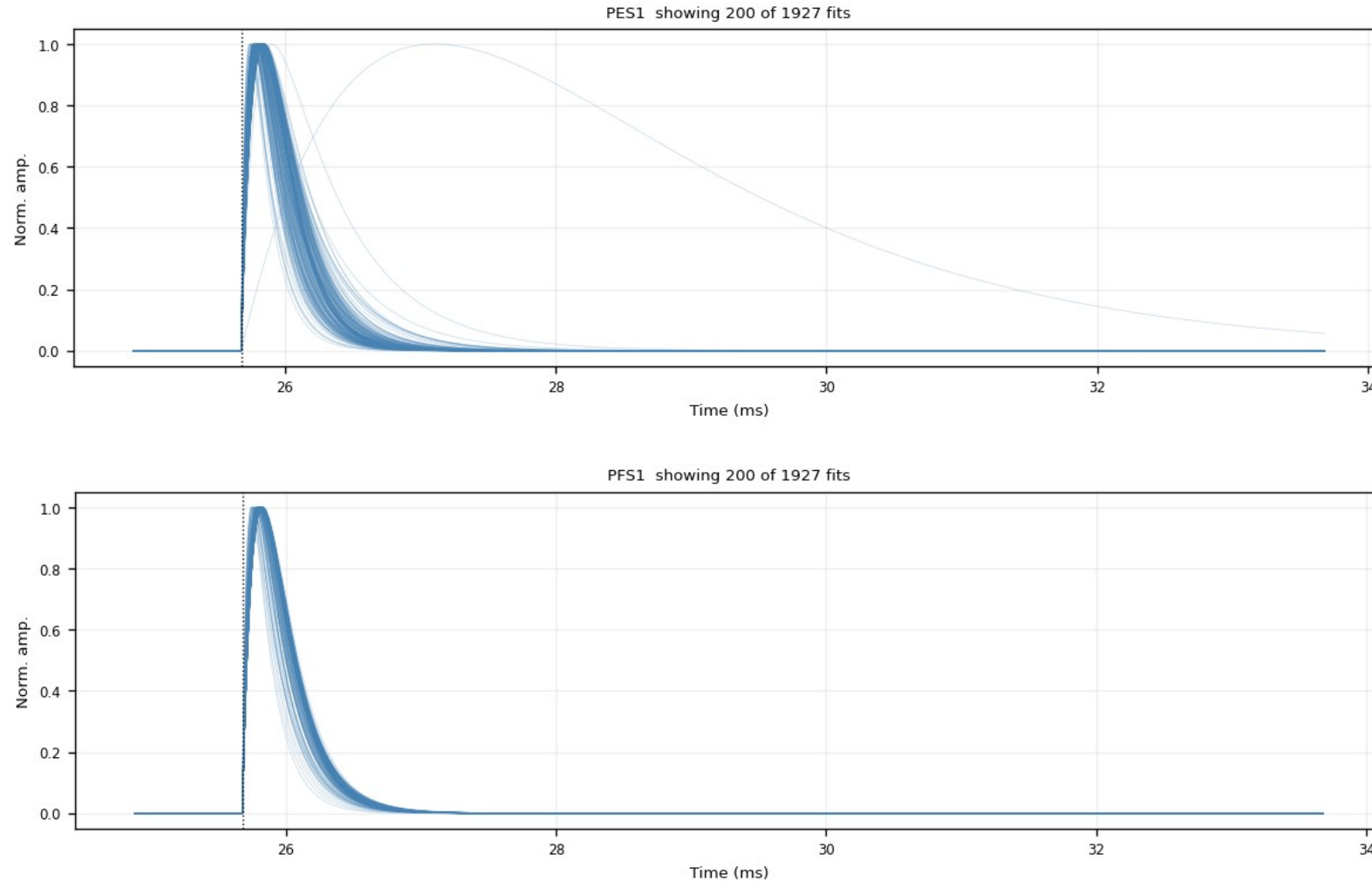
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



# Backup · Z7 · fitted\_curves\_overlay: after $\text{NRMSE} \leq 0.4$ (3/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

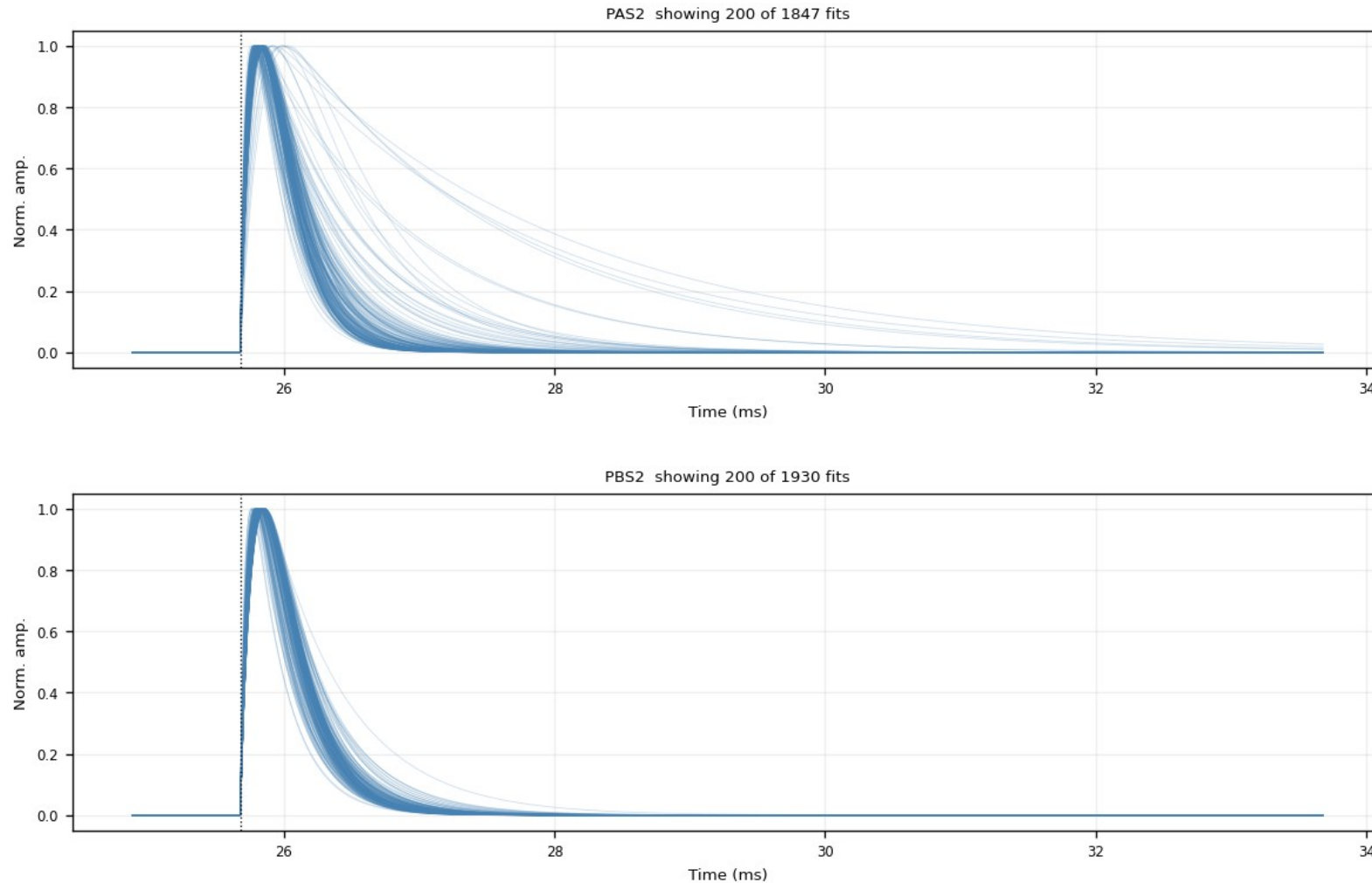
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



# Backup · Z7 · fitted\_curves\_overlay: after $\text{NRMSE} \leq 0.4$ (4/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

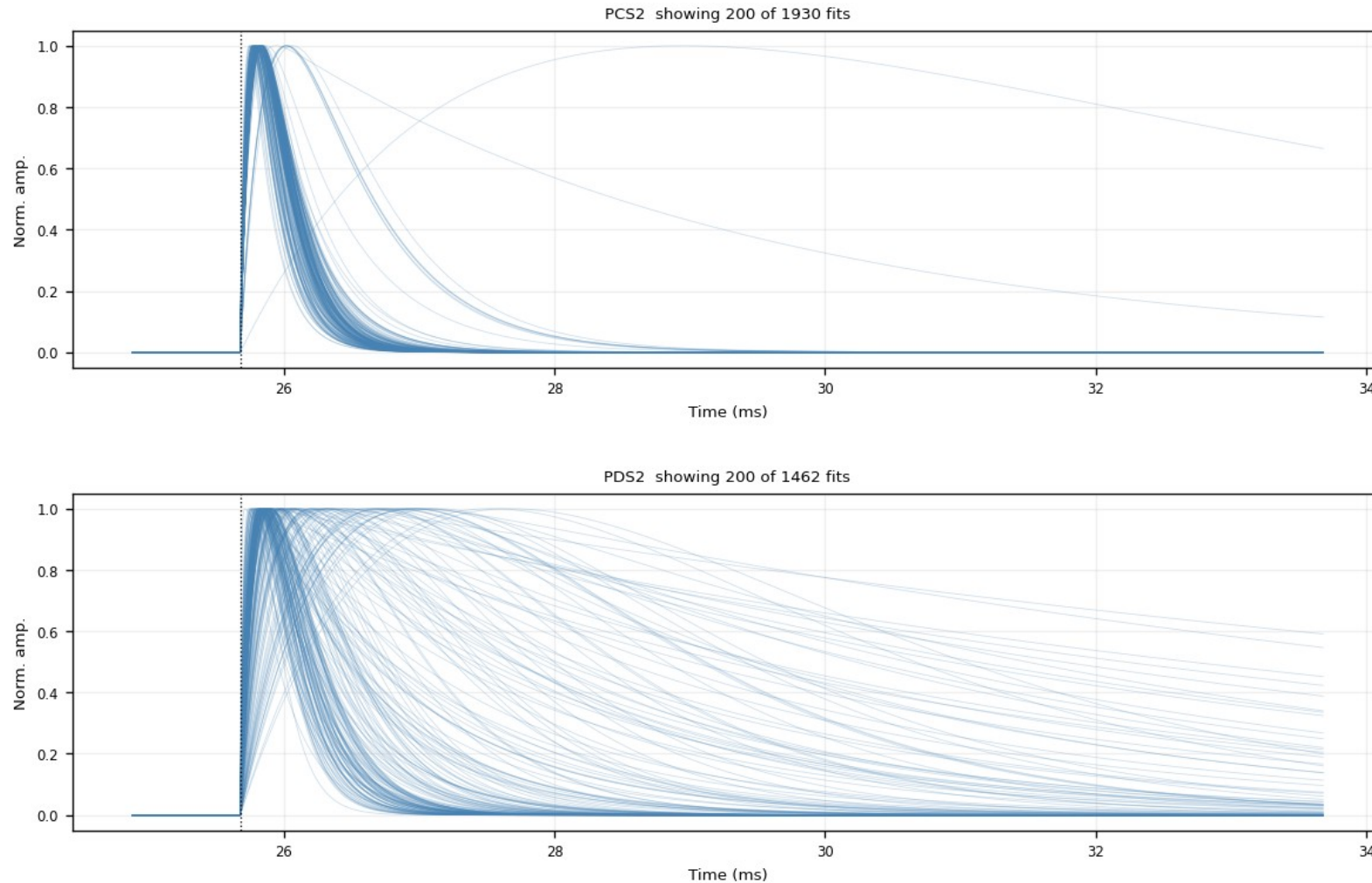
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



# Backup · Z7 · fitted\_curves\_overlay: after $\text{NRMSE} \leq 0.4$ (5/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

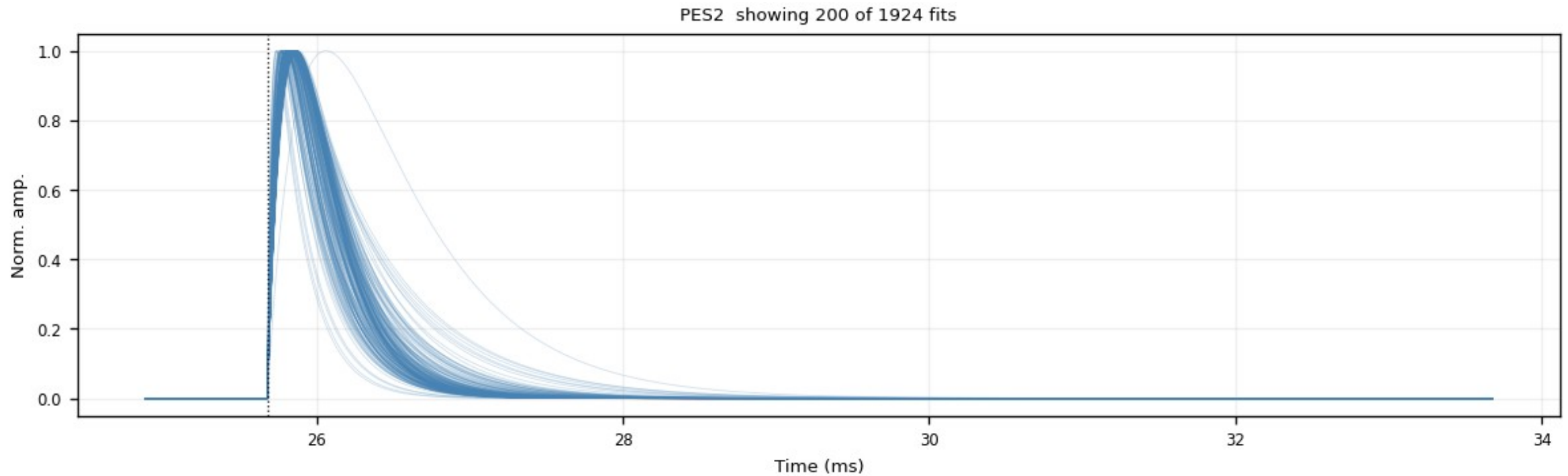
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



## Backup · Z7 · fitted\_curves\_overlay: after $\text{NRMSE} \leq 0.4$ (6/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.

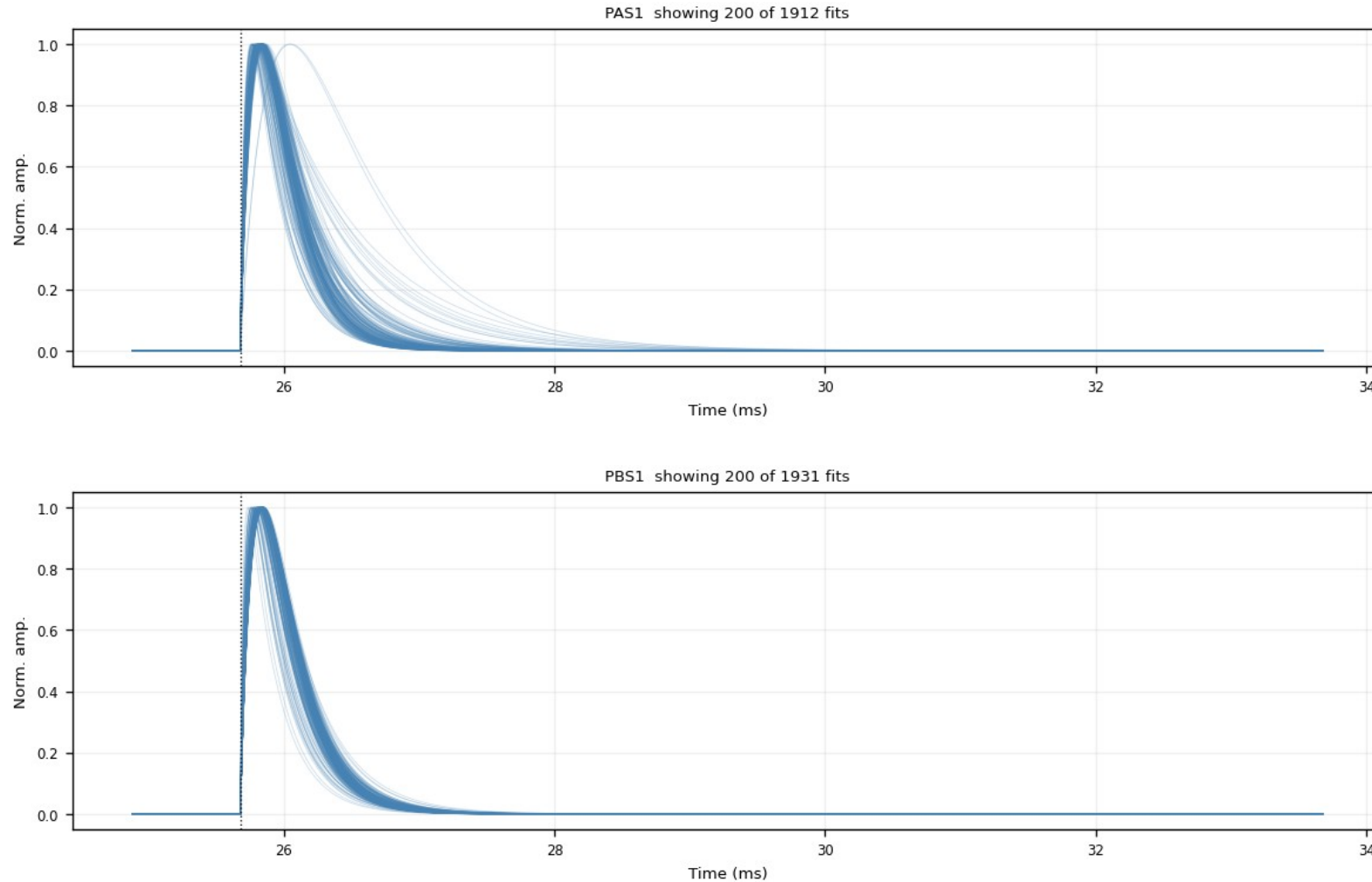




# Backup · Z7 · fitted\_curves\_overlay: after both cuts (1/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png, channel panels 1–2 of 11, top to bottom

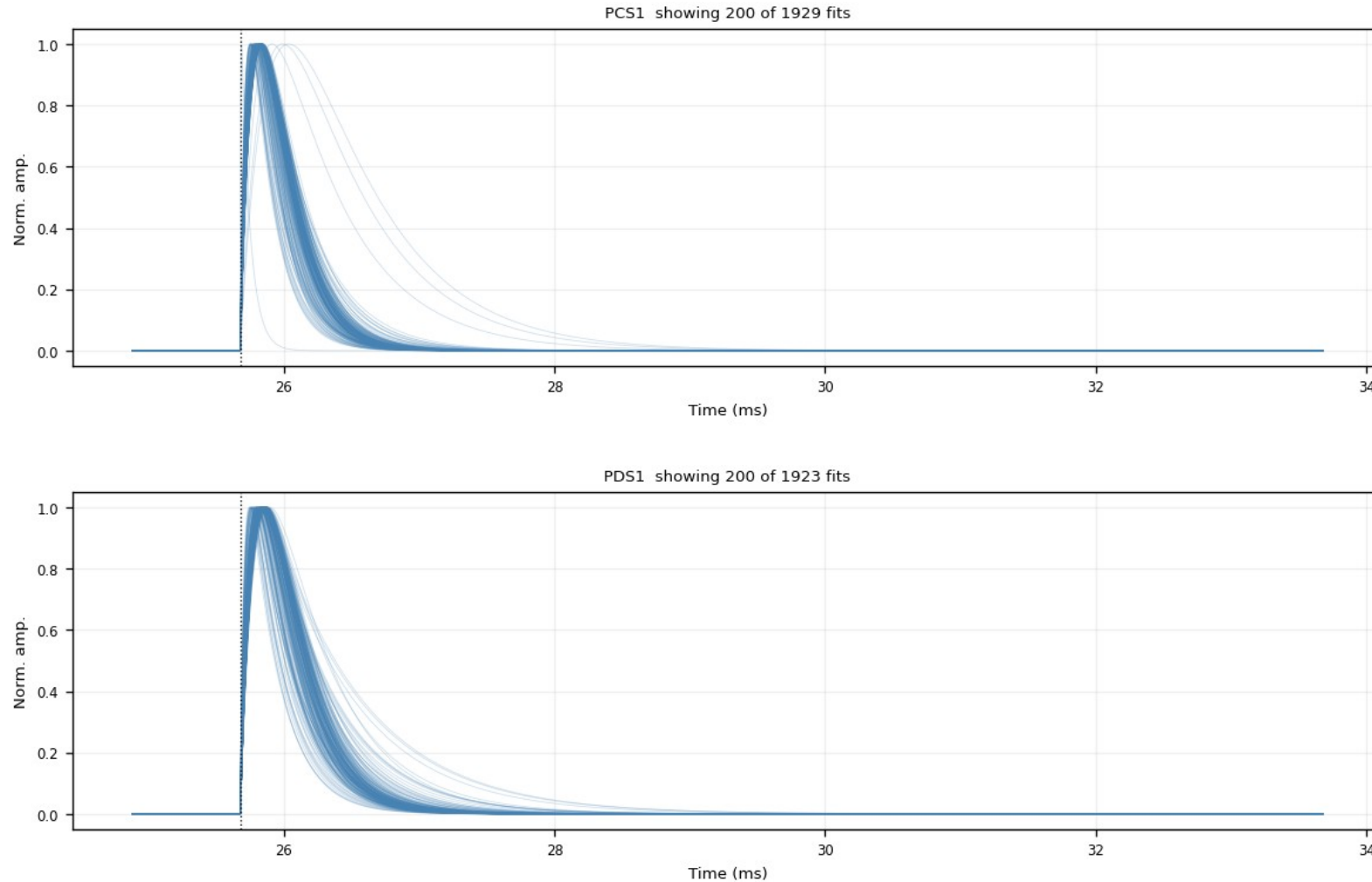
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms, the exact PCA input population: the clean fast-pulse bundle only.



# Backup · Z7 · fitted\_curves\_overlay: after both cuts (2/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png, channel panels 3–4 of 11, top to bottom

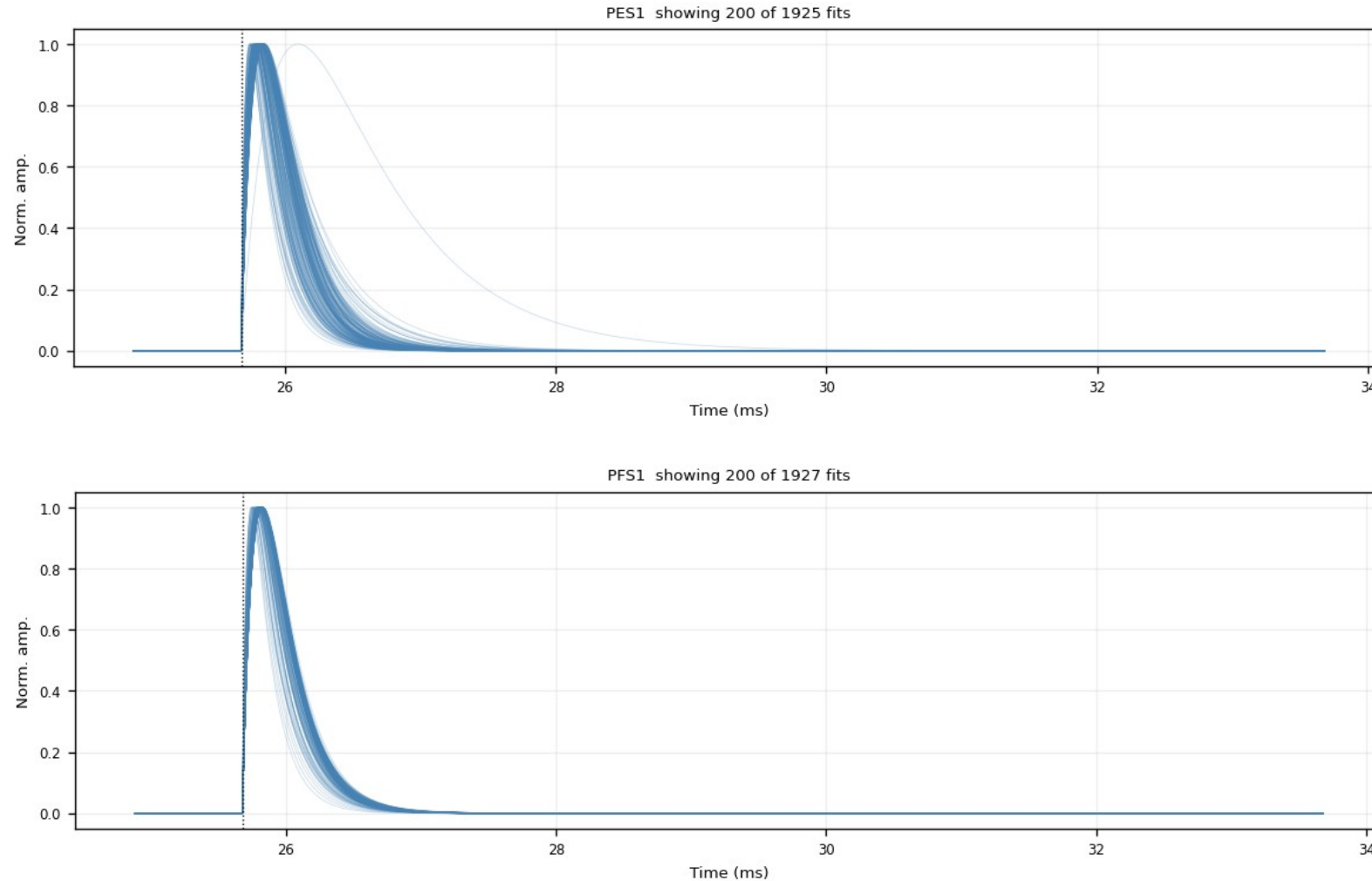
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms, the exact PCA input population: the clean fast-pulse bundle only.



# Backup · Z7 · fitted\_curves\_overlay: after both cuts (3/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png, channel panels 5–6 of 11, top to bottom

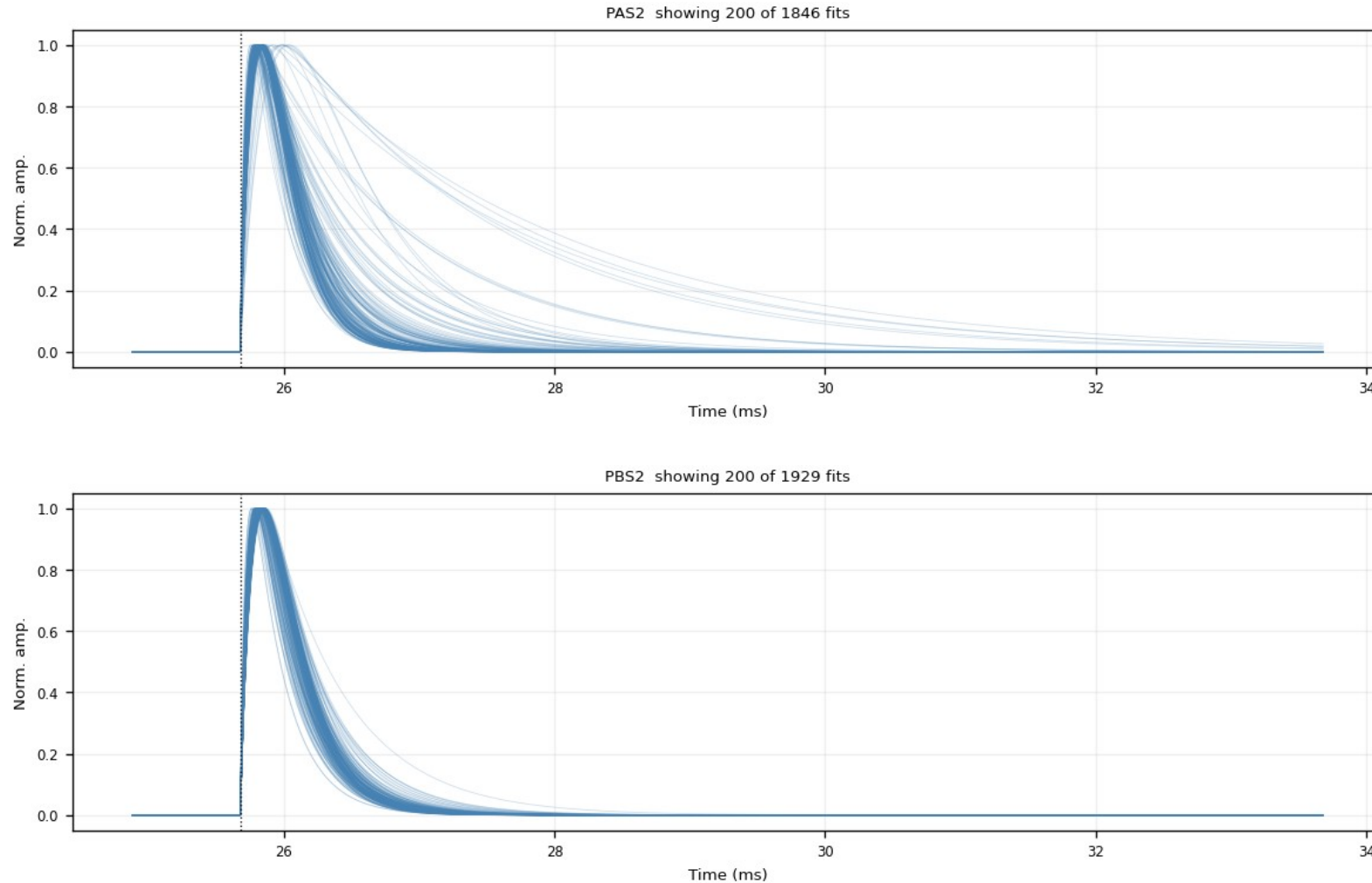
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms, the exact PCA input population: the clean fast-pulse bundle only.



# Backup · Z7 · fitted\_curves\_overlay: after both cuts (4/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png, channel panels 7–8 of 11, top to bottom

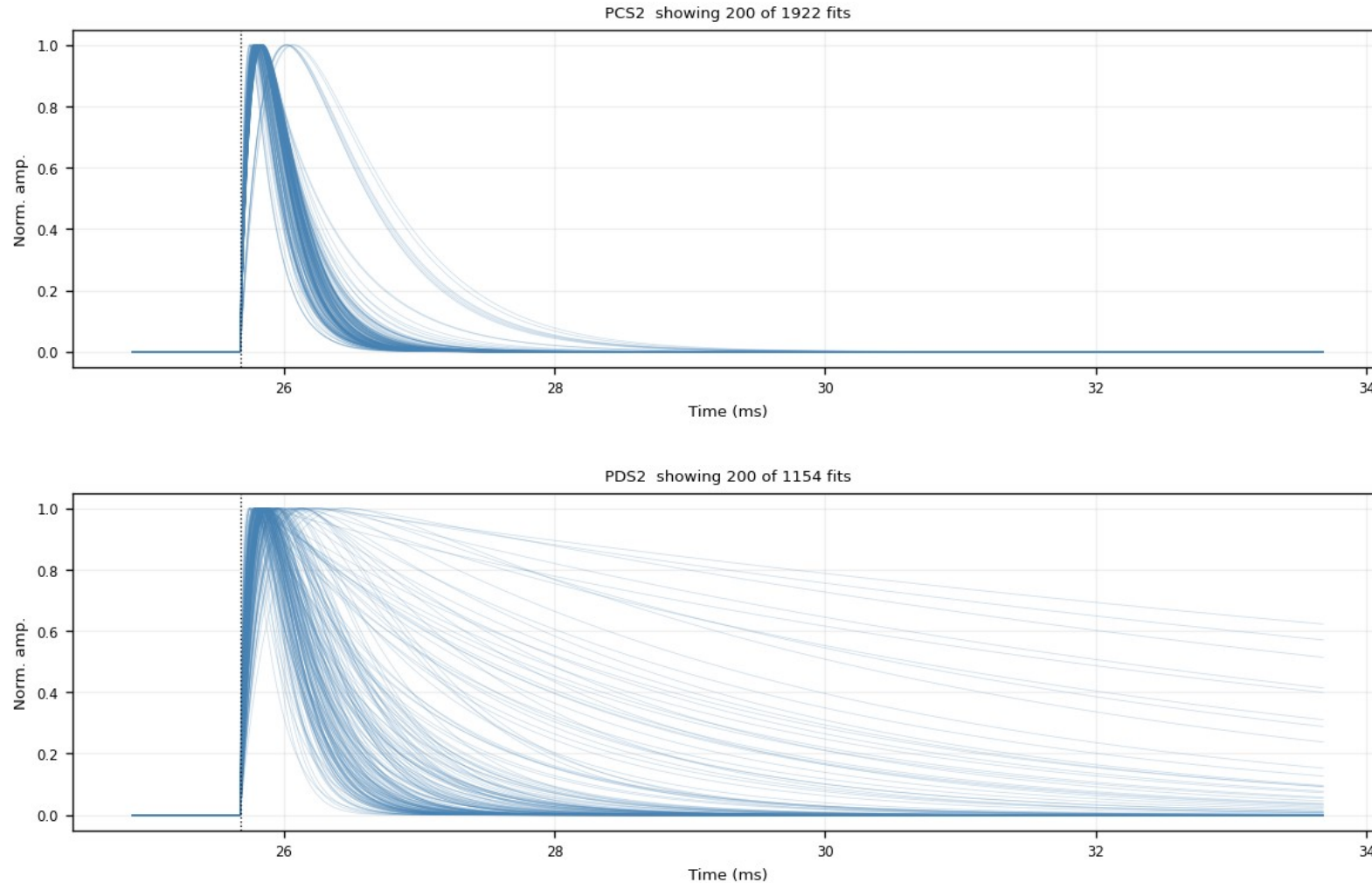
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms, the exact PCA input population: the clean fast-pulse bundle only.



# Backup · Z7 · fitted\_curves\_overlay: after both cuts (5/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png, channel panels 9–10 of 11, top to bottom

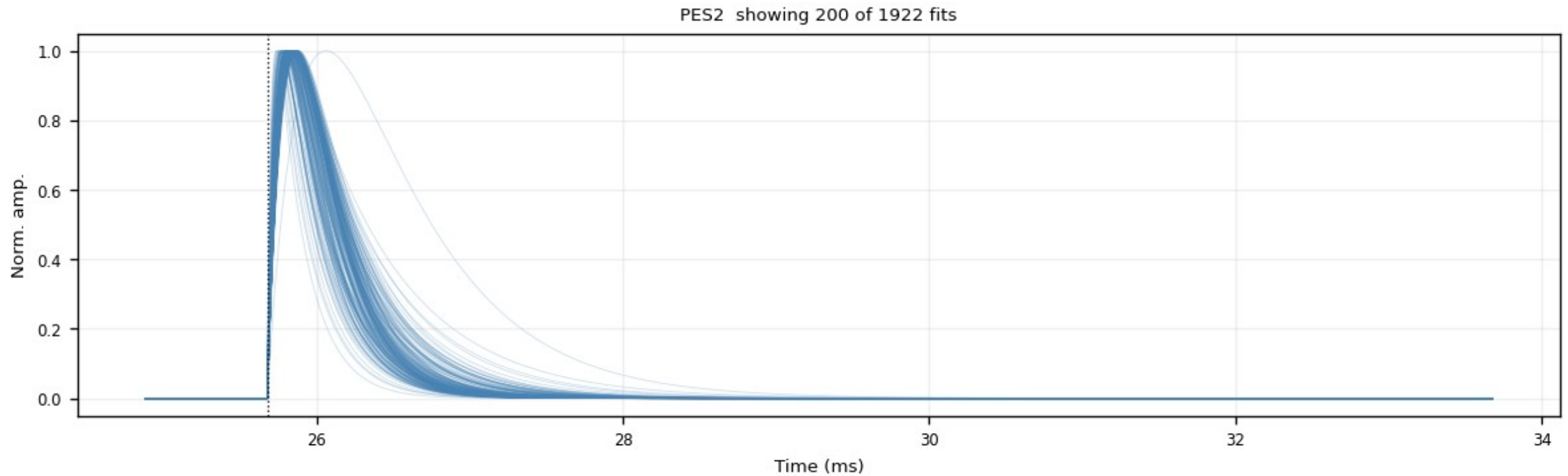
After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms, the exact PCA input population: the clean fast-pulse bundle only.



# Backup · Z7 · fitted\_curves\_overlay: after both cuts (6/6)

figure: zip7\_fitted\_curves\_overlay\_nrmse0.4\_trise0.30ms.png, channel panels 11–11 of 11, top to bottom

After  $\text{NRMSE} \leq 0.4$  AND  $\tau_{\text{rise}} \leq 0.3$  ms, the exact PCA input population: the clean fast-pulse bundle only.

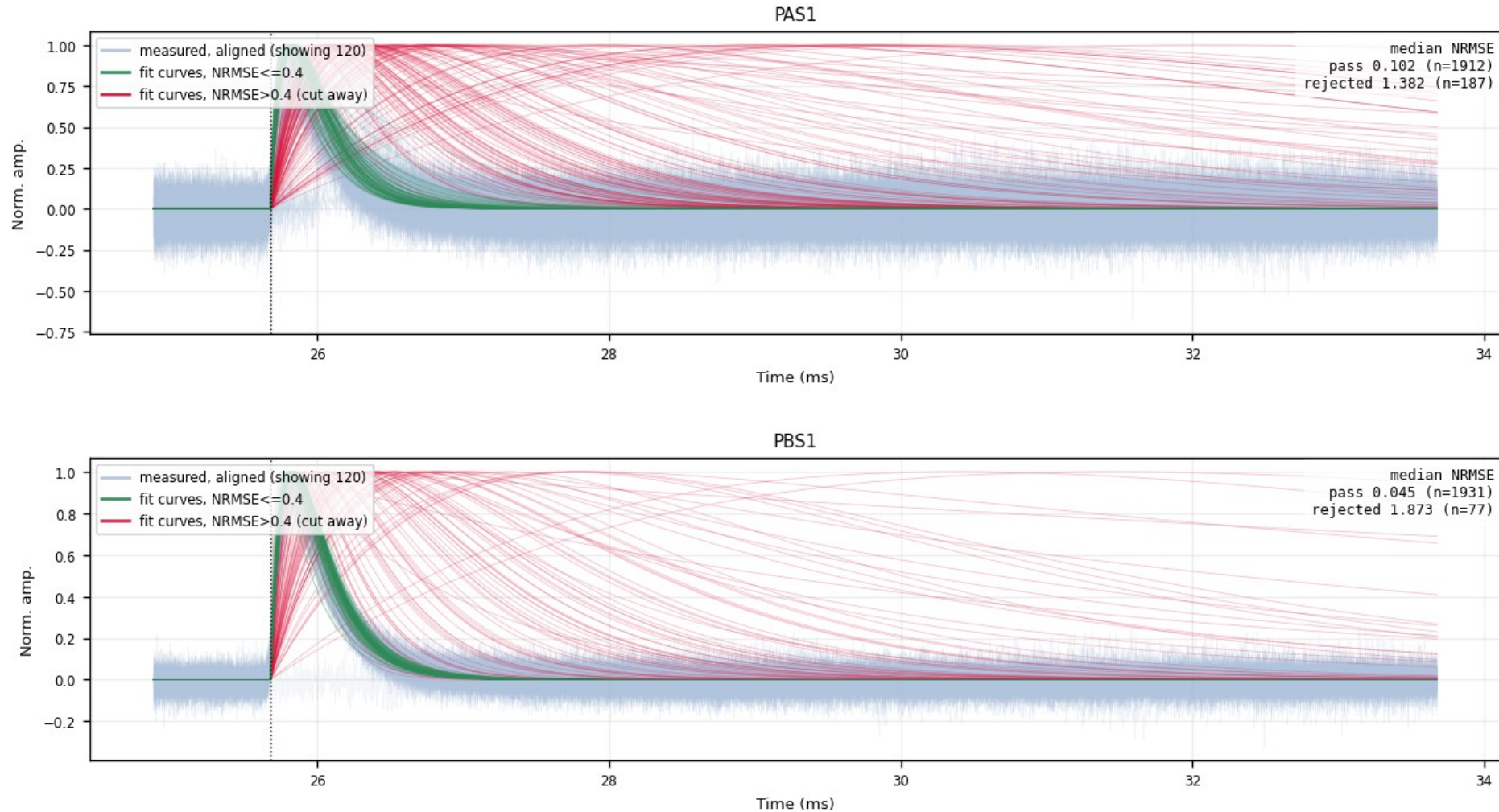




# Backup · Z7 · overlay\_fan\_cut: kept vs rejected fits (1/6)

figure: zip7\_overlay\_fan\_cut\_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

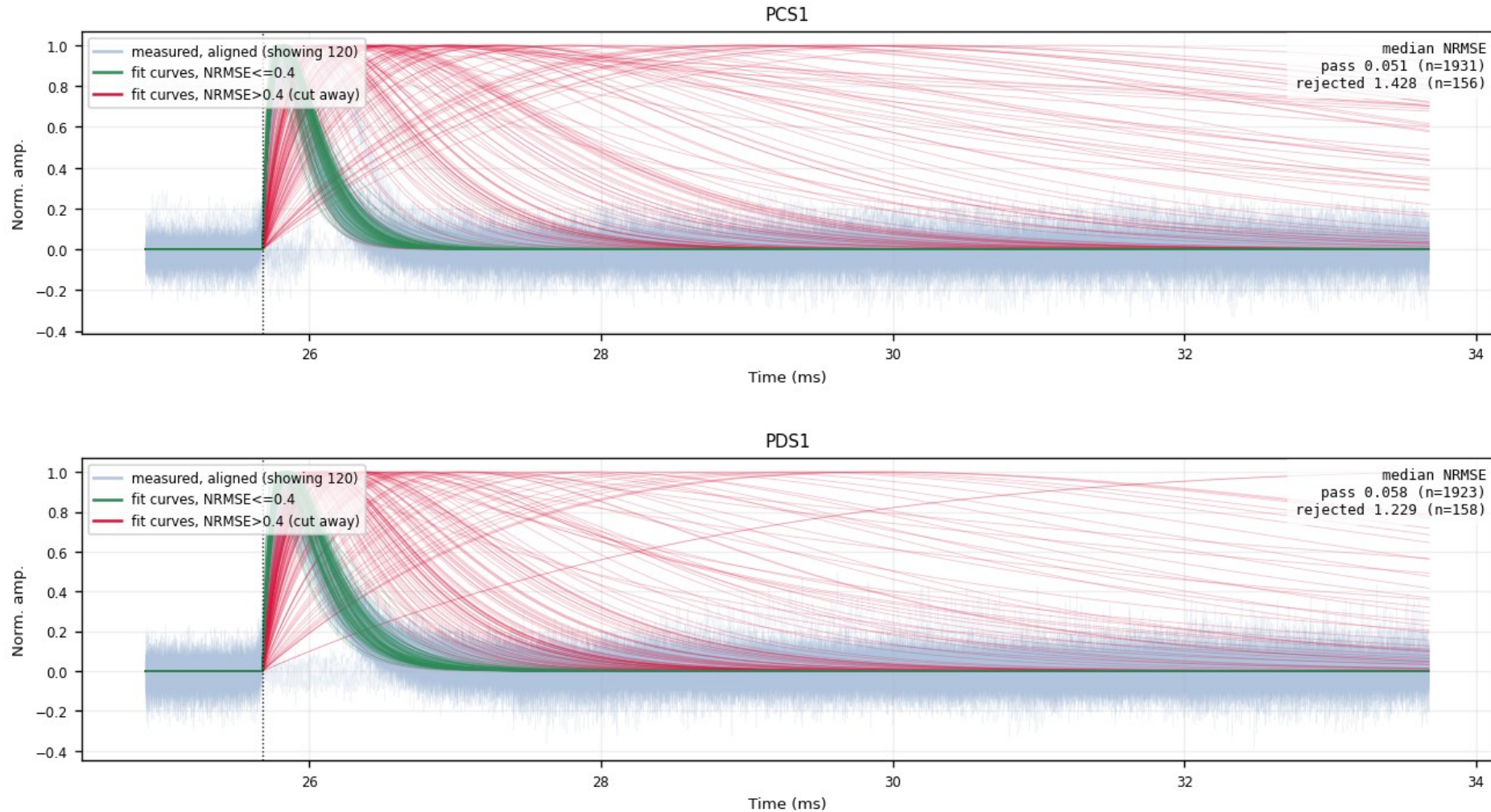
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



# Backup · Z7 · overlay\_fan\_cut: kept vs rejected fits (2/6)

figure: zip7\_overlay\_fan\_cut\_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.

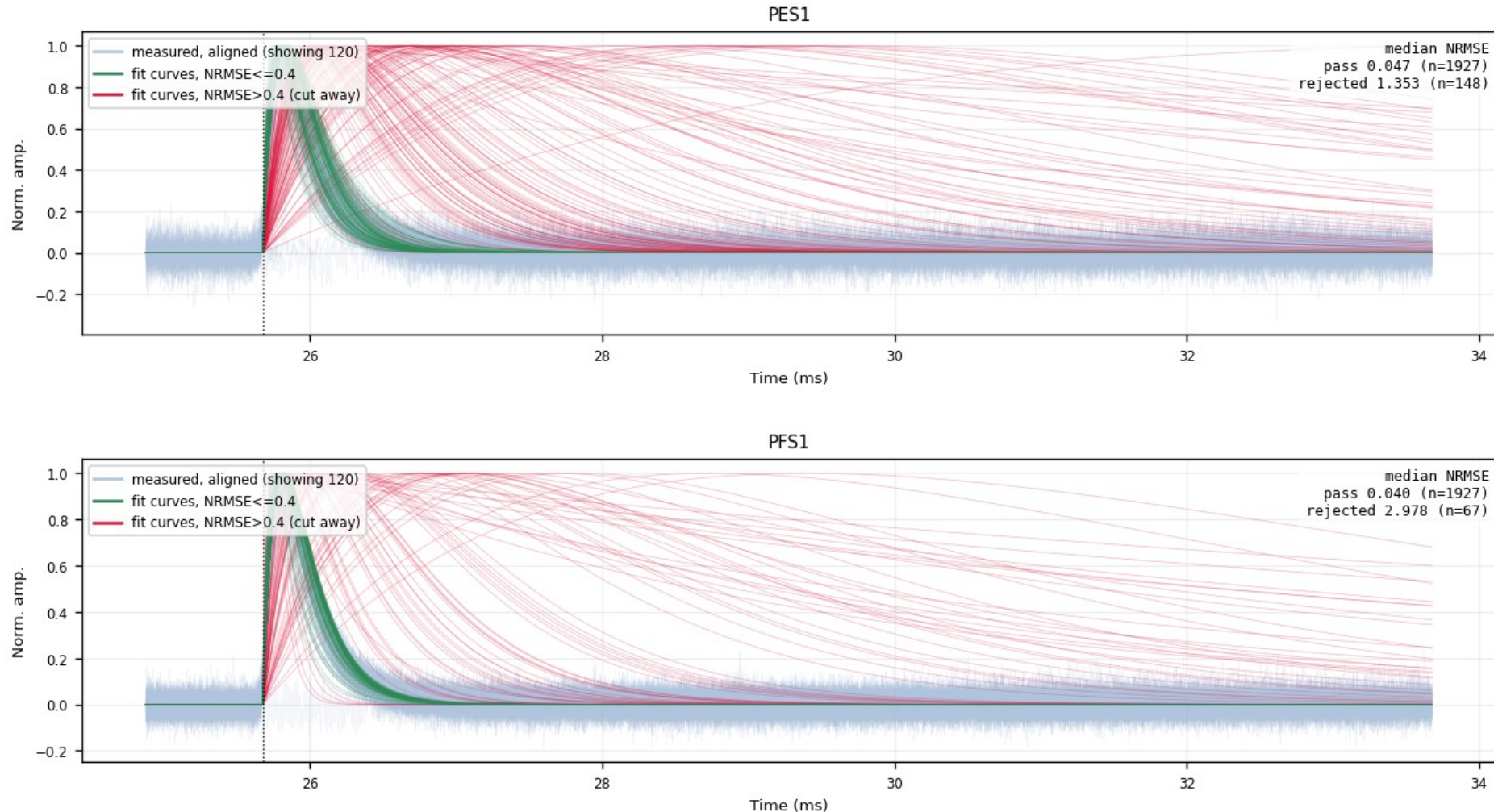




# Backup · Z7 · overlay\_fan\_cut: kept vs rejected fits (3/6)

figure: zip7\_overlay\_fan\_cut\_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

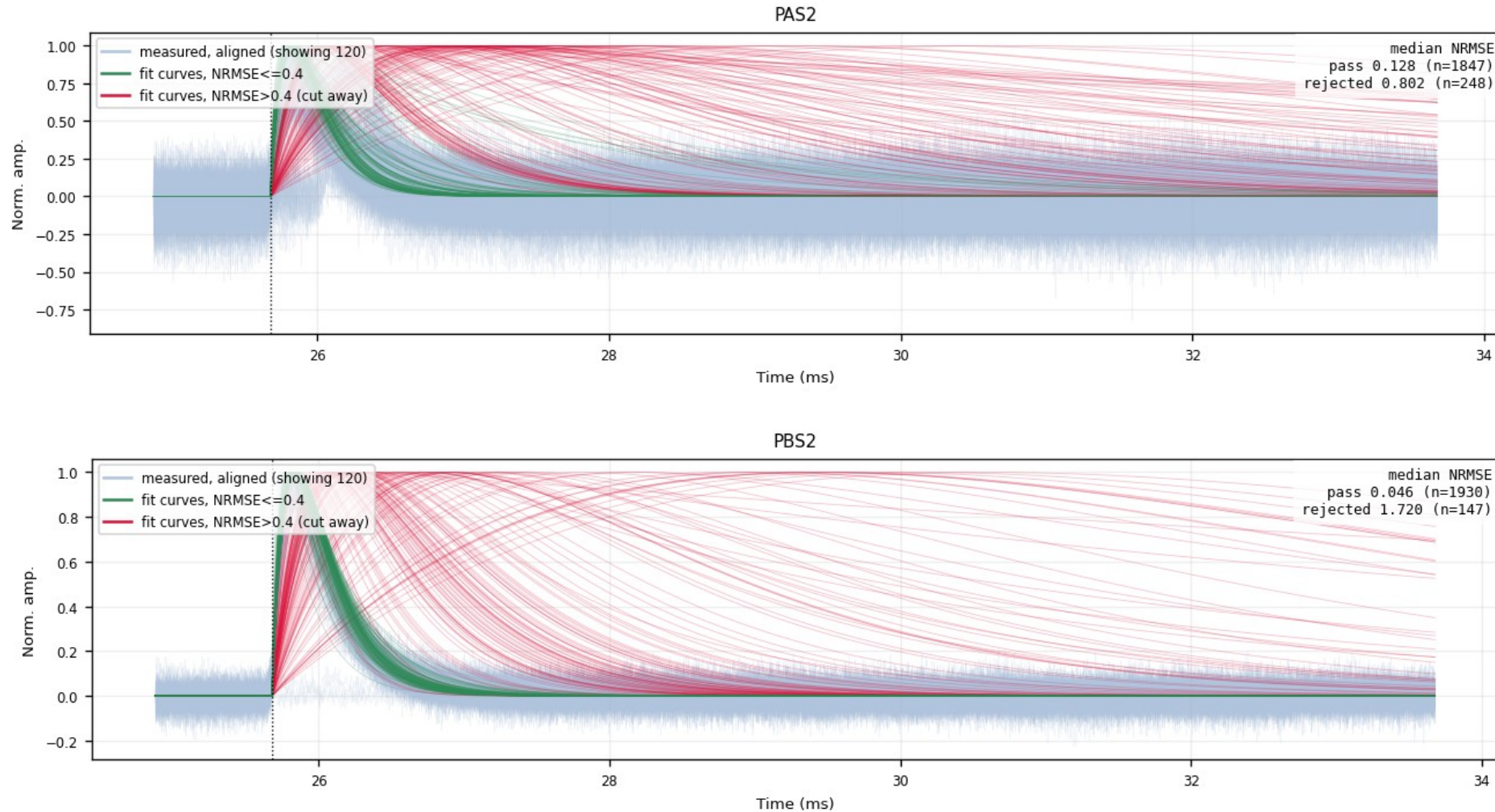
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



# Backup · Z7 · overlay\_fan\_cut: kept vs rejected fits (4/6)

figure: zip7\_overlay\_fan\_cut\_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.

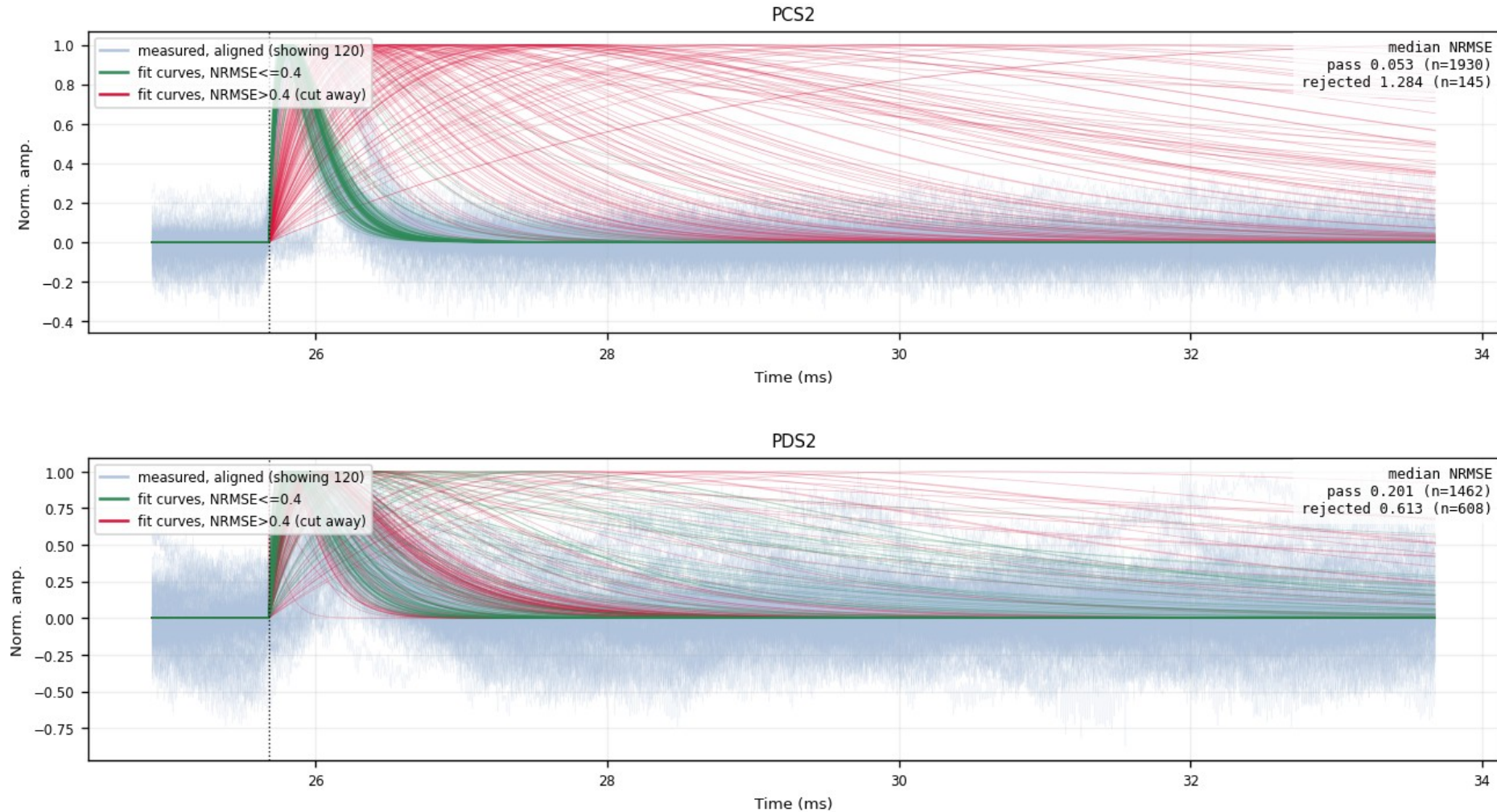




# Backup · Z7 · overlay\_fan\_cut: kept vs rejected fits (5/6)

figure: zip7\_overlay\_fan\_cut\_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

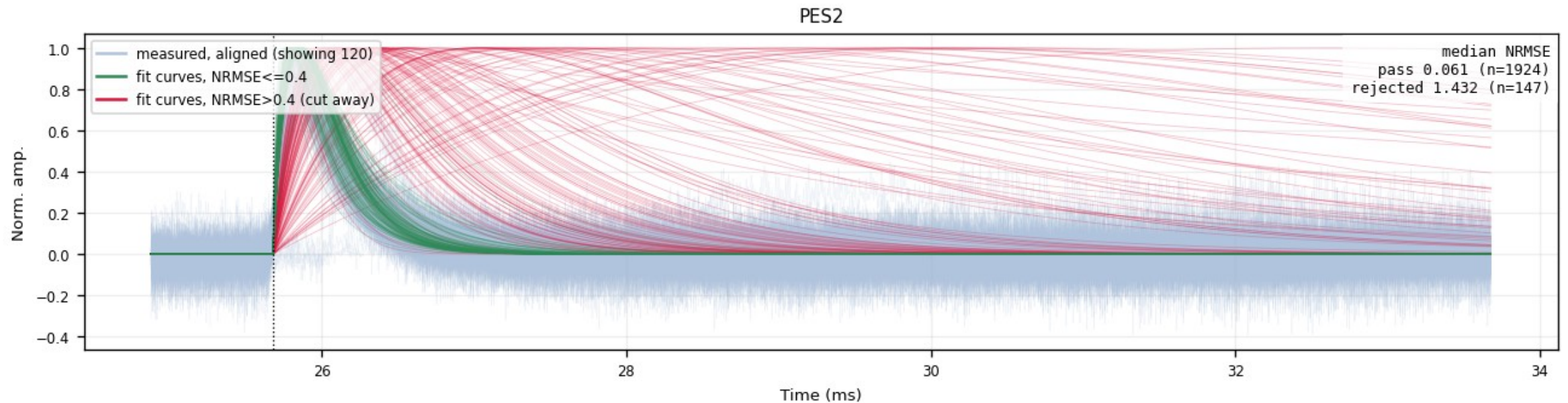
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



# Backup · Z7 · overlay\_fan\_cut: kept vs rejected fits (6/6)

figure: zip7\_overlay\_fan\_cut\_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

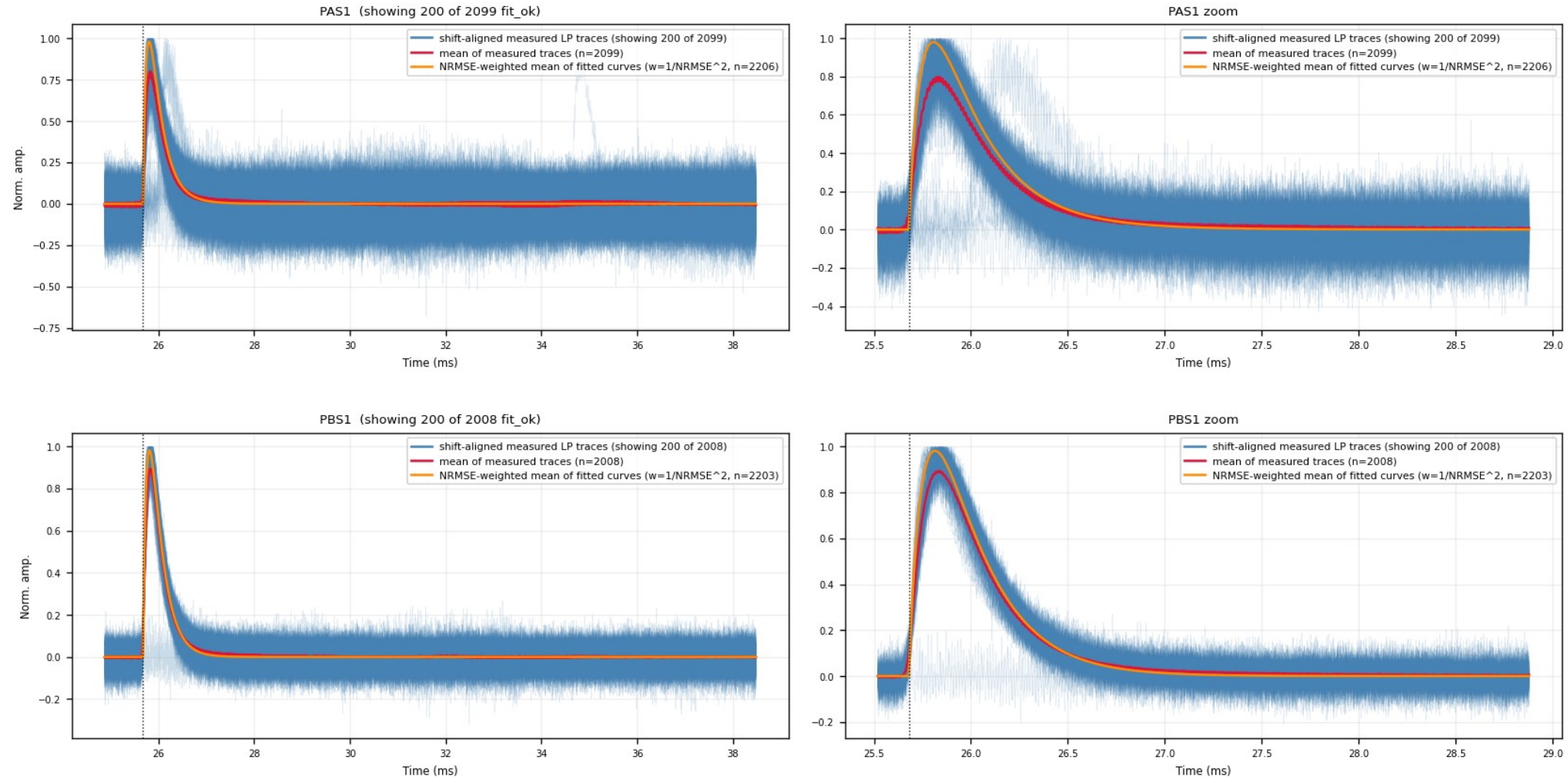
Gray-blue = aligned measured traces; green = fits with  $\text{NRMSE} \leq 0.4$  (kept); red =  $\text{NRMSE} > 0.4$  (cut away). Median NRMSE and counts of both populations stamped per channel.



# Backup · Z7 · aligned\_overlay: aligned measured traces (1/6)

figure: zip7\_lp\_aligned\_overlay.png, channel panels 1–2 of 11, top to bottom

Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.

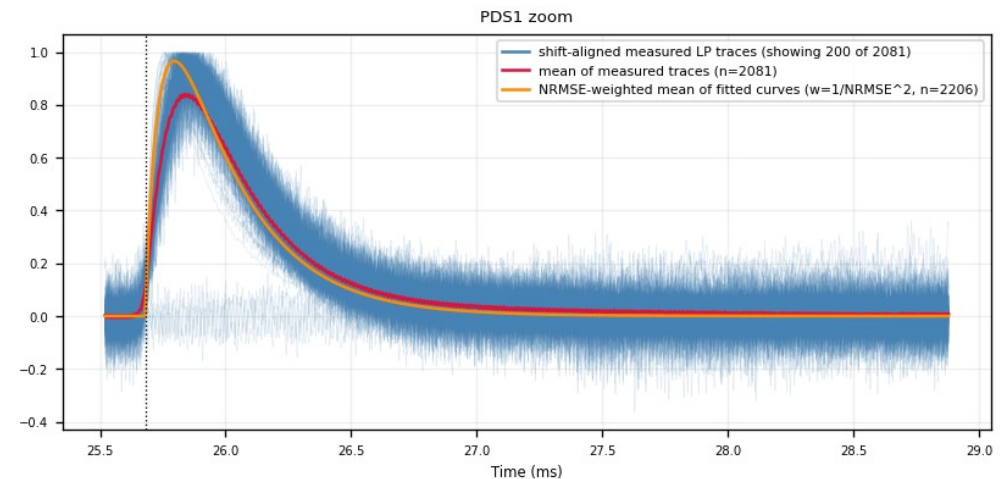
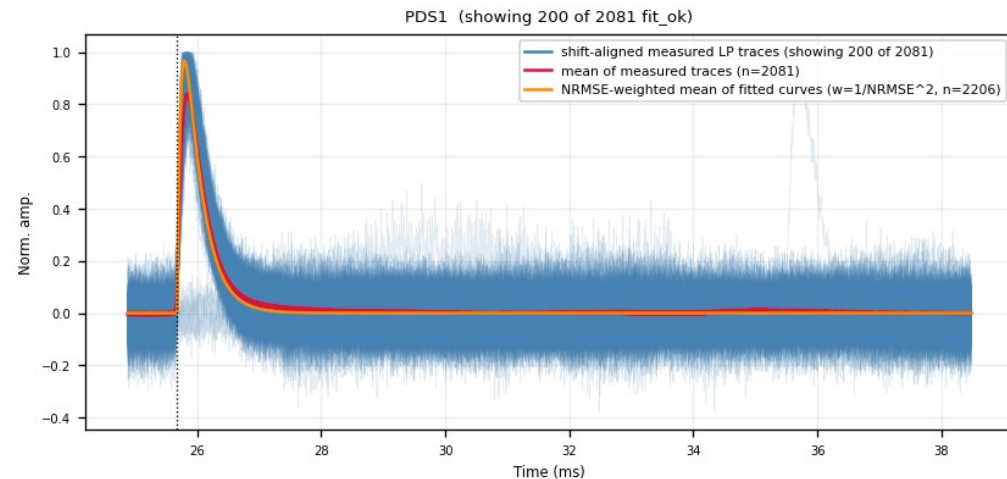
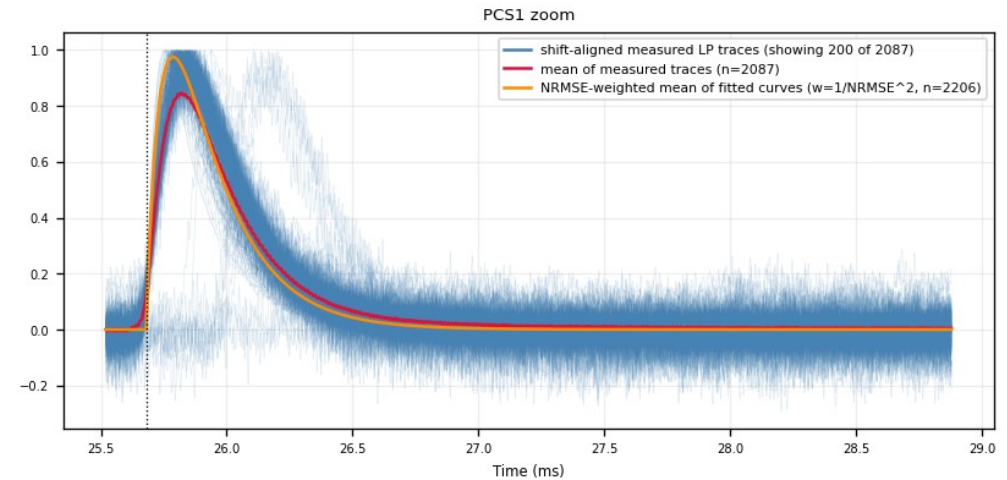
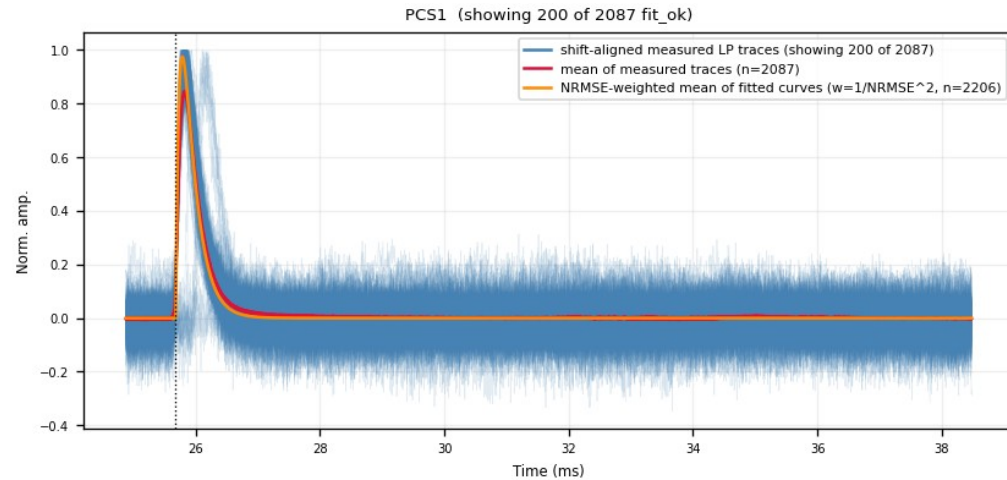




# Backup · Z7 · aligned\_overlay: aligned measured traces (2/6)

figure: zip7\_lp\_aligned\_overlay.png, channel panels 3–4 of 11, top to bottom

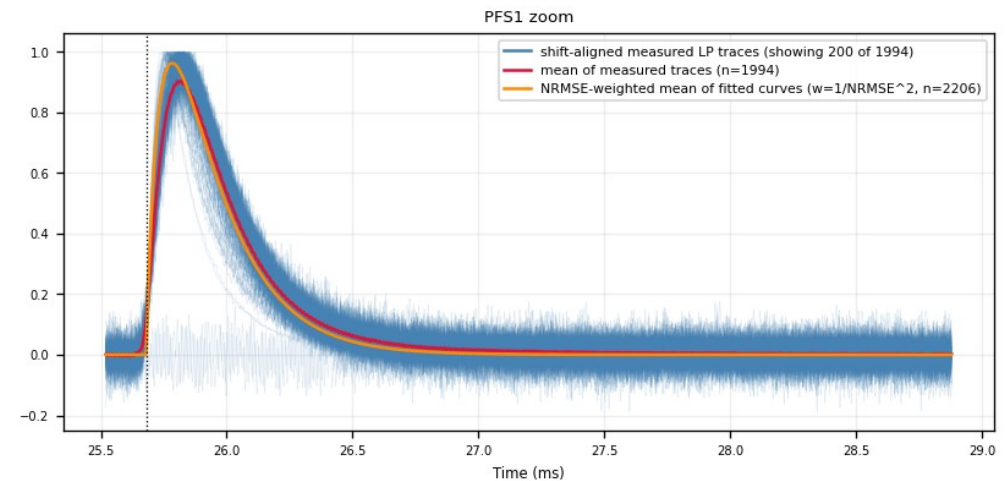
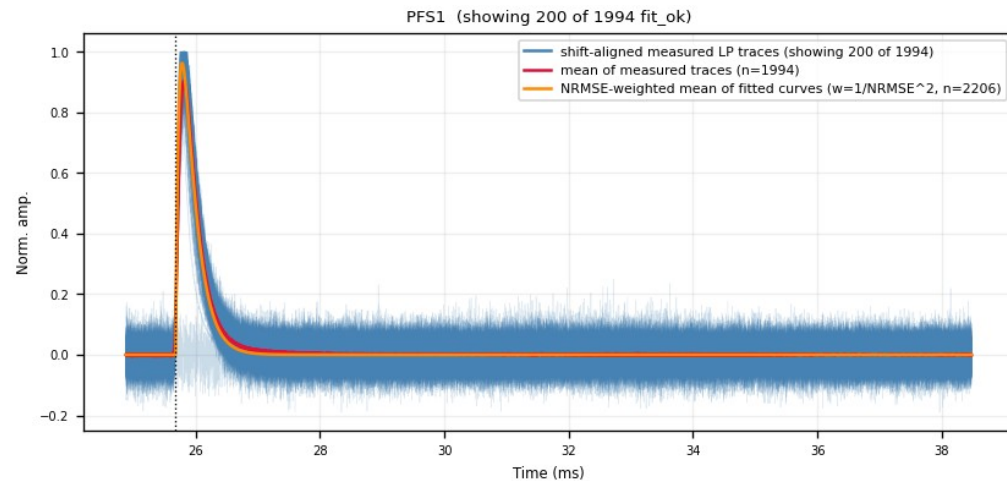
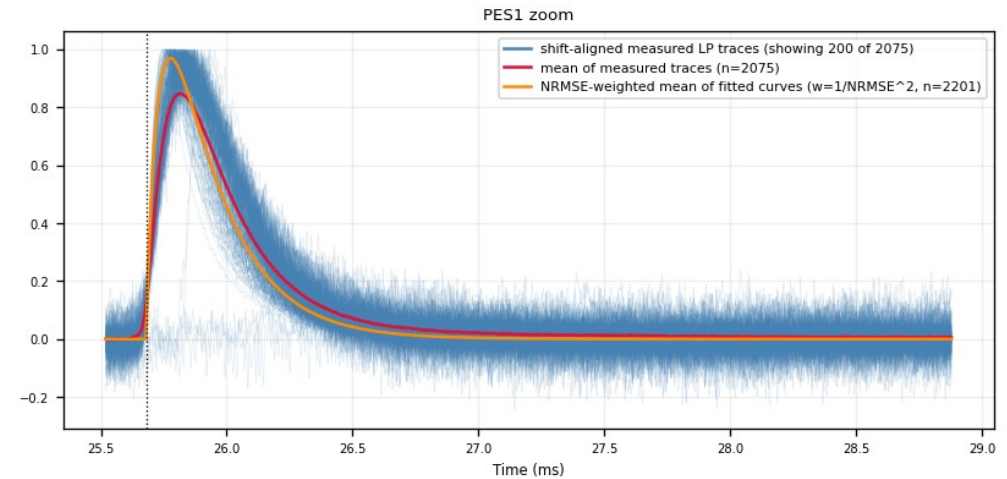
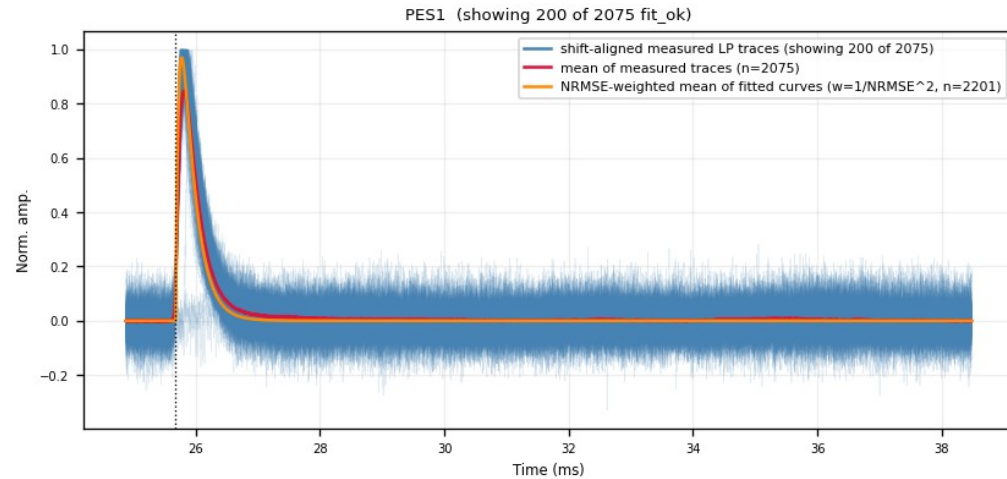
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



# Backup · Z7 · aligned\_overlay: aligned measured traces (3/6)

figure: zip7\_lp\_aligned\_overlay.png, channel panels 5–6 of 11, top to bottom

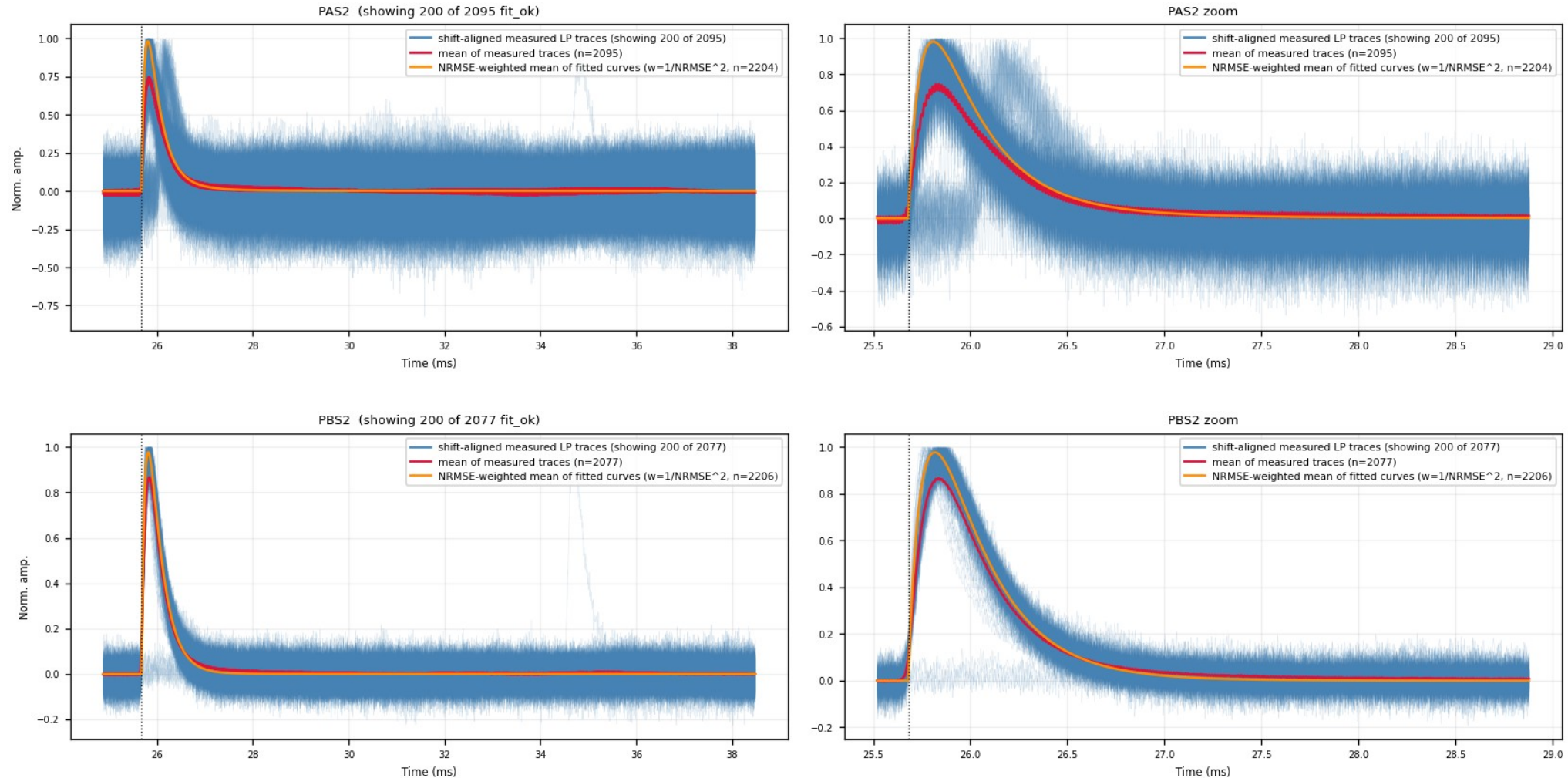
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



# Backup · Z7 · aligned\_overlay: aligned measured traces (4/6)

figure: zip7\_lp\_aligned\_overlay.png, channel panels 7–8 of 11, top to bottom

Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.

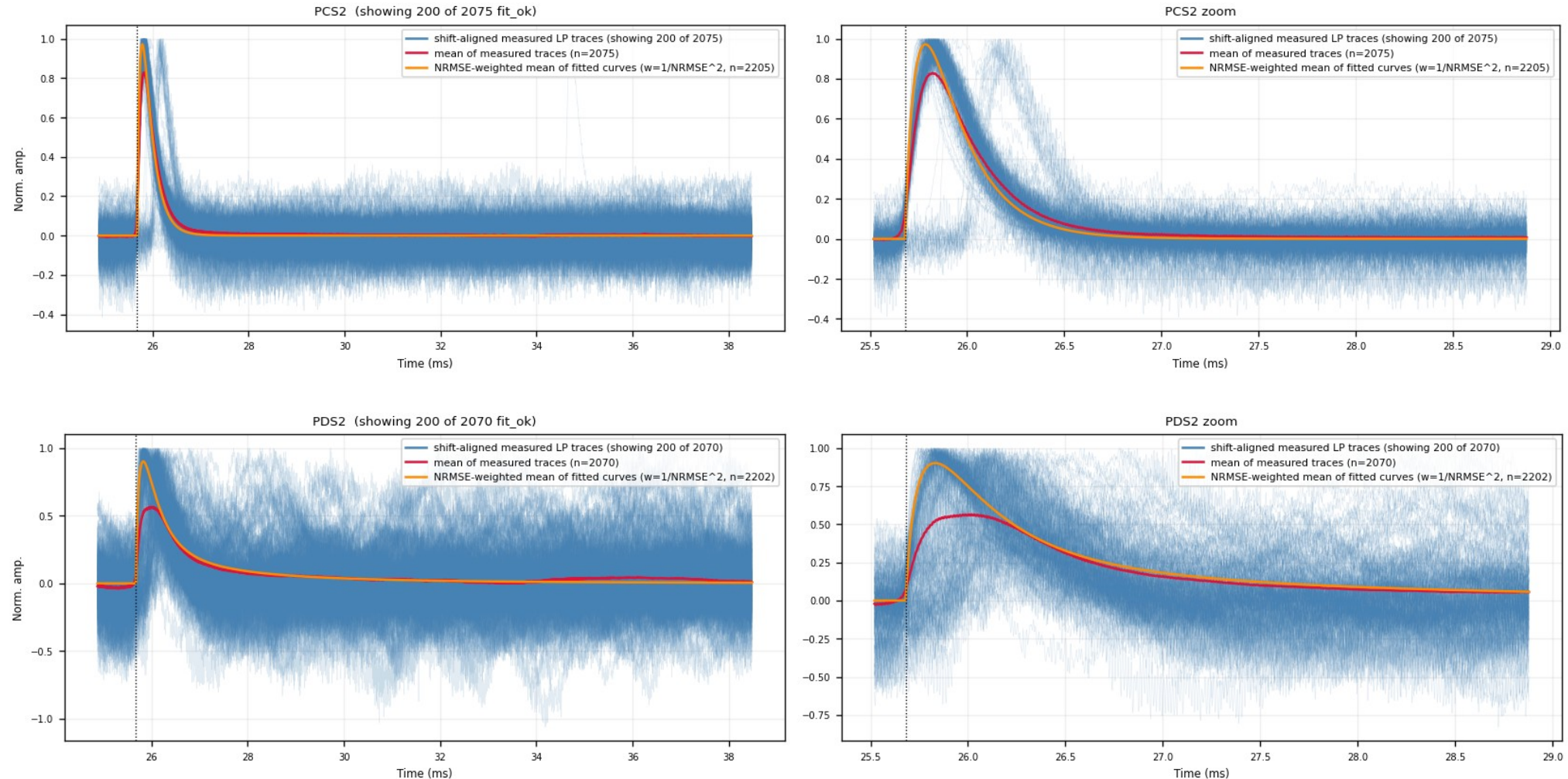




# Backup · Z7 · aligned\_overlay: aligned measured traces (5/6)

figure: zip7\_lp\_aligned\_overlay.png, channel panels 9–10 of 11, top to bottom

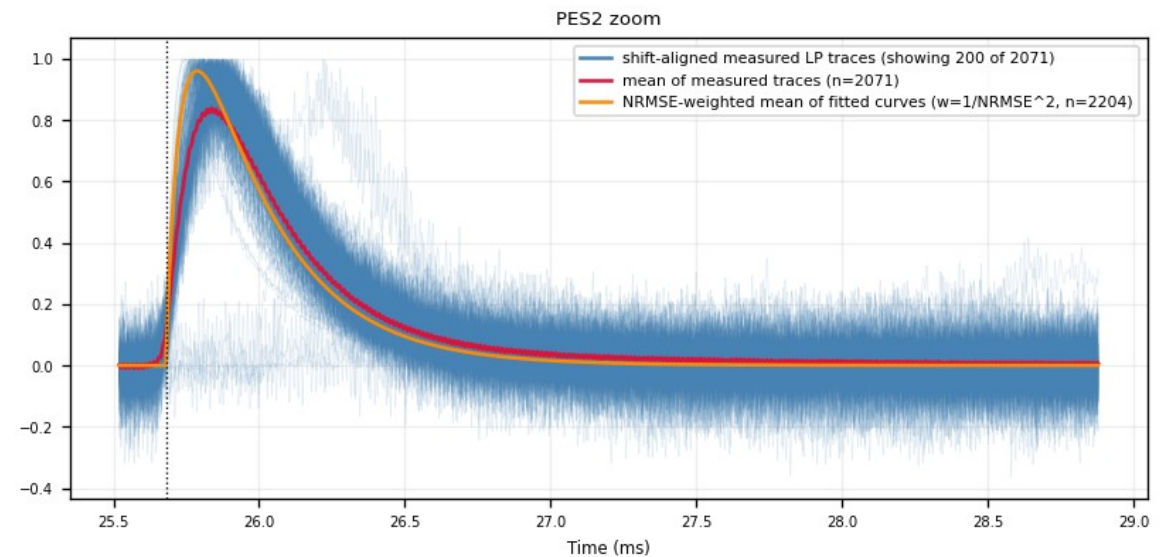
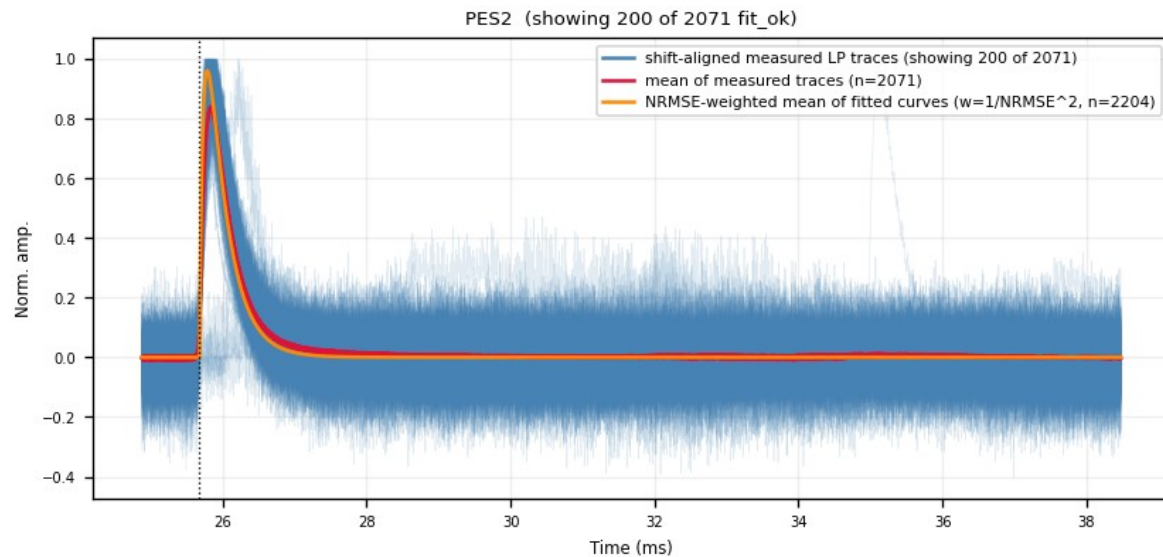
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



# Backup · Z7 · aligned\_overlay: aligned measured traces (6/6)

figure: zip7\_lp\_aligned\_overlay.png, channel panels 11–11 of 11, top to bottom

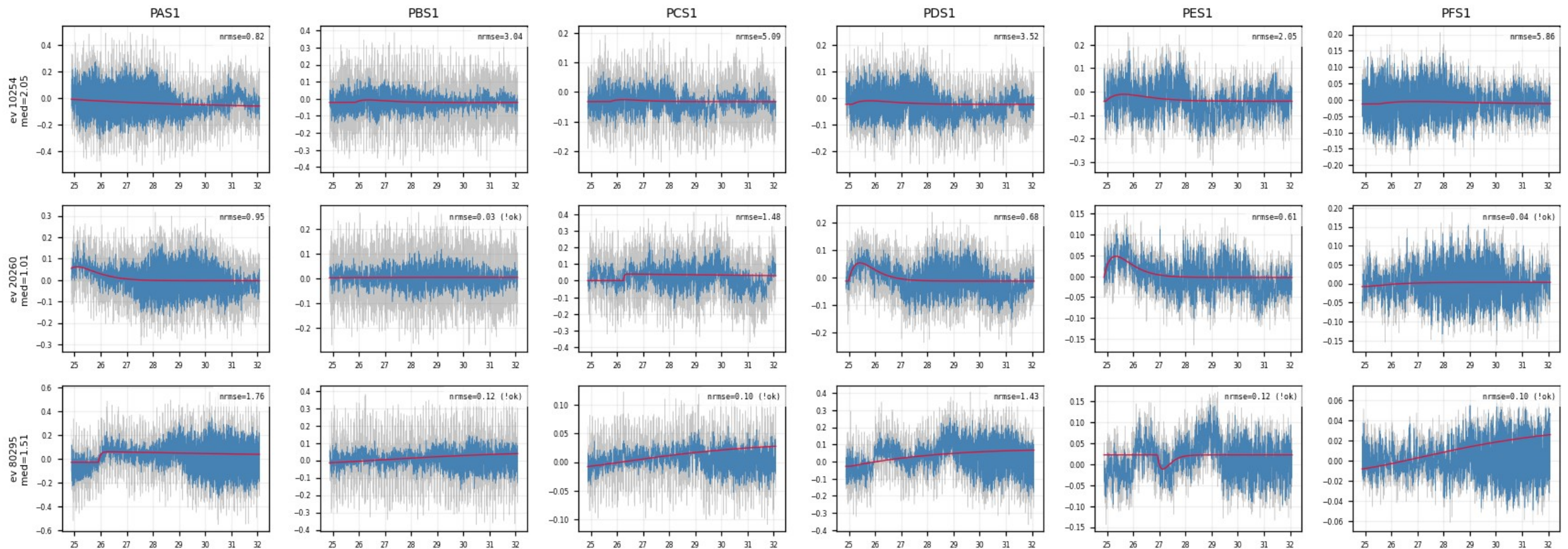
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit\_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (1/10)

figure: zip7\_slow\_rise\_events.png, event rows 1–3 of 15, left half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.

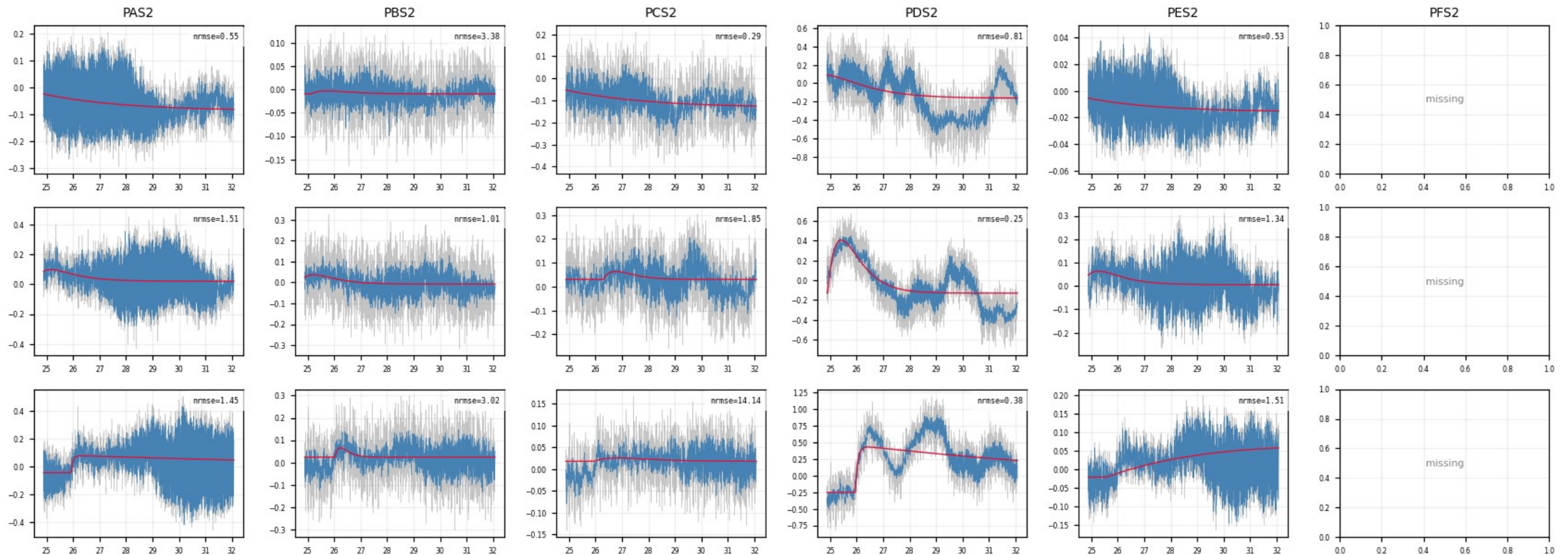




# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (2/10)

figure: zip7\_slow\_rise\_events.png, event rows 1–3 of 15, right half of the channels

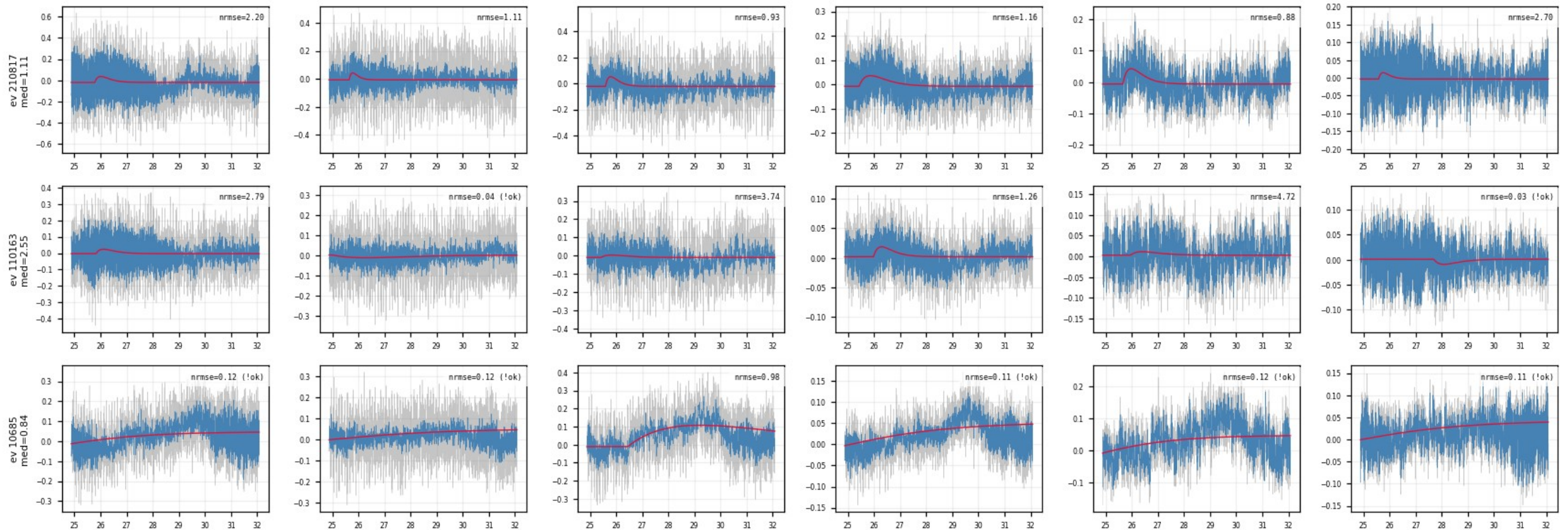
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (3/10)

figure: zip7\_slow\_rise\_events.png, event rows 4–6 of 15, left half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.

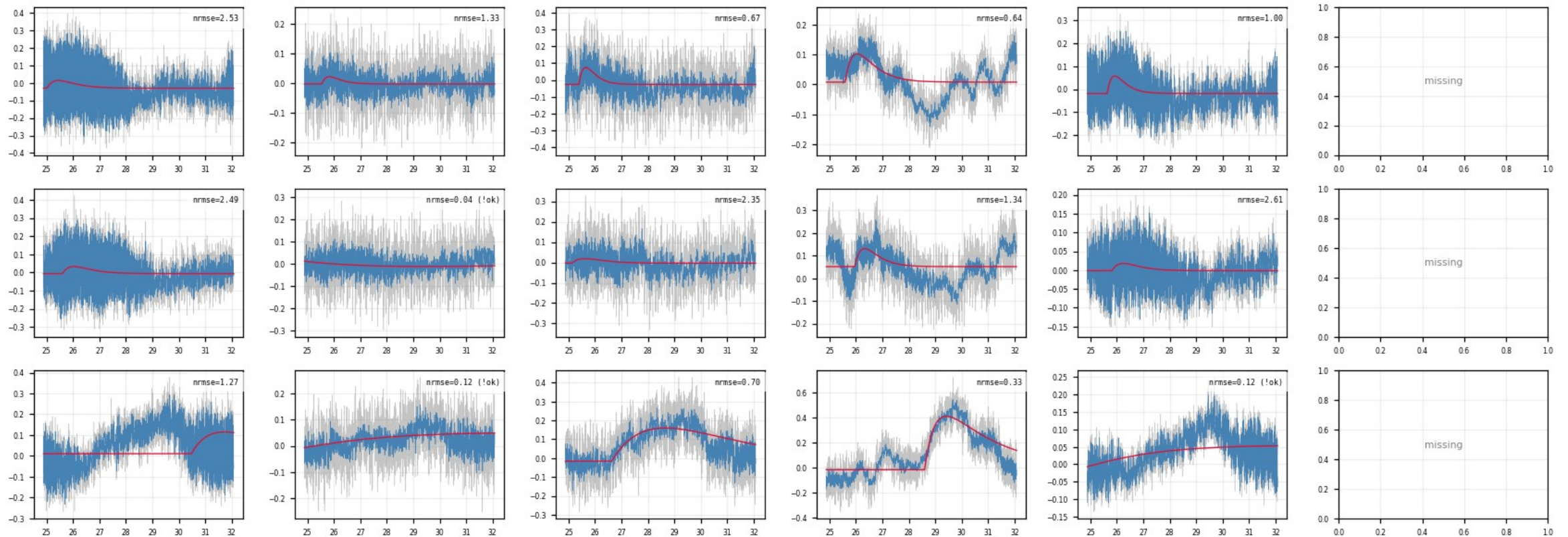




# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (4/10)

figure: zip7\_slow\_rise\_events.png, event rows 4–6 of 15, right half of the channels

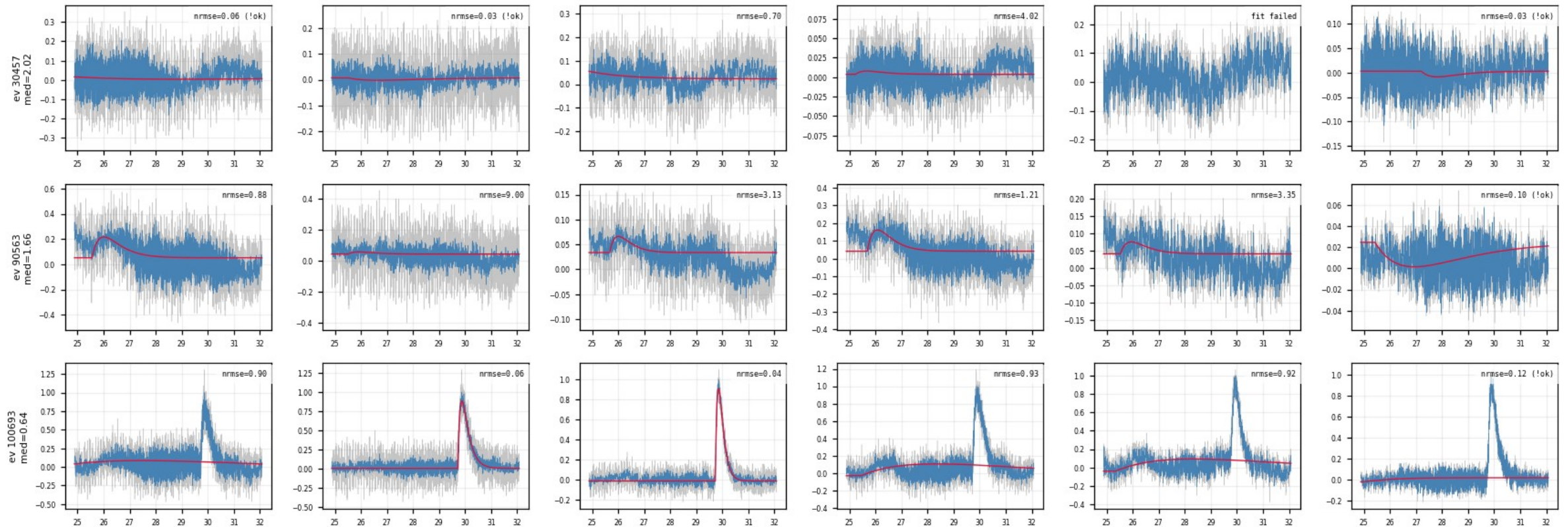
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (5/10)

figure: zip7\_slow\_rise\_events.png, event rows 7–9 of 15, left half of the channels

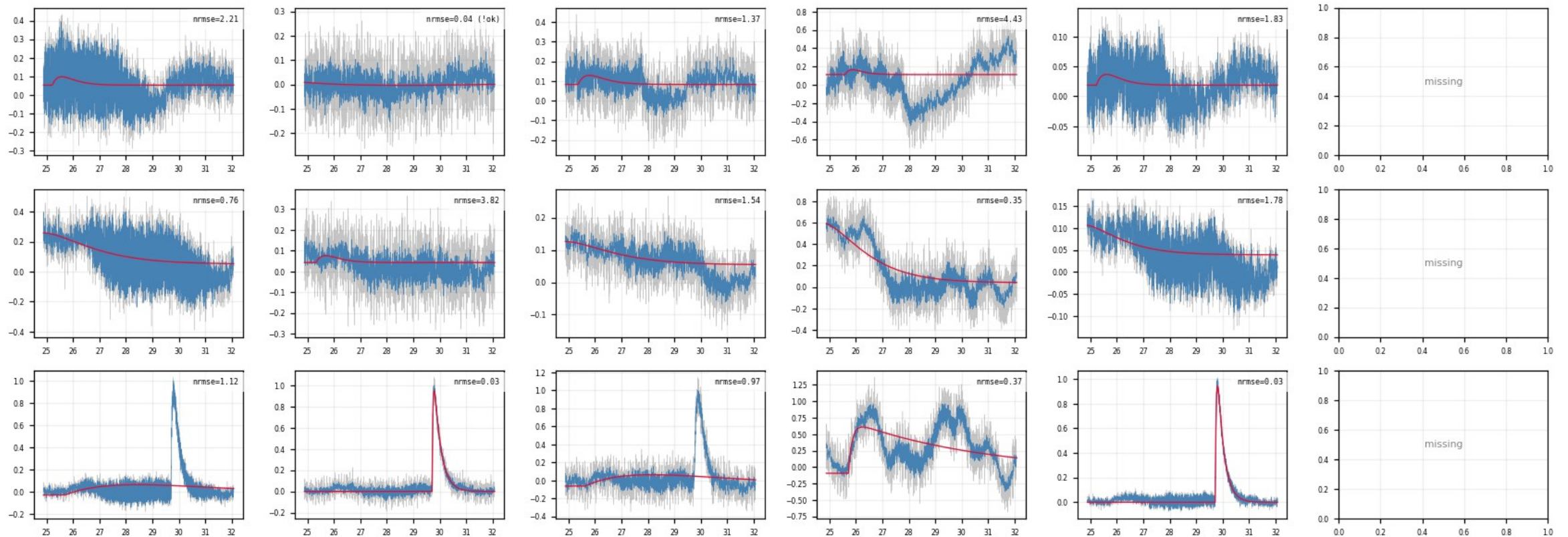
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (6/10)

figure: zip7\_slow\_rise\_events.png, event rows 7–9 of 15, right half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.

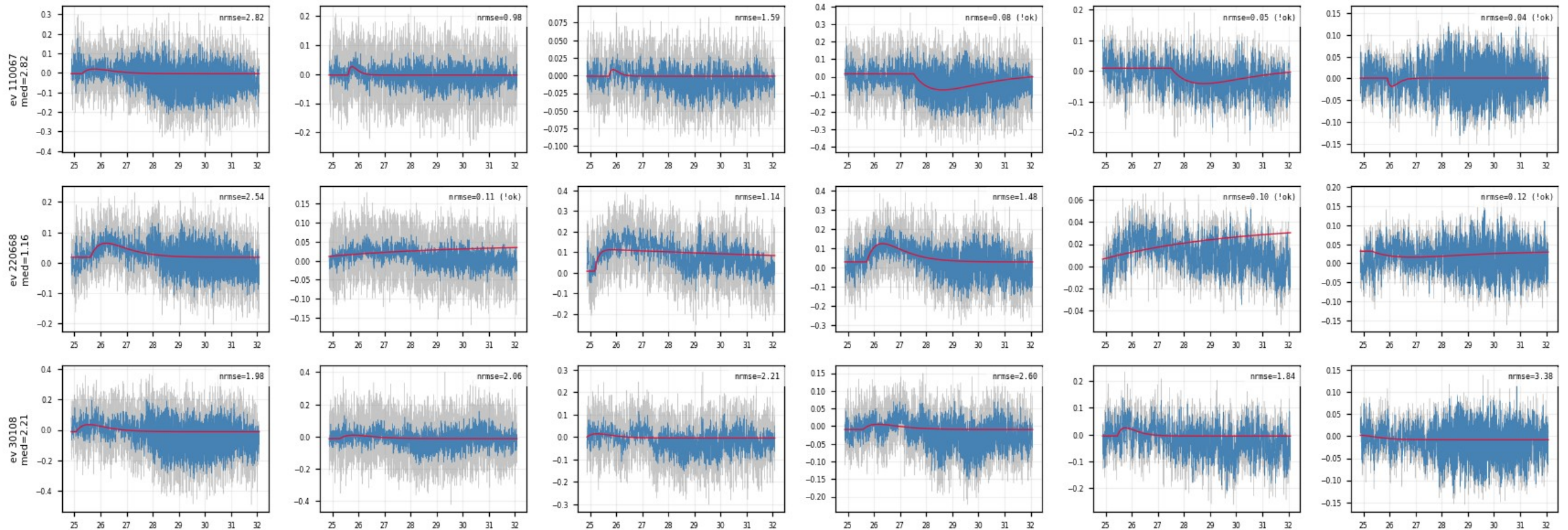




# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (7/10)

figure: zip7\_slow\_rise\_events.png, event rows 10–12 of 15, left half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.

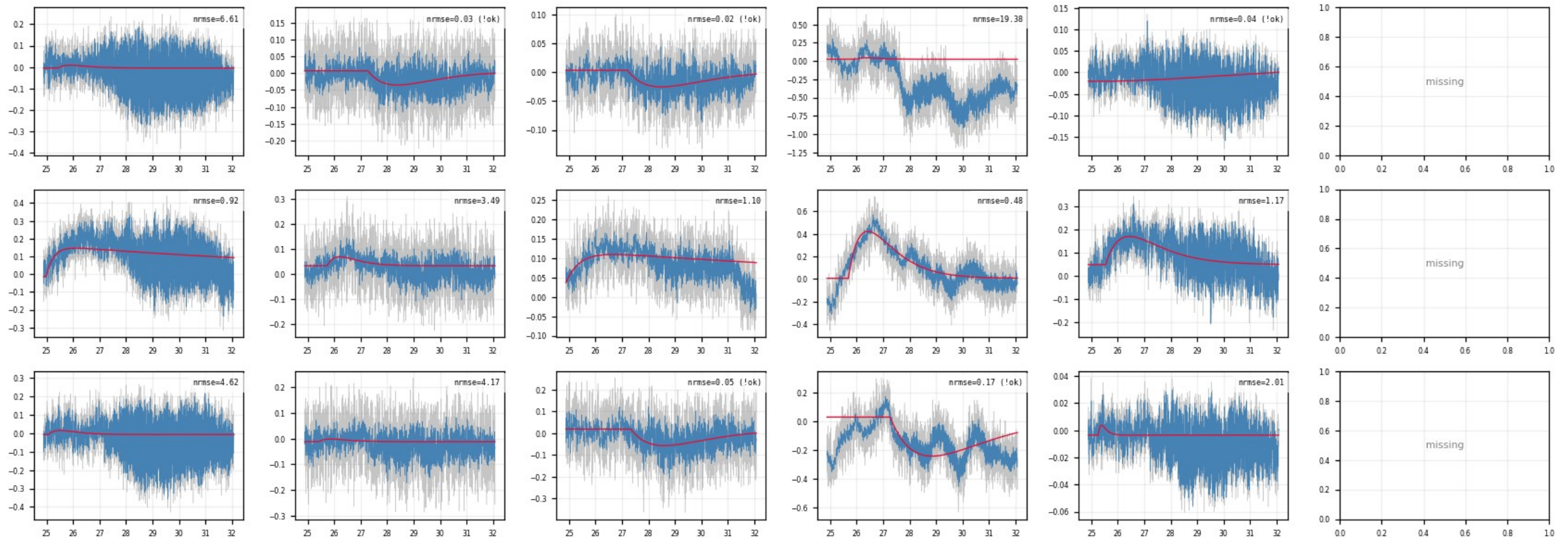




# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (8/10)

figure: zip7\_slow\_rise\_events.png, event rows 10–12 of 15, right half of the channels

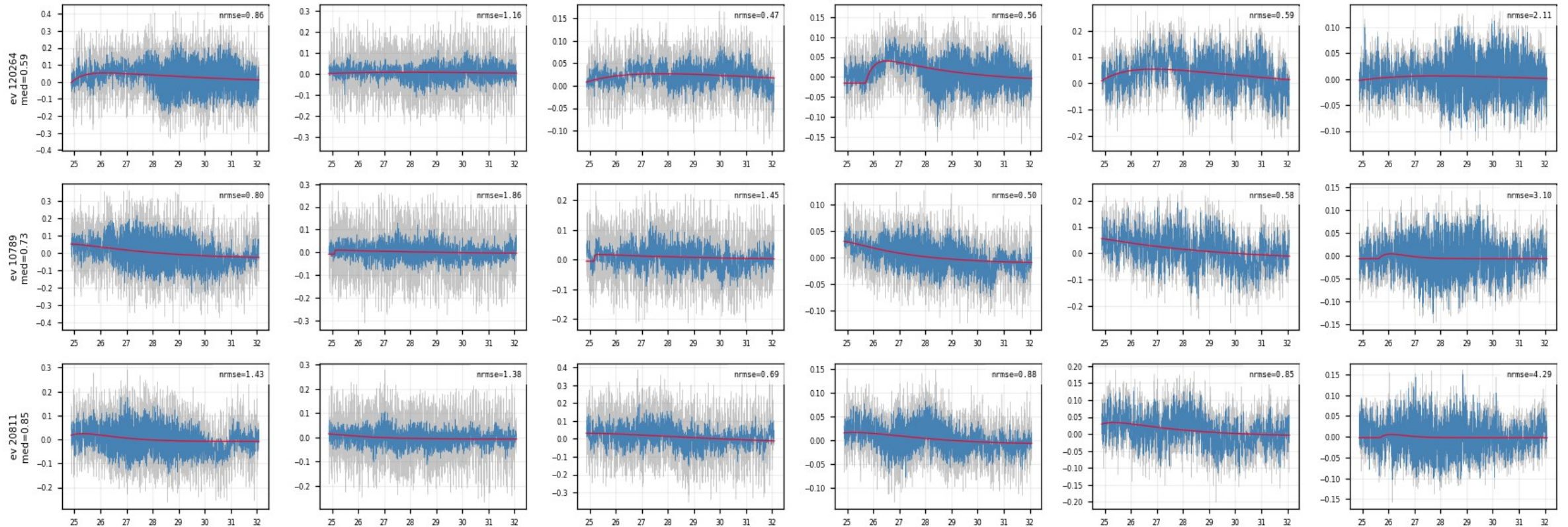
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (9/10)

figure: zip7\_slow\_rise\_events.png, event rows 13–15 of 15, left half of the channels

Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.

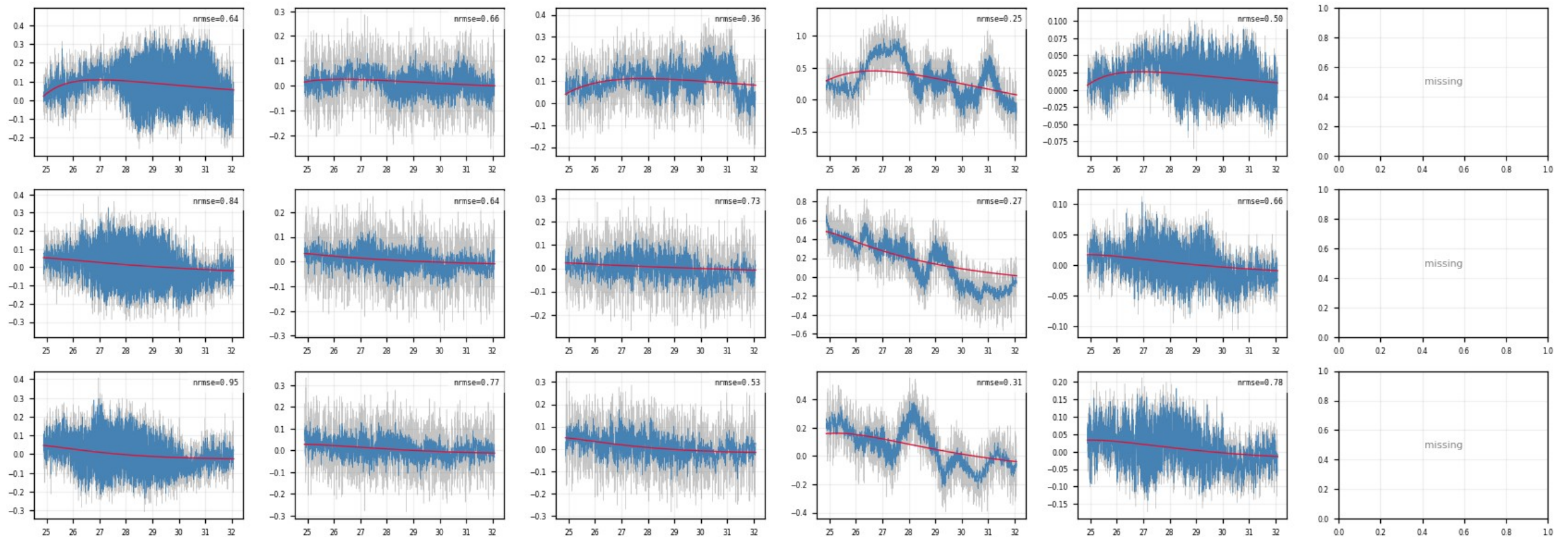




# Backup · Z7 · slow\_rise\_events: NRMSE-rejected events (10/10)

figure: zip7\_slow\_rise\_events.png, event rows 13–15 of 15, right half of the channels

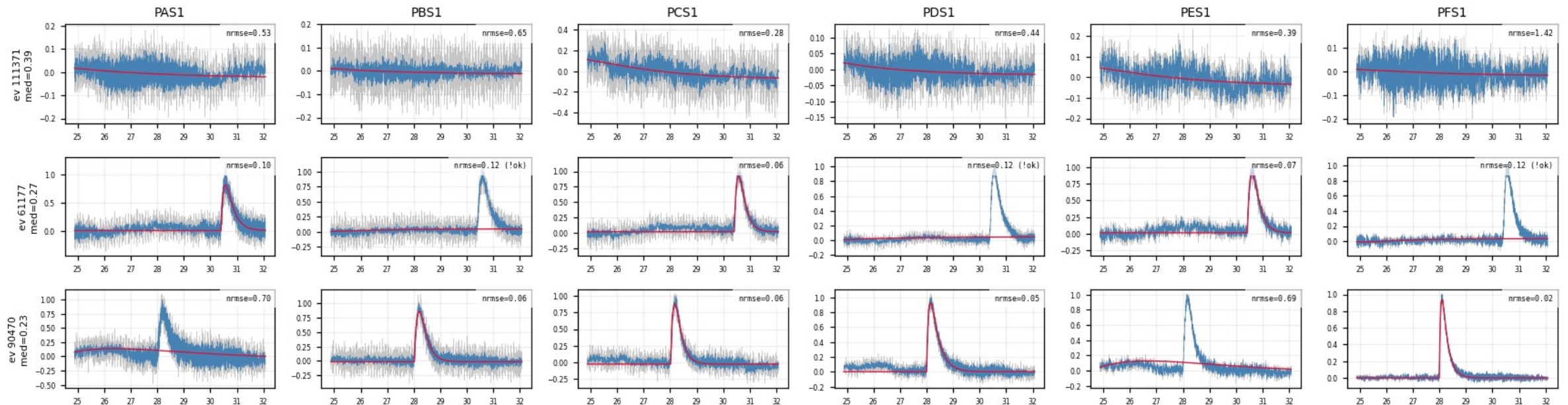
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



# Backup · Z7 · shadow\_events: well-fit slow-rise events (1/2)

figure: zip7\_shadow\_events.png, event rows 1–3 of 3, left half of the channels

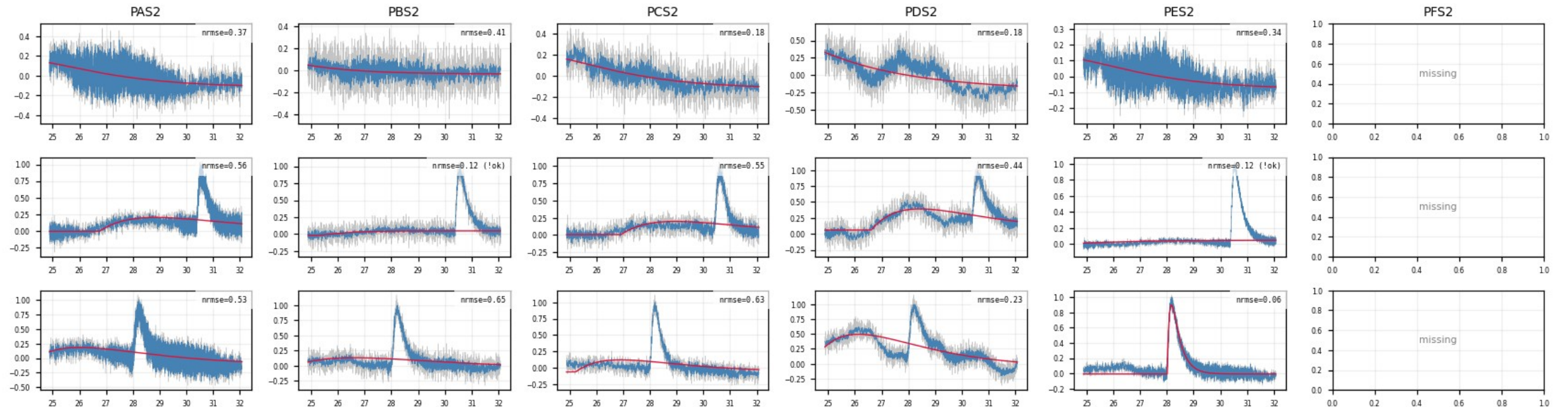
Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses; kept, and exactly what the NxM multi-templates are for.



# Backup · Z7 · shadow\_events: well-fit slow-rise events (2/2)

figure: zip7\_shadow\_events.png, event rows 1–3 of 3, right half of the channels

Median NRMSE  $\leq 0.4$  AND median  $\tau_{\text{rise}} > 0.2$  ms: genuinely slow, well-fit pulses; kept, and exactly what the NxM multi-templates are for.

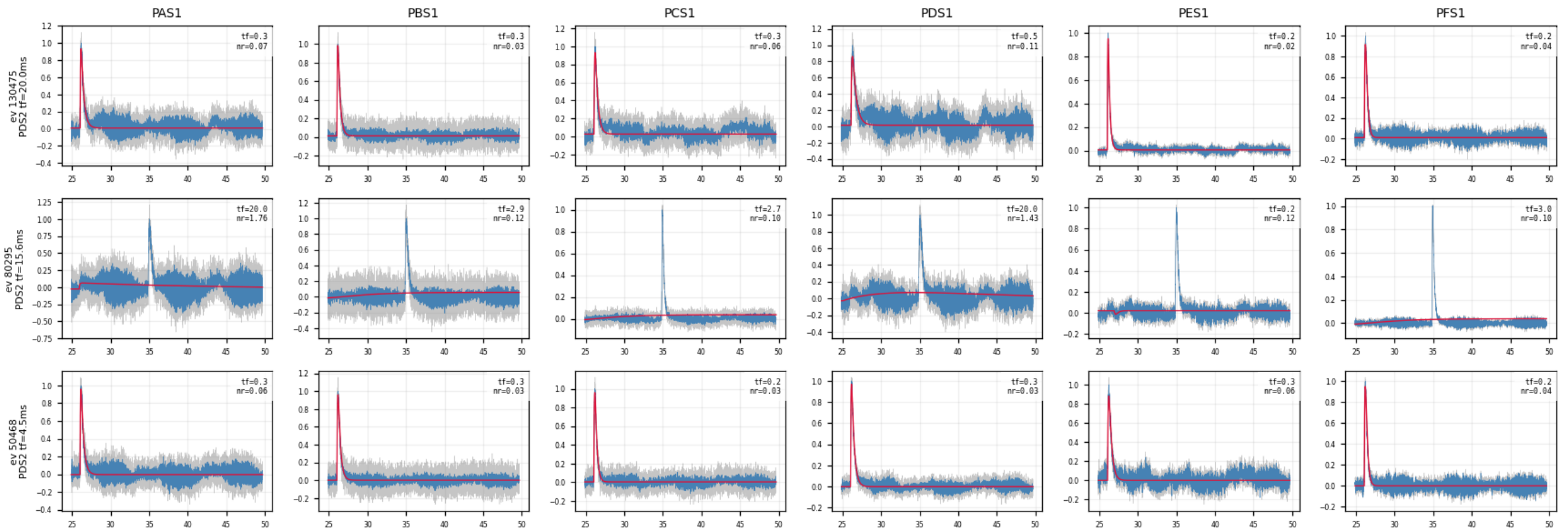




# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (1/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 1–3 of 10, left half of the channels

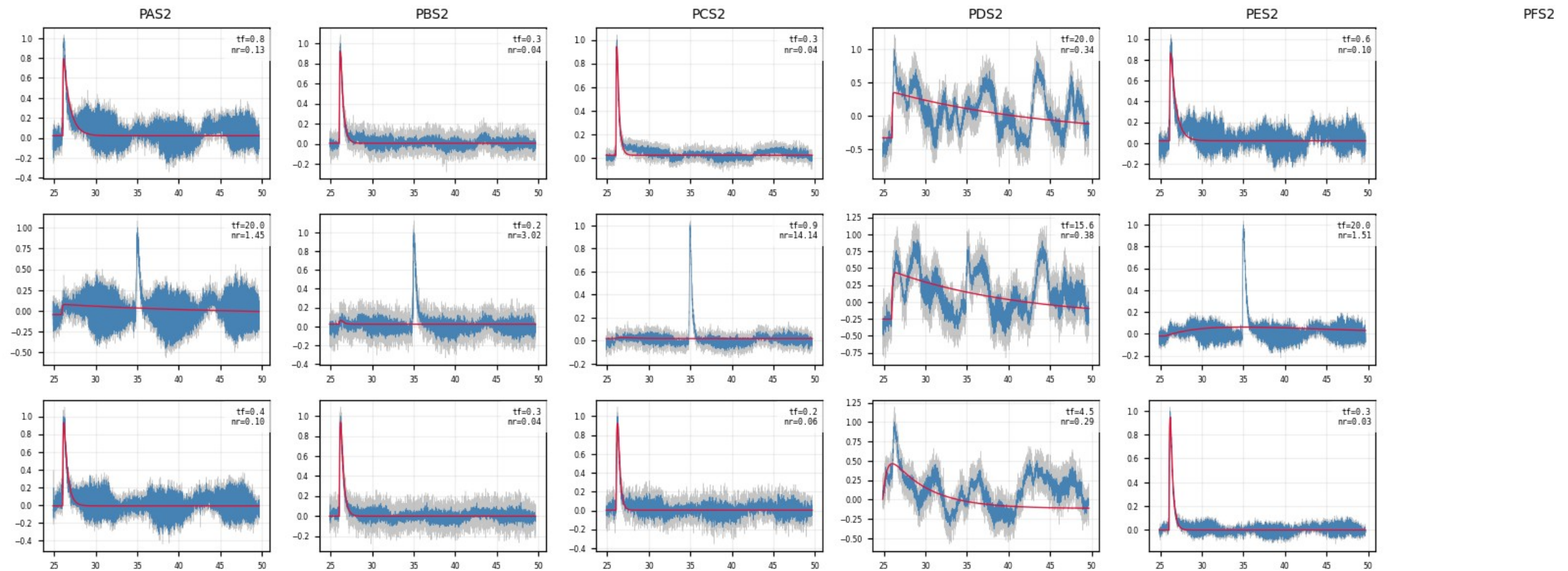
10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (2/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 1–3 of 10, right half of the channels

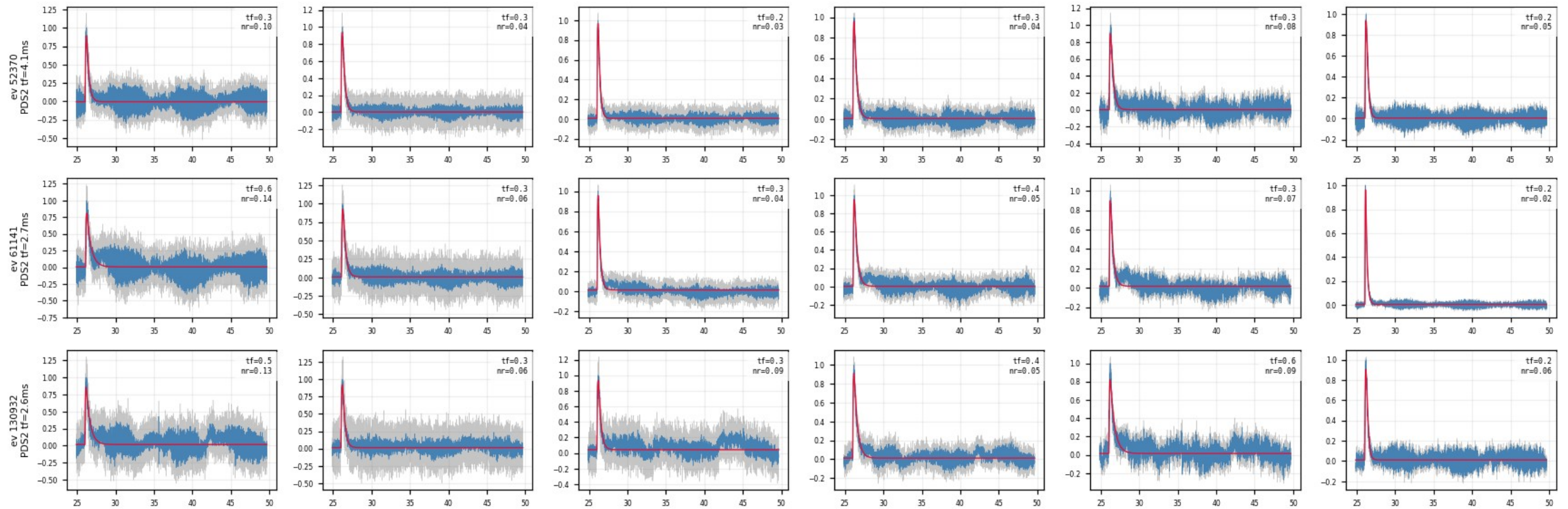
10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (3/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 4–6 of 10, left half of the channels

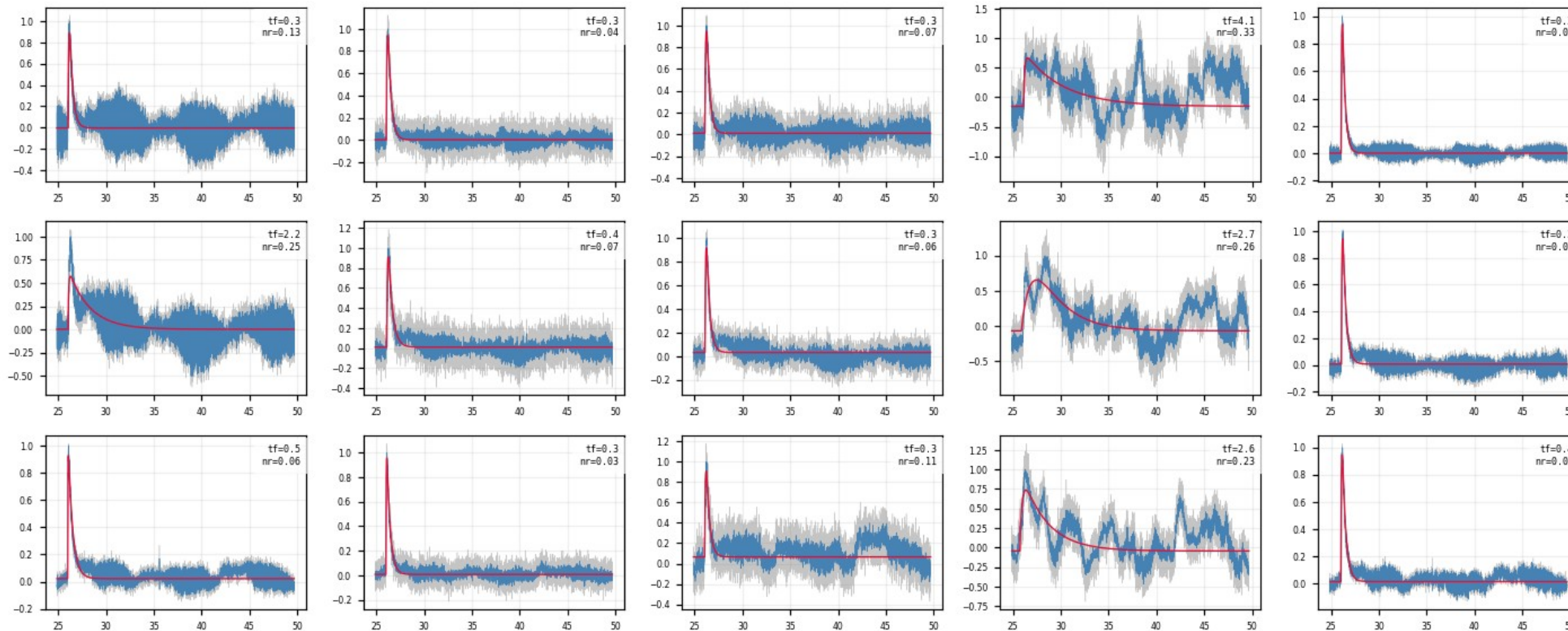
10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (4/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 4–6 of 10, right half of the channels

10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.

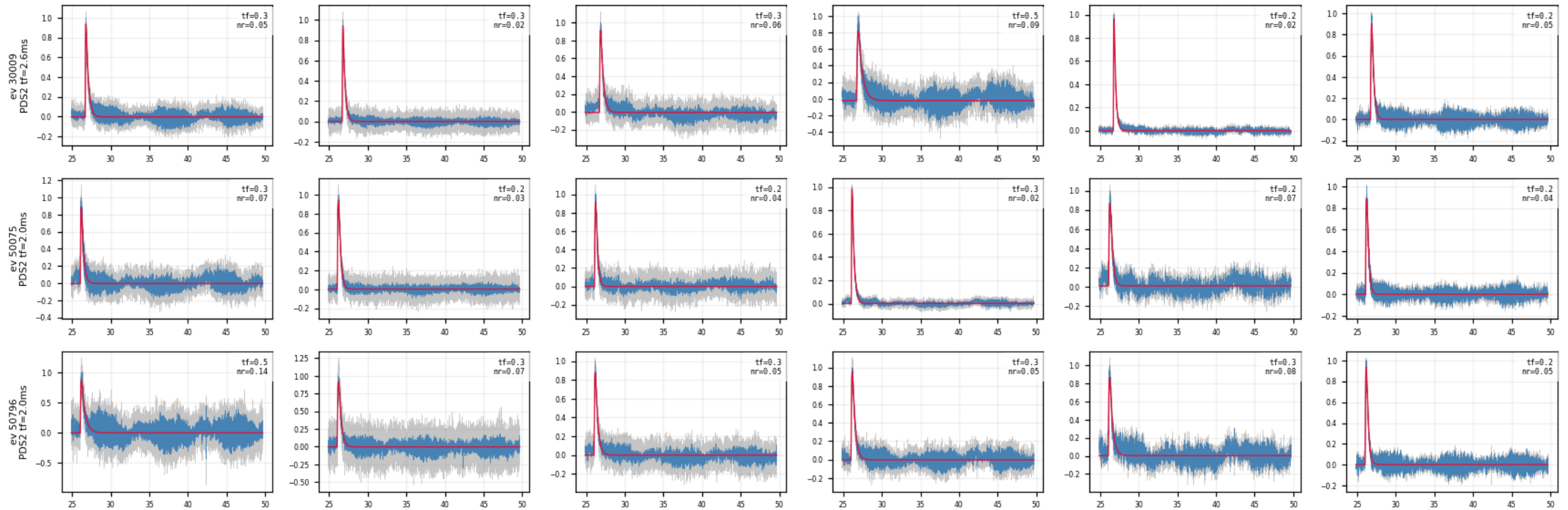




# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (5/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 7–9 of 10, left half of the channels

10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.

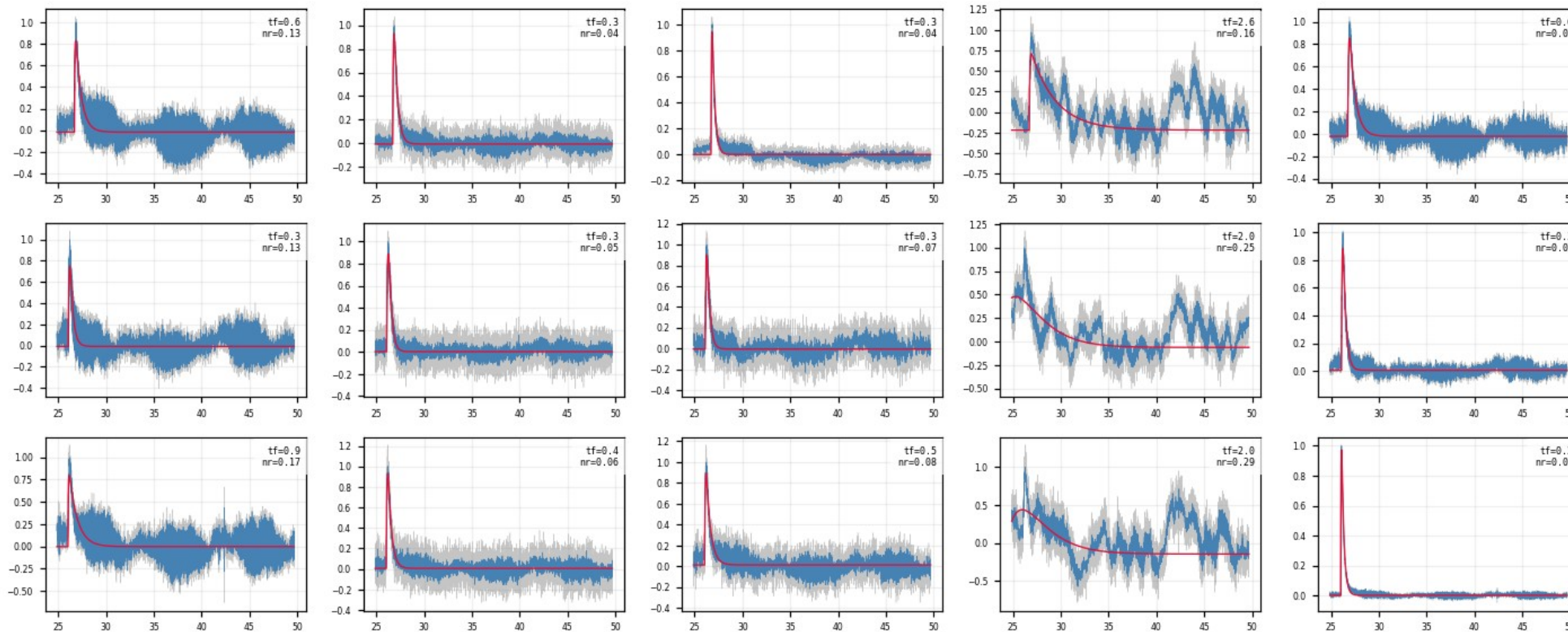




# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (6/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 7–9 of 10, right half of the channels

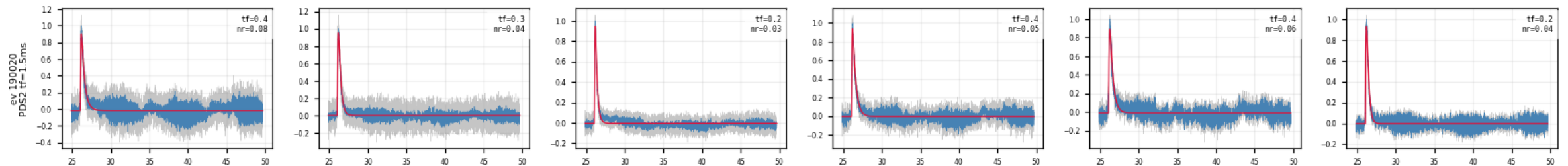
10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (7/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 10–10 of 10, left half of the channels

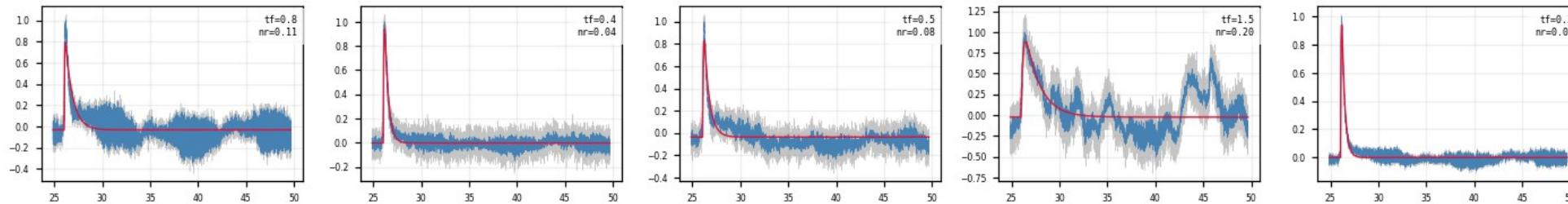
10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



# Backup · Z7 · slow\_fall\_events: long- $\tau$ \_fall study (PDS2) (8/8)

figure: zip7\_PDS2\_slow\_fall.png, event rows 10–10 of 10, right half of the channels

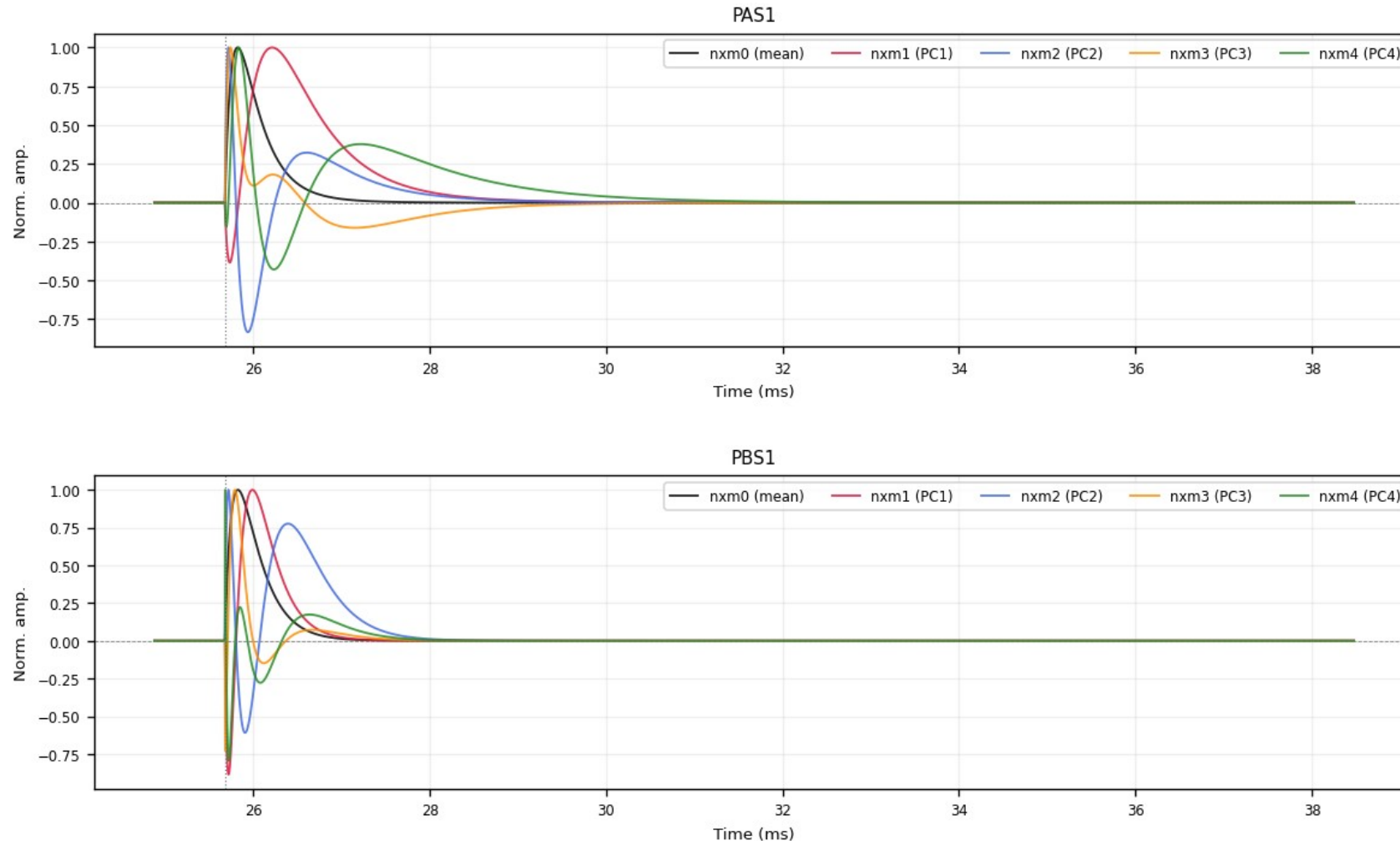
10 random events with  $\tau_{\text{fall}} > 1.5$  ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



# Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (1/6)

figure: zip7\_pca\_templates.png, channel panels 1–2 of 11, top to bottom

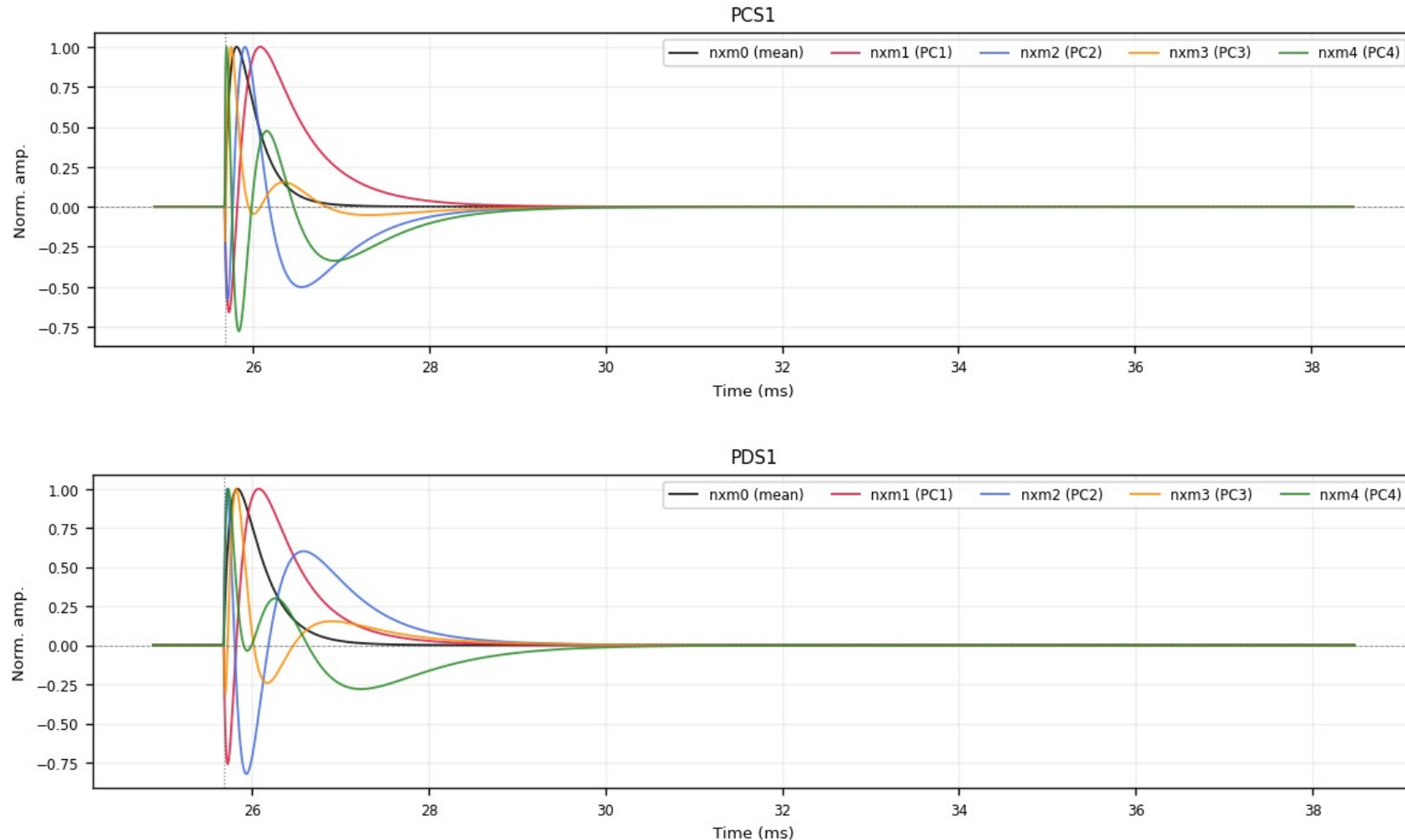
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



# Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (2/6)

figure: zip7\_pca\_templates.png, channel panels 3–4 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.

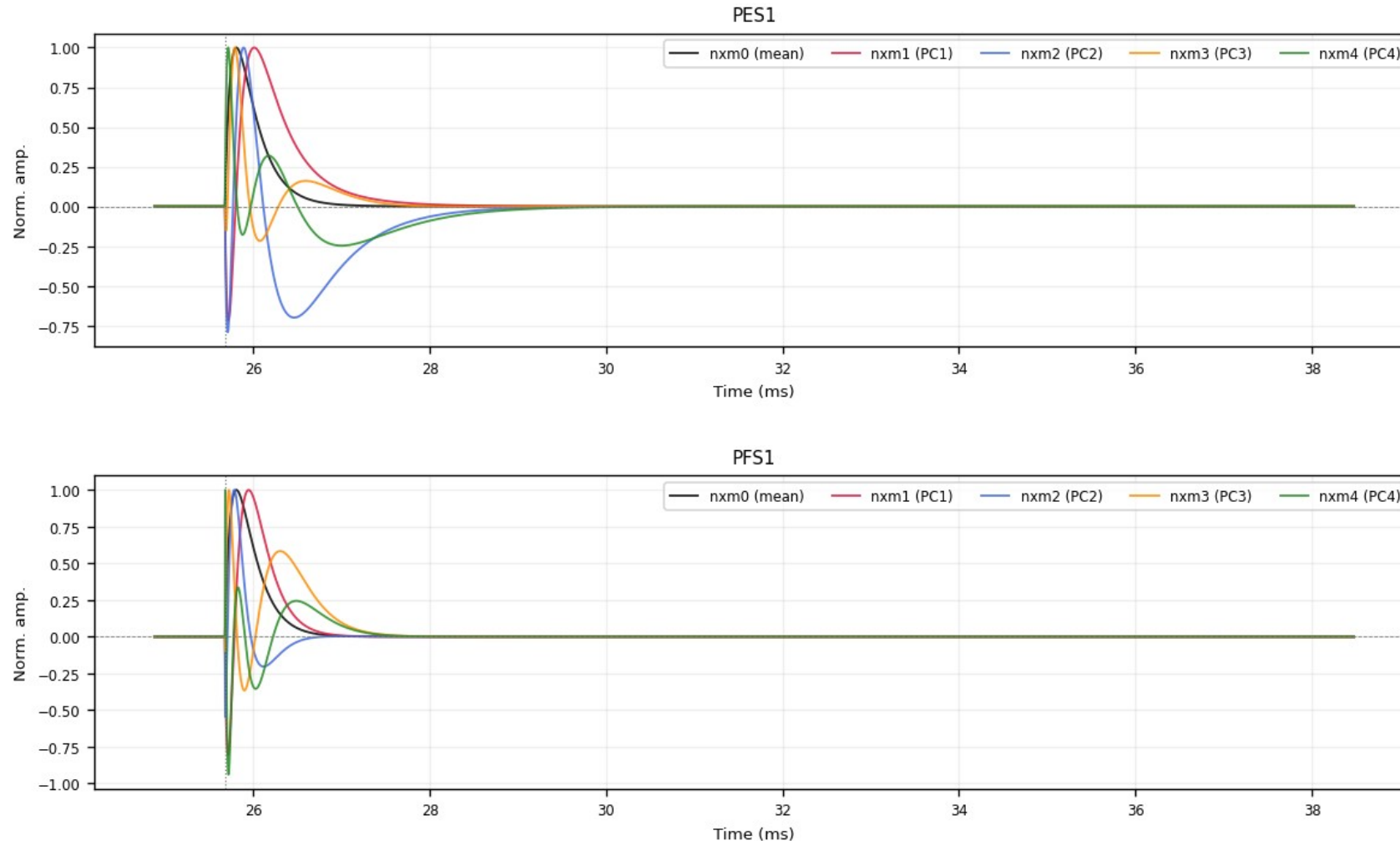




# Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (3/6)

figure: zip7\_pca\_templates.png, channel panels 5–6 of 11, top to bottom

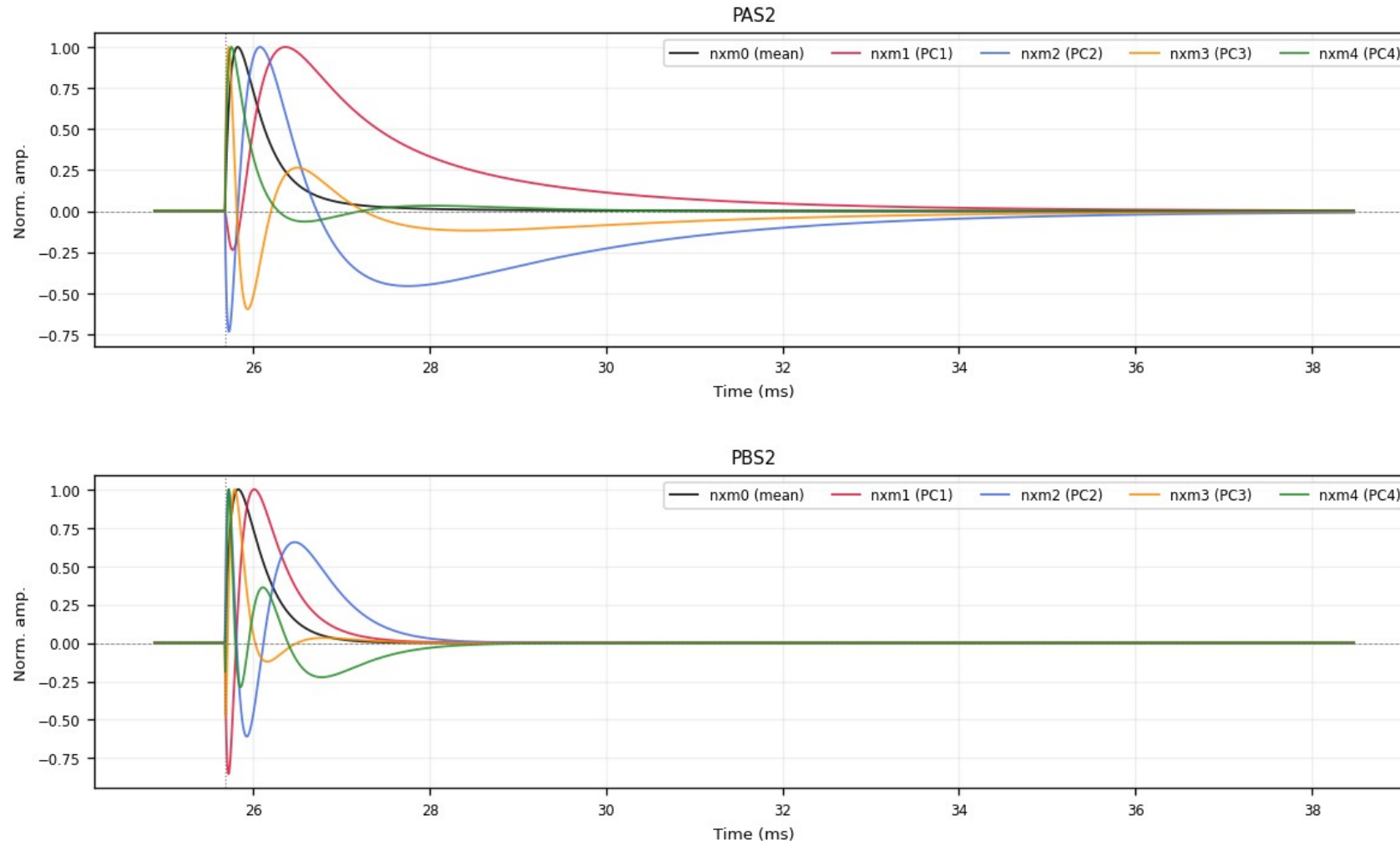
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



# Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (4/6)

figure: zip7\_pca\_templates.png, channel panels 7–8 of 11, top to bottom

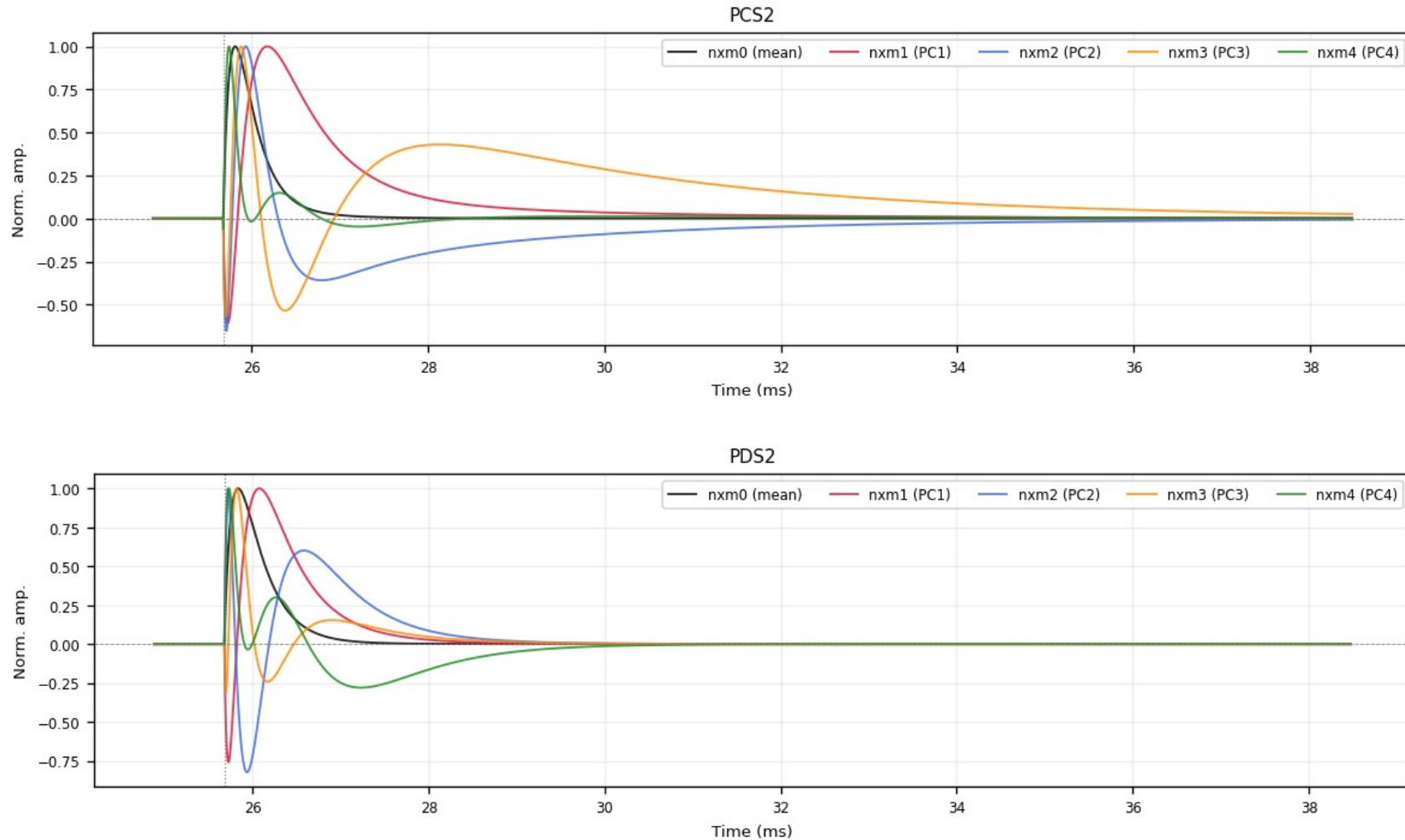
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



# Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (5/6)

figure: zip7\_pca\_templates.png, channel panels 9–10 of 11, top to bottom

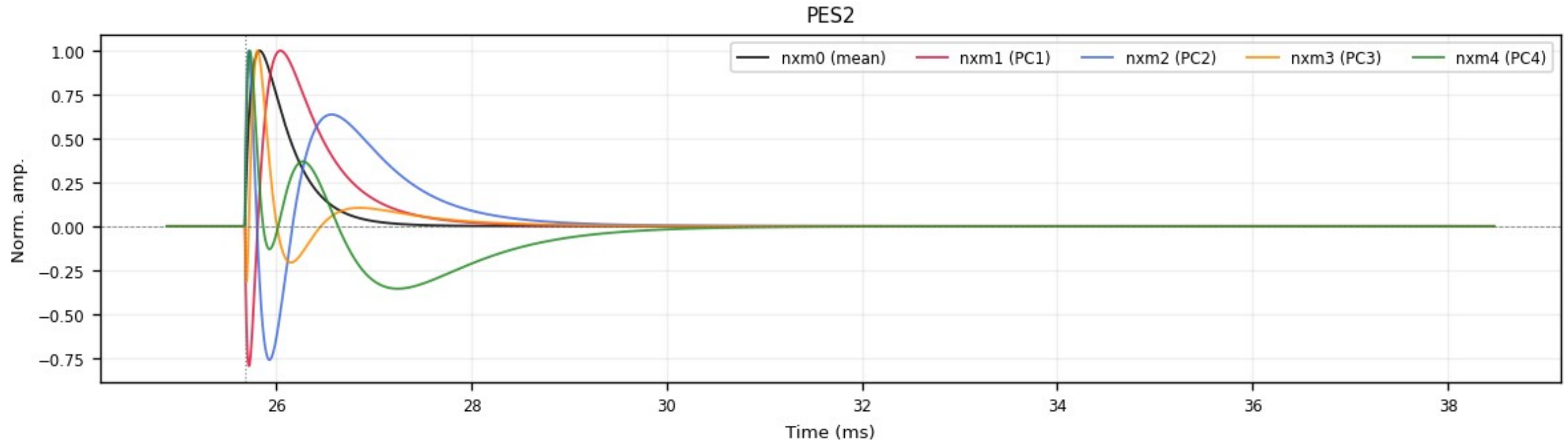
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



# Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (6/6)

figure: zip7\_pca\_templates.png, channel panels 11–11 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



# Backup · Z7 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip7\_\*.png

Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates: PT = all channels, PS1 / PS2 = side-1 / side-2 only.

